

An Agent-Based Model for the Social Collapse of Calhoun’s Universe 25 Experiment

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Abstract

We ask which *local* interaction rules suffice to reproduce, qualitatively, the social collapse of Calhoun’s Universe 25, where a mouse colony with unlimited resources stopped reproducing and died out. We study a lattice agent-based model in which each individual carries a social state, partly inherited, that deteriorates under crowding, and we evaluate seventeen observed patterns run by run against null models. Crowding and a long-lived demography alone already produce the envelope of the collapse: a peak well below capacity, the end of natality, largely the social failure of late-born cohorts, and extinction. Four local rules (a juvenile plasticity window, social learning, crowd aversion and refuges of limited capacity) add coexisting crowded and isolated deteriorated classes, absent from most null models; the persistence of the isolated class follows from the refuge design. The full set of patterns holds in 99.5% of the runs; with a small background mortality or spread initial ages it falls to 55–63%, only through a generic timing criterion. The simulated dynamics is a boom of one or two generations rather than Calhoun’s multi-generational sequence.

1 Introduction

Beginning in July 1968, John B. Calhoun followed a population of house mice in a closed enclosure that he called *Universe 25* [5]. Food, water and nest space were in excess, predators and epidemic disease were excluded, and emigration was impossible. From four founding pairs the colony grew roughly exponentially. It stopped growing at about 2200 animals, well below the 3840 the nest boxes could shelter. The last young to survive weaning were born shortly after the peak, and the population then aged out without ever recovering. Along the way Calhoun described a breakdown of social behaviour:

- withdrawn males gathered in pools at the centre of the floor;
- nursing females attacked and abandoned their young;
- there appeared the unwounded, non-reproducing “beautiful ones”.

Small groups moved to uncrowded enclosures during the decline did not reproduce, even with partners raised without crowding (Marsden’s work, reported in [5]; see also [23]). Calhoun read the collapse as a failure of social roles and of the maturation of complex behaviour, not of resources: “density per se was not the major factor” [5]. Section 3 lists the reported facts with page references.

The interpretation of the experiment has always been contested. Its popular message, “density causes pathology”, departed from Calhoun’s own emphasis on social roles and design [26, 25, 1]. Studies of human crowding found weak or no effects of density per se [11, 10]. For modelling, the decisive limitation is that Universe 25 was a single realization. Calhoun himself called the study “observation and reconstruction of a process; it was historical” [5]. A model tuned to the published curve therefore risks fitting one sample path of a strongly stochastic process [16].

We ask instead:

Which *local* interaction rules, with unlimited resources and no rule that depends on the total population, generate robustly across stochastic realizations the part of Calhoun’s sequence that runs from rapid growth to extinction? That part includes deceleration with social deterioration, cessation of reproduction near the peak, a decline without recovery, and differentiated classes of deteriorated animals.

The latency phase and the growth from four founding pairs are part of Calhoun’s sequence, but we do not claim them: our founder protocol reproduces them only partially (Section 9). Our core model is a lattice model in which each mouse carries an age, a sex and a continuous social state $s_i \in [0, 1]$. The social state deteriorates under local crowding, scales conception and pup survival, and is partly inherited. Movement is biased towards acquaintances. We then add a few local rules, each with a neutral value that switches it off and recovers the core model, and each motivated by an observation of the original report (Section 4.2). Most have one principal parameter; the refuge rule R7 has five and was designed after seeing the first candidate (Section 9.1):

- a juvenile window for social plasticity (R1);
- social learning of competence (R2);
- crowd aversion of deteriorated animals (AV);
- refuges of limited capacity preferred by deteriorated animals (R7).

Rule names follow the simulation code; rules tested and discarded are listed in Section 4.2.

Our contributions are the following. We separate throughout what is preregistered from what is post hoc, and what the rules *produce* from what they *encode*.

1. We translate Calhoun’s report, checked against the original, into seventeen falsifiable patterns, five refinements and four anti-patterns. Each has an operational criterion evaluated run by run, following pattern-oriented modelling [14]. We report the discriminating power of every essential criterion against four null models (no crowding damage; full inheritance without social learning; the long-lived demography without new rules; and the calibrated candidate without its four rules) and two partial ablations (no attraction, and the candidate without refuges) (Table 6). The preregistered forms of three criteria (O4, O10, O11) pass in null models; the primary evaluator, not preregistered, redefines them. O4 and O10 remain generic. O11 discriminates when the late cohorts are scored at six weeks of age, but not when they are scored at the end of the plasticity window. An additional desirable pattern (O10b, not preregistered) asks for a growth phase with socially competent adults.
2. We show that the core model without the rules, with a short life span (the short-lived control), is not Calhoun-like. About half of its colonies die in the first collapse, and the others oscillate with recovering natality. The mean-field analysis of that model traces the oscillations to spatial pockets of healthy individuals and to unconditional recovery. Removing the plasticity window from the calibrated model brings back recovery and recurrent natality at fast learning rates.

3. We separate what the rules add from what the demography, the crowding damage and the design of the rules already imply.

- *The generic envelope.* A long-lived demography alone already drives every colony to extinction, so extinction is *not* a result of the rules. With crowding damage and the calibrated attraction, the candidate without its four rules also has a low peak with free space (it passes the primary O4 in 200 of 200 runs, with a peak at 0.45 of the damage-threshold capacity and 74% of the cells empty), an end of natality and no rebound. Scored at the end of the plasticity window, its late-born cohorts also fail to acquire social competence (O11 in 0.755 of the runs, and 0.81 in the long-lived control), through accumulated crowding damage. These patterns verify that the model has a boom and its end; they are not evidence for the rules.
- *Consistency checks of the design.* Some patterns follow from the rules by construction, and we report them as checks of the implementation, not as findings. Transplanted late-phase animals never found a colony (O12), because under the plasticity window R1 an adult cannot recover and conception is proportional to the social state of both partners, which is essentially zero in these animals. The isolated class is persistent (O13p) because the refuge capacity equals the isolation threshold. There are no social deaths (A4) and the decline is senescent (O8) because social mortality is switched off.
- *What the rules add and the null models lack.* Two patterns: the coexistence of crowded and isolated deteriorated classes (O13), which needs crowd aversion or refuges and is a calibrated property of a nearly sessile regime; and the persistence of the isolated class (O13p), which is a design property. Both discriminate against three of the four null models, not against full inheritance without social learning. The failure of the late cohorts (O11) is not counted here. Scored at six weeks of age, as in the primary evaluator, it discriminates against all four null models, but that is partly definitional: pups are born below the deterioration threshold and are still inside the plasticity window. Scored at the end of the window, C1 still passes (0.995), the dependence on the birth state disappears, and O11 discriminates against neither the long-lived nor the rule-free null model (0.81 and 0.755), whose late cohorts end even lower than those of C1 (median $\bar{s}$ 0.07 and 0.06 against 0.13). The failure of the late cohorts is therefore mostly crowding damage.
- *What the dynamics is.* A boom of one or two generations, followed by a plateau without births or deaths and extinction by senescence. About 70% of the births are to the founding animals, and the net reproductive number falls from 3.8 to 0.4 in a single generation (reference ensemble, 200 runs, Appendix D.7): pups are born below the deterioration threshold and learn from adults who have already deteriorated. The phases appear in Calhoun’s order, but B and C are compressed, and we do not claim a cascade over many generations. The ratio of the deterioration time to the generation time, measured on animals past the plasticity window so that newborns do not dilute it, is $\Gamma = t_{\text{det}}/T_{\text{gen}} \approx 1.1$ (1.2 on the founders), against about 3.5 in Universe 25 with a different operational definition; it is a heuristic comparison of time scales, not the cause of the shallow boom. With initial ages spread over 4–52 or 4–80 weeks the classes, the low peak and the failure of the late cohorts survive, but the timing of the end of natality relative to the peak (O5) does not (item 6).

4. We identify which rules are needed, conditionally on the evaluation thresholds. With the class threshold of 10%, isolation at $m \leq 2$, persistence of four weeks and the ensemble threshold of 0.8, the minimal set for the primary outcome is R1, R2 and R7 (candidate C1’, primary

outcome in 0.835 [0.78, 0.88] of 200 runs, borderline because the lower Wilson bound is below 0.8). With a class threshold of 15% it falls to 0.33, so minimality is not a property of the rules alone. Crowd aversion (AV) is a contributing rule: adding it (candidate C1) makes the primary outcome more reliable, through the generic criterion O5, and the decline senescent (O8); it *reduces* the free space at the peak (16% of the cells empty with AV, 52% without). The contrast between C1 and C1' was not preregistered.

5. We test seven preregistered hypotheses (plan fixed on 1 October 2026, before the production runs; docs/preregistered_plan.md). With the primary criteria the null hypothesis is rejected for six after Holm correction and the quantitative prediction holds for four (Q1, Q3, Q4, Q7); with the preregistered criteria the counts are five and four, and Q2 is not rejected. The only change that produces the rejection of Q2 is the post hoc choice of the end of natality in O5 (the 99th percentile of the births instead of the last birth), made after exploratory runs in which the alternative lowered the primary outcome of C1 to about one half (Table 27). Two of the four predictions that hold are consistency checks, Q1 of R1 (above) and Q4 of the demography, and the preregistered contrast of Q3 is confounded by the demography. Q7 is confirmed only in the restricted sense stated in Section 8.4. Removing crowd aversion lowers the primary outcome significantly with the primary criteria, but by much less than predicted; the predicted dependence on the length of the plasticity window (Q6) is not supported, because with the primary O11 the window has little or no effect on the boundary; and a predicted coexistence band (Q5) is refuted. Every deviation from the preregistered plan is listed in Table 27.
6. We report robustness in four senses. Against stochasticity it is high. Locally around C1 it holds in most of the phase planes, with a nest period for pups (0.835 [0.78, 0.88], borderline) and with an asynchronous update (1.00). Globally it is limited and relative to the sampled box: the mean primary outcome over two 15-factor Latin hypercubes is 0.040 [0.026, 0.059], with 2.0% of the points Calhoun-like, and the Sobol' indices are conditional on the factors fixed after screening. Demographically the primary outcome falls to 0.55 with a background mortality of 0.003 per week and to 0.63 with initial ages spread over 4–80 weeks. Natality still ends with the population near its maximum, but the plateau starts earlier, so the last births fall more than half a peak time after its start and O5 becomes marginal (in 83 and 73 of the 90 and 74 runs that fail; the others fail it by the same timing clause). O5 is generic, and the patterns that the rules add (O13, O13p), as well as O11, pass in every one of these runs (Section 9.1).

The model is documented with the ODD protocol [13] (Appendix A), and the code is available in the repository <https://github.com/jfbonder/universe25-abm>.

2 Related work

Calhoun's experiments and their reception. Calhoun's programme of crowding experiments began with wild Norway rats in an outdoor enclosure. There the population stabilized at about 150 adults, far below what adult survival would have allowed, because of very high infant mortality [2, 3]. It continued with laboratory "universes" for rats and mice, of which Universe 25 is the best known [5]. His theoretical writings stress optimal group size and the saturation of social roles rather than physical density [4]. The scientific and cultural history of these experiments, including the gap between Calhoun's interpretation and its popular uptake, is treated in detail in [26, 25, 1, 9]. Earlier confined-population studies of house mice and other rodents had already reported that, with abundant food, population growth stops through reduced litter survival and reproduction mediated

by social interference [30, 20]. This is the empirical basis of the coupling between social state and newborn survival in our model.

Models of Universe 25. Quantitative modelling of the experiment is very recent. Gwizdalla and Duš [16] proposed an agent-based simulation with a daily time step and empirically motivated life tables and litter-size distributions. They calibrate the growth phase against the reported population and number of adults at day 315. The decline is produced by deviation factors that are triggered when the total population exceeds critical values, together with prescribed probabilities of litter abandonment and of becoming a “beautiful one”. They obtain the qualitative shape of the curve and approximate the timing of the peak. They also emphasize the large dispersion across replicates and the lack of a methodological basis for the choice of rules. Mantzaris [22] attributes the collapse to the progressive loss of uncertainty about the social hierarchy in a fully visible enclosure, implemented through a game-theoretic payoff matrix whose parameters change across generations. Roman [27] fits maximum-entropy unimodal curves to the population time series without a mechanistic model.

Our approach differs from these in three respects:

1. all rules are local in space and use no global population thresholds;
2. the deterioration and its maternal transmission are dynamical variables rather than prescribed probabilities;
3. validation is based on multiple qualitative patterns evaluated as distributions over stochastic replicates, not on curve fitting.

Social stress, maternal effects and population cycles. The idea that crowding acts through physiological stress on reproduction goes back to Christian’s adreno-pituitary hypothesis [6]. In the population-cycle literature, the transmission of individual quality from mother to offspring—the *maternal effect hypothesis*—produces delayed density dependence and cycles in simple two-variable models of density and mean quality [12, 19]. Field evidence shows that maternal stress reduces reproduction and is carried over to the offspring [29]. Our inherited social state is a spatially explicit, individual-based version of this mechanism. Consistent with that theory, its mean-field limit has an oscillatory regime, which we show is not the one realized in Universe 25.

Allee effects and stochastic extinction. Because reproduction requires that partners meet in the same cell, the model has a component Allee effect at low density [31, 8]. This has to be distinguished from social deterioration as an explanation for the absence of recovery. Extinction is ultimately driven by demographic stochasticity acting on an absorbing state. Mean times to extinction and their scaling with population size, quasi-stationary behaviour and extinction from the troughs of cycles are reviewed in [24]. These provide the reference for our survival analysis of extinction times.

Methodology for agent-based models. We describe the model with the ODD protocol [13] and validate it in the spirit of pattern-oriented modelling [14]. For sensitivity analysis of stochastic agent-based models we follow the recommendations of ten Broeke et al. [32]: exploratory one-factor-at-a-time analysis to uncover mechanisms, followed by variance-based global indices [28]. The number of replicates is chosen by the stability of output statistics [21]. Although our aim is qualitative, the region of parameter space compatible with the reported patterns can be characterized with approximate Bayesian computation using pattern-based summary statistics [17, 33].

3 Observations to reproduce

Because Universe 25 is a single realization, we do not fit the population curve. Instead we extract from Calhoun’s report [5] a set of qualitative, falsifiable patterns and evaluate each of them as a *distribution over stochastic replicates*, in the spirit of pattern-oriented modelling [14]. Every pattern is translated into an operational criterion on simulated time series. The criterion returns PASS, MARGINAL, FAIL or NOT APPLICABLE for each run. For an ensemble we report the fraction of PASS runs with a 95% Wilson interval. An essential (E) pattern is considered reproduced if this fraction is at least 0.8; a desirable (D) pattern if it is at least 0.5. A criterion is informative only if it can fail. For every essential pattern we therefore also report its pass rate in four null models run from the same initial condition: no crowding damage ($\gamma = 0$), full inheritance without social learning ($-R2^*$), the long-lived demography without new rules, and the calibrated candidate without its four rules; and in two partial ablations, no attraction ($\alpha = 0$) and the candidate without refuges (C0) (Table 6, Section 8.2). A pattern that passes in at least two of the four null models is *generic* and is not counted as evidence. The preregistered forms of O4, O10 and O11 pass in null models, and the primary evaluator, which is not preregistered, redefines them (marked § in Table 1): O10 is measured on adults, because pups are born with $s \leq w s_f \leq 0.2$ and dilute the mean in any growing colony; O11 does not pass when no animal is born in its window; and O4 also requires free space at the peak. O4 and O10 remain generic. O11, scored at sexual maturity (six weeks), discriminates against every null model, but in part because pups are born below its threshold; scored at the end of the plasticity window it also passes in most runs of the long-lived and rule-free null models (0.81 and 0.755; Section 8.2), so the failure of the late cohorts is mostly crowding damage. The free-space condition of O4, $f_0 \geq 0.10$, turns into a criterion what had been an informal target of the calibration, used to discard candidates (Section 5). An additional desirable pattern, O10b, asks for a growth phase with socially competent adults, as in Calhoun’s phase B. These redefinitions were made after seeing the data and are listed among the deviations from the preregistered plan (Table 27). Patterns that follow from the rules by construction are marked ‡ and reported as consistency checks. One model step is one week. Calhoun’s dates are converted to weeks and enter only through dimensionless ratios: t_{pk} is the time of the population peak, N_{pk} the peak size, and t_{B0} the time at which surviving births cease. The table states the criteria with these symbols; the primary evaluator replaces t_{pk} by the start of the plateau t'_{pk} (the first week with $N \geq 0.98 N_{max}$) and, in O5, t_{B0} by the week t_{B99} at which 99% of the births have occurred (Section 7, Table 27). Table 1 lists the patterns, and Table 2 the numbers taken from the original report. Page numbers refer to [5].

Table 1: Qualitative patterns (O) and anti-patterns (A). E = essential, D = desirable, X = exploratory. [†]O13p is not preregistered (defined after calibration) and enters the primary outcome O17p = O17 $\wedge$ O13p (Section 9.1). O17p is strict: a not-applicable essential pattern counts as a failure (except O1–O3 when $N_0 > 20$, which are excluded by design), and an evaluation error is never a pass. O17 alone counts not-applicable patterns as passes, so we do not cite it. [‡]Consistency check of the design: the pattern follows from the rules by construction (O12 from R1 and $p_{\text{conc}} \propto s_f s_m$; O13p from R7 with $c_R = 2$; O8 and A4 from $\mu_s = 0$) and is reported as a check, not as a finding (Section 9). [§]Not preregistered: the preregistered form also passes in null models (Table 18); in the primary form O4 and O10 are still generic. Primary evaluator: $t_{\text{pk}} \rightarrow t'_{\text{pk}}$ in every criterion, $t_{B0} \rightarrow t_{B99}$ in O5, f_0 measured at t'_{pk} , crowded class at $m > 8$. $K_\gamma = m_c |G|$ is the population at which every cell would sit at the damage threshold; it is not a physical capacity. Full operational definitions are in the supplementary specification.

ID		Observation in Universe 25 (page in [5])	Operational criterion
O1	E	104 days of “considerable social turmoil” before the first litters (p. 82)	founder scenario only: $t_{\text{lag}}/t_{\text{pk}} \in [0.10, 0.35]$ (Calhoun 0.19)
O2	E	Exponential growth, doubling time “about 55 days” (p. 83)	$R^2 \geq 0.90$ for $\log N$ in the growth window
O3	E	Doubling time rises to ≈ 145 days from $N = 620$ (pp. 83–84)	$r_B/r_C \geq 2$, phase C lasting $\geq 0.1 t_{\text{pk}}$
O4	E [§]	Peak of 2200 versus shelter for 3840, food for 9500 and water for 6144; 20% of nest boxes empty at the peak (p. 82)	$N_{\text{pk}}/K_\gamma \leq 0.8$ and empty fraction of cells at the peak $f_0 \geq 0.10$
O5	E	Growth “abruptly ceased” on day 560; last young surviving weaning born around day 600 (p. 85)	$N(t_{B0})/N_{\text{pk}} \geq 0.7$ and $(t_{B0} - t_{\text{pk}})/t_{\text{pk}} \leq 0.5$
O6	E	No recovery; the demise “contradicts prior knowledge” that remnant groups reinitiate growth (p. 85)	no rebound of $\geq 0.1 N_{\text{pk}}$, including slow regrowth still rising at censoring; birth-rate hysteresis $A_H \leq 0.1$
O7	E	Total demise; last male projected dead on day 1780 (p. 85)	run: extinct by T ; ensemble: $P_{\text{ext}}(T) \geq 0.9$ (Kaplan–Meier)
O8	D [‡]	Decline through senescence; mean age of survivors 776 days (p. 85)	$(T_{\text{ext}} - t_{\text{pk}})/t_{\text{pk}} \geq 1$ (Calhoun ≥ 1.84 observed, 2.2 projected)
O8b	D	Slow initial decline: $N = 2056$ on day 736 (p. 83)	$N(1.31 t_{\text{pk}})/N_{\text{pk}} \geq 0.8$ (Calhoun 0.93)
O9	D	27 survivors (23 F, 4 M), all older than 987 days (p. 83)	no young in the remnant; female fraction > 0.5
O10	E [§]	“death of societal organization by the end of Phase C” (p. 85)	adult mean social state falls below 0.5 before the peak and stays below it at the peak: $t(\bar{s}_{\text{ad}} < 0.5) < t_{\text{pk}}$, $\bar{s}_{\text{ad}}(t_{\text{pk}}) < 0.5$
O10b	D [§]	Phase B: 14 functional social groups during growth (days 104–315; pp. 83–84)	adults socially competent during growth: $t(\bar{s}_{\text{ad}} < 0.5) \geq 0.5 t_{\text{pk}}$
O11	E [§]	From mid-phase C young “prematurely rejected by their mothers”; complex behaviours fail to mature (p. 85)	social state at maturity decreasing across cohorts, and < 0.2 for the cohorts born in the second half of natality; not applicable with fewer than 30 such animals (counts as failure in O17p)

ID		Observation (cont.)	Criterion
O11b	D	Only 18% of 148 late-phase-C females had ever conceived (p. 85)	fraction ever conceived ≤ 0.3
O12	E [‡]	Small groups moved to uncrowded universes in mid-phase D failed to reproduce, even with partners raised without crowding (pp. 85–86)	transplant in silico: $P_T(D) \leq 0.1$, $P_T(B) \geq 0.5$
O13	E	Withdrawn males pooled at the centre, wounded by other withdrawn males; “beautiful ones” unwounded, never courting or fighting (pp. 84–85, 88)	instantaneous coexistence of crowded ($m > m_c$) and isolated ($m \leq 2$) deteriorated classes, each $\geq 10\%$ of adults
O13p	E ^{†‡}	As O13, with a class that persists	isolated class persistent (≥ 4 consecutive weeks)
O13b	D	Beautiful ones are a late, unsocialized cohort, “the last half of the population born” (p. 85)	isolated class born later than crowded class; $\geq 50\%$ born after $0.8 t_{pk}$ (post hoc variant: after the median birth time)
O14	D	“Behavioural sink” on some hoppers (p. 81); births from 13 to 111 per brood group (pp. 83–84)	dispersion index ≥ 2 at the peak; birth Gini above the $\alpha = 0$ control
O15	D	Conception declined, foetal resorption and pre-weaning loss increased; no surviving young after day 600 (p. 85)	infant mortality increasing, ≥ 0.9 after t_{B0}
O15b	D	Conceptions continued until about day 920 (p. 85)	$(t_{C0} - t_{B0})/t_{pk} \geq 0.1$ (Calhoun 0.57)
O16	X	Nursing females aggressive towards own young; withdrawn males attack each other (pp. 84–85)	only with aggression rules
O17	–	Robustness criterion, not an observation. Motivated by the single realization: “it was historical” (p. 88)	run is Calhoun-like: all applicable E patterns pass, no anti-pattern; ensemble fraction ≥ 0.8 . Reported only through the strict primary outcome O17p
A1		Logistic saturation at capacity	plateau within 10% of N_{pk} for ≥ 100 weeks
A2		Cycles with recovery of births	births resume to $\geq 10\%$ of maximum, or ≥ 2 rebounds
A3		Demographic collapse before social collapse	per-capita fecundity halves only after t_{pk}
A4	‡	Adult death from deterioration as main cause	social share of deaths ≥ 0.5

4 Model

The model is an individual-based, discrete-time stochastic process on a finite lattice. One time step represents one week. Each individual carries four state variables: a position, an age, a sex and a *social state* $s_i \in [0, 1]$, which summarises its capacity for normal social and reproductive behaviour ($s = 1$ fully competent, $s = 0$ completely deteriorated). Individuals also keep a memory of past contacts through a sparse affinity matrix. All rules are *local*: no individual has access to the population size or to any global statistic.

We first describe the *core model* (Section 4.1): the lattice, the affinity matrix, movement with attraction towards acquaintances, crowding damage to the social state, reproduction and senescent mortality. Section 4.2 introduces four additional rules, R1, R2, AV and R7; each has a neutral value that switches it off and leaves the core model unchanged. Section 4.3 defines the four calibrated candidates C0, C1', C1 and C2 and the controls against which they are compared, and Section 4.4

Table 2: Numbers reported in [5] (days after colonization, DAC) and their conversion to weeks. The model has no weaning period: “last young surviving weaning” is compared with the end of surviving births in the model (t_{B0} , pups that survive birth), which is an earlier event in the life of a pup.

Event	DAC	weeks	page
4 pairs of 48-day-old BALB/c mice introduced (9 July 1968)	0	0	82
First litters (end of phase A)	104	14.9	82
End of phase B: 620 weaned (150 adults in 14 groups, 470 juveniles)	315	45	83–84
Peak, $N = 2200$	560	80	83, 85
Last young surviving weaning	~ 600	85.7	83, 85
$N = 2056$	736	105	83
Last conception	~ 920	131	85
122 survivors (22 M, 100 F)	1444	206	85
27 survivors (23 F, 4 M), all older than 987 days	1588	227	83
Projected death of last male	1780	254	85

the dimensionless groups, the scales and the regimes we expect. The full ODD description is in Appendix A. Fig. 1 summarises the ingredients and Table 3 lists all parameters.

4.1 Core model

Space and affinity. Individuals live on an $L \times L$ lattice G ($L = 30$, $|G| = 900$) with hard boundaries. Any number of individuals may share a cell; m_i denotes the number of occupants of the cell of i , itself included. Every pair of individuals at Chebyshev distance ≤ 1 at the end of a week reinforces its affinity, $F_{ij} \leftarrow (1 - \eta)F_{ij} + \eta$. All other pairs decay, $F_{ij} \leftarrow (1 - \delta)F_{ij}$, and pairs with $F_{ij} < \varepsilon$ are forgotten ($\eta = 0.2$, $\delta = 0.01$, $\varepsilon = 10^{-6}$). The affinity memory is therefore long, $1/\delta = 100$ weeks, and is built only by proximity.

Movement. Every week each individual moves to one of the nine cells y of its Moore neighbourhood, its own cell included, cells outside G excluded. All individuals move simultaneously, using the positions at the start of the week. The probability of choosing y is proportional to

$$W_i(y) = [1 + \alpha A_i(y)] \underbrace{e^{-\kappa(1-s_i)^p n_y}}_{\text{aversion}} \underbrace{[1 + \pi(1 - b_i)^q \mathbf{1}_{\mathcal{R}}(y)] [1 - \mathbf{1}_{\mathcal{F}}(y)]}_{\text{refuges}}, \quad A_i(y) = \sum_{j \in y} F_{ij}, \quad (1)$$

where n_y is the number of *other* individuals in y . A refuge cell is *full* if it already holds $n_y \geq c_R$ other individuals; $\mathcal{R}$ is the set of non-full refuges and $\mathcal{F}$ the set of full ones, so that $\mathbf{1}_{\mathcal{R}}(y) = 1$ if y is a non-full refuge and 0 otherwise, and $\mathbf{1}_{\mathcal{F}}(y) = 1$ if y is a full refuge and 0 otherwise. In the core model only the first factor is present ($\kappa = 0$, no refuges, $\mathcal{R} = \mathcal{F} = \emptyset$): the parameter $\alpha \geq 0$ sets the attraction towards acquaintances. The aversion and refuge factors are the additional rules of Section 4.2.

Social state. After moving, the social state is updated with the local occupancy,

$$s_i \leftarrow \min\left\{1, \max\{0, s_i + \varrho_i - \gamma(m_i - m_c)_+\}\right\}, \quad (2)$$

where $\gamma(m_i - m_c)_+$ is the damage inflicted by each occupant in excess of the community size $m_c = 8$ ($\gamma = 0.1$). In the core model the recovery term is a constant, $\varrho_i = r$ ($r = 0.01$). Deterioration is thus a purely local, density-triggered process, and its convexity in the local occupancy makes it act as a switch (Section 6).

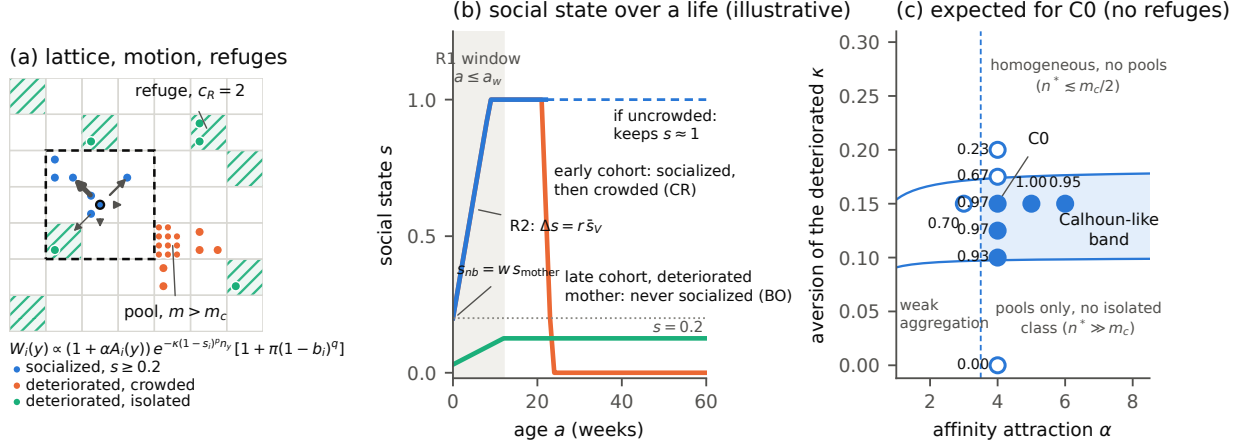


Figure 1: Schematic of the model. **(a)** A focal agent (ringed) chooses a cell of its Moore neighbourhood (dashed) with the weight of Eq. (1); arrow width is proportional to the weight. Deteriorated individuals gather in pools ($m > m_c$) or occupy refuges (hatched, capacity $c_R = 2$). **(b)** Illustrative trajectories of s versus age, from Eqs. (2)–(5) with fixed environments (not a simulation). A pup starts at $w s_{\text{mother}}$ and gains competence only inside the window $a \leq a_w$ (R1), at a rate set by its neighbours (R2). **(c)** Phase diagram expected for C0 (no refuges). Crowded and isolated classes coexist in the shaded band, $5.5 \lesssim n^* = 1/\kappa - 1/\alpha \lesssim 10$ with $\alpha \gtrsim 3.5$. Points are calibration ensembles of C0 labelled with f_{O17} ; filled markers have $f \geq 0.8$. With refuges the band disappears (Q5, refuted; Section 8).

Reproduction. Every reproductive female ($a_{\min} \leq a \leq a_{\text{rep}}$) that shares her cell with at least one reproductive male chooses one of them uniformly at random, and conceives with probability

$$p_{\text{conc}} = p_0 s_f s_m, \quad p_0 = 0.3. \quad (3)$$

The litter size $\ell \in \{3, \dots, 10\}$ is drawn from a fixed distribution with mean $\bar{\ell} = 6.55$. Each pup survives birth with probability $q_0 s_f$ ($q_0 = 0.95$), so deteriorated mothers lose their young. Newborns are placed in the mother’s cell, have a 1:1 sex ratio, and inherit $s_{\text{nb}} = w s_f + (1 - w) s_0$. The core model has full inheritance, $w = 1$; this is its central hypothesis, that the lack of maternal care propagates to the offspring. There is no gestation delay, and a female may give birth every week.

Mortality. Every individual dies at the end of the week with probability

$$p_{\text{death}}(a, s) = \left[1 + e^{-k(a - a_{\max}[1 - \mu_s(1 - s)])} \right]^{-1}, \quad k = 0.15. \quad (4)$$

The factor $\mu_s \geq 0$ would advance the senescence of deteriorated individuals. Every model in this paper, candidates and controls, has $\mu_s = 0$, so adult death is purely senescent. Calhoun attributes the decline to senescence and describes the beautiful ones as “physically healthy” [5]. Anti-pattern A4 and the senescent character of the decline (O8) therefore hold by construction: they are consistency checks of the design, not results. There is no mortality before senescence. The population therefore cannot decrease before the first senescent deaths: once natality stops it sits on a plateau, and $t_{\text{pk}} = \arg \max N$ falls on the last birth that precedes the first death (exactly so in 69% of the runs of C1). A constant background hazard μ_{bg} , independent of age and s , is a reported variant; it acts at every age, pups included. With $\mu_{\text{bg}} = 0.003$ per week the plateau shrinks from 51 to 17 weeks

and the primary outcome falls to 0.55 [0.48, 0.62], through O5 alone, which becomes MARGINAL rather than failing (83 MARGINAL and 7 FAIL among the 90 runs that do not pass). Natality still ends at the population maximum, $N(t_{B99}) \approx 0.99 N_{pk}$. What moves is the start of the plateau: deaths now balance the last births, so t'_{pk} comes earlier and the last 1% of the births falls more than $0.5 t'_{pk}$ after it. The coexisting classes (O13, O13p), the patterns that discriminate against the null models, pass in every run, and so does O11, also with $\mu_{bg} = 0.008$, where O5 fails in 70 of 200 runs (Section 8.5).

Longevity, order of events, initial condition. The candidates use $a_{max} = 120$ and $a_{rep} = 80$ weeks (the short-lived control below uses 60 and 50). These values match the menopause at about 560 days and the frequent longevity of about 800 days in Universe 25 [5], and they allow a decline governed by ageing (O8). Each week the order is: movement, affinities, update of s , reproduction, ageing, mortality. The initial condition is dispersed: $N_0 = 200 + 200$ individuals of age 4 weeks with $s = 1$, at random positions. Non-synchronous variants draw the initial ages uniformly over 4–52 or 4–80 weeks; the primary outcome is then 0.93 and 0.63, again through O5 alone and by the same mechanism (with 4–80 weeks, 73 of the 74 runs that do not pass are MARGINAL, with $N(t_{B99}) \geq 0.97 N_{pk}$; Section 8.5). The founder protocol is in Appendix D.6.

4.2 Minimal additional rules

The core model reproduces the explosive growth and a density-triggered loss of social state, and with the long life span nearly every colony dies too (Section 8). With a short life span ($a_{max} = 60$) it does not reproduce three facts of Universe 25: the irreversibility of the collapse, the failure of the young born during the crowded phase, and the coexistence of distinct classes of deteriorated animals. With the long-lived demography used as a null model the first two are largely there: the colony does not rebound (O6 and O7 in 0.99, 1.00 without the rules at $\alpha = 4$) and, scored at the end of the plasticity window, its late cohorts fail too (O11 in 0.81, and in 0.755 without the rules; Section 8.2). Only the coexistence of classes is missing. R1 is still needed once refuges exist, because refuges create pockets of low density in which, without an irreversible window, deteriorated animals recover and breed again: without R1 natality resumes (O6 and O7 in 0.22, O17p 0.21), and C1 without R1 and R2 also fails them (0.35, with O13p in 0.05), whereas without R1 and without refuges the collapse stays irreversible in most runs (O6 and O7 in 0.85 of 40 runs without R1 and R7; 1.00 without any rule; Appendix C.4). R1 is what lets a persistent isolated class coexist with an irreversible collapse. A fourth fact, the low peak with free space, needs no new rule. With the weak attraction $\alpha = 1.4$ of the long-lived control (Section 4.3) the peak reaches 0.85 of K_γ , with 59% of the cells empty, so O4 fails only through the peak density. With the calibrated $\alpha = 4$ and none of the rules below, the peak stays at 0.45 with 74% of the cells empty, and the primary O4 passes in every run (Table 6). The low peak with free space is therefore a property of strong attraction with crowding damage, not of the rules. We add one rule for each of the three facts (R1, R2, and AV or R7 for the classes), justified by an observation of the original report [5]. Each rule has one principal parameter, but not only one: R2 also fixes the partial inheritance (w, s_0), and R7 has five parameters (f_R, c_R, π, q and the basis b) plus the choice of a hard capacity, and was designed after C0 had failed to produce a persistent isolated class (Section 5). Whether a rule is *necessary* depends on the others. In C0 each of R1, R2 and AV is necessary (Section 5.2). With refuges (R7), AV is no longer necessary for the primary outcome; it then only shapes the spatial organisation (Section 8 and the paragraph on AV below). The minimal set for the primary outcome is therefore R1 + R2 + R7, which defines the candidate C1' (Section 4.3). This minimality is conditional on the evaluation thresholds (class threshold 0.10, isolation $m \leq 2$, persistence ≥ 4 weeks, ensemble threshold 0.8; Section 8.3): C1' passes the primary

outcome in 0.835 [0.78, 0.88] of the runs, with a lower bound below the threshold.

R1: critical window (a_w). Recovery is only possible for $a \leq a_w = 12$ weeks: $\varrho_i = 0$ for older individuals, while crowding damage acts at every age. An adult is therefore a ratchet, $\dot{s}_i \leq 0$. *Motivation.* Behaviours that failed to develop in youth were not acquired later. Animals moved from the mid-third of phase D to new universes at very low density, “even placing them with adequate sex partners ... that had matured in uncrowded conditions”, showed no recovery of reproductive behaviour (H. Marsden’s work reported in [5]; see also [23]). R1 makes a deteriorated cohort unable to recover by individual means. Since conception scales as $s_f s_m$, an adult with $s \simeq 0$ can neither recover nor reproduce, whatever its partner. The failure of phase-D transplants (O12) is therefore a consistency check of R1, which was motivated by the same observation, and not an independent result.

R2: juvenile socialisation by neighbours (λ , with partial inheritance w). Recovery is *learned* from the social environment:

$$\varrho_i = r[(1 - \lambda) + \lambda \bar{s}_{V(i)}], \quad \bar{s}_{V(i)} = \frac{1}{|V(i)|} \sum_{j \in V(i)} s_j, \quad (5)$$

where $V(i)$ contains the other individuals of the 3×3 block centred on i , and $\bar{s}_{V(i)} = 0$ for an isolated individual. We use $\lambda = 1$, $r = 0.1$, and partial inheritance $s_{nb} = w s_f$ with $w = 0.2$, $s_0 = 0$. With R1, Eq. (5) acts only on juveniles: a pup starts with a fraction of its mother’s competence and must acquire the rest from competent neighbours before age a_w . *Motivation.* “By midway in phase C essentially all young were prematurely rejected by their mothers. They started independent life without having developed adequate affective bonds” [5]. Competence is transmitted socially. Once competent adults become scarce, pups have nobody to learn from, and the state $s = 0$ is absorbing for the population, not just for an individual. In C1 this transmission is shallow: it fails in a single step. Crowding deteriorates the founders during their own adult life, within about one generation time. Their offspring, about 70% of all births, are born below the threshold and learn little, because their neighbours are mostly other pups and founders that are already deteriorating. The net reproductive number falls from 3.8 daughters per founder female to 0.4 in the first generation born, and 97% of the births belong to the first two generations (Appendix D.7). R2 does not produce a cascade over many generations; it makes the failure of the first generations born during the boom irreversible for the colony. With $w = 0.2$ and $s_0 = 0$ every pup is born with $s_{nb} \leq 0.2$, that is, at or below the threshold that the patterns use for “deteriorated”. Part of the failure of the crowded-born cohorts is therefore definitional: with $s_0 = 0.25$ pups are born at $s_{nb} \approx 0.4$, and O11, which reads the state at the age of maturity ($a_{min} = 6$ weeks), passes in only 8% of the runs, so that the primary outcome falls to 0.08 (Section 8.5). The essential part of R2 is social learning, not inheritance. With refuges, $w = 0$ and $r = 0.15$ give the primary outcome in every run (Section 8). We keep $w = 0.2$ for continuity with the inheritance hypothesis of the core model.

AV: aversion of the deteriorated (κ). Deteriorated individuals avoid occupied cells through the factor $e^{-\kappa(1-s_i)^p n_y}$ of Eq. (1), with $\kappa = 0.15$. The exponent $p = 4$ is fixed, so that the aversion only matters for $s \lesssim 0.3$. The attraction α does *not* depend on s . For a deteriorated individual facing a cell with n acquaintances ($A \simeq n$), the product $(1 + \alpha n)e^{-\kappa n}$ is maximal at

$$n^* = \frac{1}{\kappa} - \frac{1}{\alpha}, \quad (6)$$

which is 6.4 for the calibrated values, while cells of strangers are always avoided. Deteriorated individuals therefore split between groups of a few times m_c , where crowding damage continues, and almost empty cells. *Motivation.* Calhoun distinguishes withdrawn males, “aggregated in large pools near the centre of the floor” and wounded “as a result of attacks by other withdrawn males”, from the beautiful ones, which “never engaged in fighting”, carried “no wound or scar tissue” and “never attempted to enter” social life [5]. In the model the damage term $\gamma(m - m_c)_+$ inside a pool is damage inflicted by other deteriorated individuals, and isolated individuals receive none (objective O13). Making α itself decrease with s instead homogenises the population and raises the peak (Section 5). *Status.* AV is necessary in C0, where it alone generates the isolated class. With refuges it is not necessary for the primary outcome: without AV it passes in 0.835 [0.78, 0.88] of the runs, against 0.995 [0.97, 1.00] (Section 5.2). AV then controls the use of space and the decline. It *reduces* the free space: without it the deteriorated animals gather in a few pools and half of the lattice is empty at the peak ($f_0 = 0.52$), with it $f_0 = 0.16$ (Universe 25: about 20% of nest boxes). It also makes the decline senescent (O8 0.625 against 0.21). Neither enters the preregistered outcome, which is why the minimal candidate C1’ omits AV. The low peak with free space itself needs no rule (see above).

R7: refuges with preference and hard capacity (π). A fixed random fraction $f_R = 0.3$ of the cells are refuges of capacity $c_R = 2$. Nobody may enter, or stay in, a refuge that already holds c_R other individuals. The capacity is *hard*: after each movement step residents keep their place, entrants are admitted in random order up to the capacity, and the rejected ones return to their previous cell; admission is iterated until no refuge exceeds its capacity, and pups that would be born in a full refuge are placed in a neighbouring cell that is not a refuge. A simpler single-pass admission, which does not count returning animals and newborns and lets a refuge hold up to 11 individuals, is a reported variant; it leaves the primary outcome unchanged (0.995 with either rule; Section 8.5). Deteriorated individuals prefer refuges through the factor $1 + \pi(1 - b_i)^q$ of Eq. (1), where b_i is the state that drives the retreat. In C1, $b_i = s_i$ with $q = 4$ and $\pi = 100$. The total refuge capacity is $f_R c_R |G| = 540$ places, about 20% of the adult population at the peak. *Motivation.* The Universe 25 enclosure offered 256 nest boxes, each holding about 15 mice, and would not have limited shelter below 3840 animals. At the peak about 20% of the nest boxes were usually unoccupied, because many mice chose to crowd together. Withdrawn females retreated to the high, less preferred nests [5]. *Mapping.* We use a single mapping. Every cell is a resting site, the analogue of a nest box, and m_c plays the role of the comfortable occupancy (15 per box in Universe 25, $m_c = 8$ in the model), so that $K_\gamma = m_c |G|$ is the analogue of the 3840 places and f_0 of the fraction of empty boxes. The refuges are the analogue of the high, less preferred boxes (the upper one or two of the four levels of each tunnel, 25–50% of the boxes, against $f_R = 0.3$). Their capacity $c_R = 2$ is *not* that of a nest box: it is the isolation threshold $m \leq 2$ of O13, a design choice (Section 5). Neither $|G|$ nor m_c was calibrated to Universe 25, so densities are compared with Calhoun’s only in sign (Appendix D.8).

Both the preference and the hard capacity are necessary with synchronous moves. Without a preference ($\pi = 0$) the refuges stay empty, because a deteriorated individual surrounded by acquaintances prefers an occupied neighbouring cell. With a soft capacity, simultaneous moves produce stampedes (Appendix D.5). The hard capacity is a remedy for the collisions of the synchronous update: with a random sequential update a soft capacity also produces a persistent isolated class (O17p 0.995, $\rho_{pk} = 0.43$ against 0.36; Section 8.5).

Variant of R7 in C2: retreat driven by maternal socialisation. In C2 the retreat is driven by the competence of the mother at birth, $b_i = h_i \equiv s_f(t_{\text{birth}})$, with $h_i = 1$ for founders, $q = 1$ and $\pi = 1000$. Pups of deteriorated mothers retreat, and socialised adults that later deteriorate do not,

so they end up crowded. *Motivation.* The beautiful ones belonged to “the last half of the population born”; they were young animals that never entered social life, not adults that withdrew [5]. Basing the retreat on juvenile competence (the value of s at age a_w) fails. Under R2 almost every cohort born after the first ends the window with $s < 0.1$, so there is no gradient between cohorts.

Rules tested and rejected. Several rules were unnecessary or destroyed some pattern, and are not part of any candidate:

- long-range jumps;
- an attraction that decreases with s ;
- territoriality and defeat of males;
- social mortality;
- bonds restricted to competent pairs;
- movement sub-steps.

Mating in the Moore neighbourhood (`mate_radius` = 1) only matters for the founder protocol.

4.3 Calibrated candidates

We consider four candidates:

C0 core model + longevity + R1 + R2 + AV. This is the minimal model that satisfies the essential patterns evaluable with $N_0 = 400$ when O13 is read instantaneously (O17 in 0.92 of the runs).

C1' core model + longevity + R1 + R2 + R7 ($b = s$), without AV. This is the minimal set of rules for the preregistered primary outcome O17p (all essential patterns and a *persistent* isolated class). See Section 5.2.

C1 C1' + AV, equivalently C0 + R7. AV adds the spatial organisation: it spreads the deteriorated animals over the lattice (the empty fraction at the peak falls from 0.52 to 0.16) and makes the decline senescent. C1 is the reference of the production design.

C2 C0 + R7 with the retreat driven by maternal socialisation ($b = h$). It makes the isolated class a later-born cohort.

The presets are `calhoun_C0`, `calhoun_C1p`, `calhoun_C1` and `calhoun_C2`, and Table 3 lists their parameters.

Controls. A rule is switched off by its neutral value: $a_w = \infty$ for R1; $\lambda = 0$ with full inheritance ($w = 1$) and individual recovery at the core rate $r = 0.01$ for R2; $\kappa = 0$ for AV; no refuge cells ($f_R = 0$) for R7. Two controls are the core model alone, with all four rules switched off, attraction $\alpha = 1.4$ and the dispersed initial condition:

Short-lived control a short life span, $a_{\max} = 60$ and $a_{\text{rep}} = 50$ weeks;

Long-lived control the life span of the candidates, $a_{\max} = 120$ and $a_{\text{rep}} = 80$ weeks.

Table 3 gives both in full. The null models of Section 8.2 are run in a separate ensemble and are defined relative to C1: no crowding damage ($\gamma = 0$); $-R2^*$, C1 with R2 switched off; the long-lived control; and “no rules”, C1 with all four rules switched off, which is the long-lived control with the calibrated attraction $\alpha = 4$. Two partial ablations of C1 complete the comparison: $\alpha = 0$ (no attraction) and C0 (no refuges).

Table 3: Parameters of the short-lived (SL) and long-lived (LL) controls, which are the core model with the four rules switched off, and of the calibrated candidates C0, C1', C1 and C2. Time in weeks. A dash means that the rule is inactive. The code name is the field of **Params** in the **u25** package.

Symbol	Meaning	Code name	SL	LL	C0	C1'	C1	C2
<i>Space, motion and affinity</i>								
L	lattice side (hard walls)	<code>grid_size</code>	30	30	30	30	30	30
α	attraction to acquaintances	<code>alpha</code>	1.4	1.4	4.0	4.0	4.0	4.0
η, δ	affinity reinforcement, decay	<code>eta, delta</code>	0.2, 0.01	0.2, 0.01	0.2, 0.01	0.2, 0.01	0.2, 0.01	0.2, 0.01
κ, p	aversion of the deteriorated	<code>crowd_aversion, _power</code>	–	–	0.15, 4	–	0.15, 4	0.15, 4
<i>Social state</i>								
m_c	community size	<code>m_c</code>	8	8	8	8	8	8
γ	damage per excess occupant	<code>gamma</code>	0.1	0.1	0.1	0.1	0.1	0.1
r	recovery / learning rate	<code>rec</code>	0.01	0.01	0.1	0.1	0.1	0.1
a_w	R1 critical window	<code>plast_age_max</code>	∞	∞	12	12	12	12
λ	R2 weight of neighbours	<code>social_rec_lambda</code>	0	0	1	1	1	1
w, s_0	inheritance $s_{\text{ub}} = w s_f + (1 - w) s_0$	<code>inherit_weight, s_newborn_base</code>	1, –	1, –	0.2, 0	0.2, 0	0.2, 0	0.2, 0
<i>Refuges</i>								
f_R, c_R	refuge fraction, capacity (hard)	<code>refuge_frac, _capacity</code>	–	–	–	0.3, 2	0.3, 2	0.3, 2
π, q	refuge preference	<code>refuge_pref, _power</code>	–	–	–	100, 4	100, 4	1000, 1
b_i	state driving the retreat	<code>refuge_pref_basis</code>	–	–	–	s_i	s_i	h_i (mother)
<i>Demography</i>								
N_0	initial individuals (age 4, $s = 1$)	<code>n_males+n_females</code>	400	400	400	400	400	400
p_0	conception probability	<code>p0</code>	0.3	0.3	0.3	0.3	0.3	0.3
$q_0, \bar{\ell}$	neonatal survival, mean litter	<code>neonatal_survival</code>	0.95, 6.55	0.95, 6.55	0.95, 6.55	0.95, 6.55	0.95, 6.55	0.95, 6.55
$a_{\text{min}}, a_{\text{rep}}$	reproductive ages	<code>repro_age_min, _max</code>	6, 50	6, 80	6, 80	6, 80	6, 80	6, 80
a_{max}, k	senescence age, steepness	<code>a_max, k_mort</code>	60, 0.15	120, 0.15	120, 0.15	120, 0.15	120, 0.15	120, 0.15
μ_s	social advance of senescence	<code>social_mortality_factor</code>	0	0	0	0	0	0

4.4 Dimensionless groups, scales and expected regimes

The qualitative behaviour depends on a few combinations of parameters; the values quoted are for C1. The scales of the model and their counterparts in Universe 25 are listed in Appendix D.8.

- $\Pi_\gamma = \gamma/r = 1$: crowding damage per excess neighbour relative to the weekly learning rate.
- $\Lambda = r a_w = 1.2$: the largest competence a juvenile can learn inside the window. A socialised lineage sustains itself only if $w + \Lambda \langle \bar{s}_V \rangle \gtrsim 1$. Fast growth lowers the competence of the neighbours $\langle \bar{s}_V \rangle$, because novices outnumber teachers (a *learning dilution*). Without crowding the colony goes extinct below a boundary in (w, Λ) and grows without bound at the calibrated values (Q7). When a_w is varied as well, r and a_w do not act through their product alone: the extinction transition is located only to within the spacing of the grid, and the local weight of a_w relative to r is about one half between $a_w = 6$ and 12 weeks and about one between 12 and 24 (Section 8.4). Crowding is the trigger and R2 the amplifier.
- $n^*/m_c \approx 0.8$ (Eq. (6)): the preferred group size of a deteriorated individual relative to the damage threshold. Without refuges it decides between pools, isolation, or both; with refuges it only marks the onset of homogenisation, near $n^* \approx 3\text{--}4$ (Q5).
- $\phi_R = f_R c_R |G|/N_{\text{ad}}(t_{\text{pk}}) \approx 0.2$: refuge capacity per adult at the peak. It bounds the persistently isolated fraction, $\Phi_{\text{BO}}^{\text{pers}} \leq \phi_R$. The bound holds at all 50 points of the refuge plane, so a class of $\geq 10\%$ needs $f_R c_R \gtrsim 0.3$; in that plane a persistent class appears only for $f_R c_R \geq 0.4$ and a strong preference (Section 8.5).
- $n_0 = N_0/|G| = 0.44$ and the Damköhler number $\text{Da} = t_{\text{mix}}/t_{\text{det}}$. Here $t_{\text{mix}} = L^2/(4D_{\text{eff}})$ is the time an individual needs to move across the lattice, with D_{eff} the mean squared displacement over 16 weeks divided by 4×16 , and $t_{\text{det}} = t_{1/2} - t_{\text{on}}$ is the time the animals past the critical window (age $\geq a_w$) need to lose half of their competence ($t_{1/2}$, the first week with their

mean $s < 0.5$) after crowding sets in (t_{on} , the first week with 5% of the individuals in cells with $m > m_c$). These animals no longer learn, and newborns do not enter their mean until they leave the window, so t_{det} measures deterioration and not the dilution by pups born with $s \leq 0.2$. In C1 $D_{\text{eff}} \approx 0.13$ cells² per week (8 exploratory runs), $t_{\text{mix}} \approx 1.7 \times 10^3$ weeks and $t_{\text{det}} \approx 10$ weeks ($t_{\text{on}} = 6$, $t_{1/2} = 16$; reference ensemble), so $\text{Da} \approx 1.7 \times 10^2$. Deterioration is correlated over $\xi = (4D_{\text{eff}}t_{\text{det}})^{1/2} \approx 2$ cells, the size of the contact neighbourhood. In founder colonies the relevant transport is the colonisation front: the filling time $t_{\text{fill}} \approx t_{\text{pk}} \approx 170$ –215 weeks against $t_{\text{det}} \approx 20$ –30 gives $\text{Da}_{\text{fill}} \approx 10$ (Appendix D.6). In Universe 25 mice crossed the enclosure daily and the social organisation took a whole phase C, about 35 weeks, to die, so $\text{Da} \lesssim 4 \times 10^{-3}$. The spatial structure of the model therefore emerges among animals that are almost sessile on the time scale of deterioration, which is not Calhoun’s regime. Colonies grow from local nuclei, so ρ_{pk} falls with n_0 , from 0.40 at $N_0 = 400$ to 0.03 for eight founders.

- $\Gamma = t_{\text{det}}/T_{\text{gen}} \approx 1.1$: deterioration time relative to the generation time, $T_{\text{gen}} \approx 9$ weeks (the interval between the median birth weeks of the first and second generations); the interquartile range over runs is 1.0–1.29, and the founders, a closed cohort, give 1.2. The mean over all animals of age $\geq a_{\text{min}} = 6$ would give 0.8, but that mean is diluted by the first generation, which enters it with $s \leq 0.2$ while still inside the window: when it falls below 0.5, in week 13, the founders still have $\bar{s} = 0.66$ and are 39% of those adults. Deterioration therefore takes of the order of one generation. The value depends on how T_{gen} is defined: with overlapping cohorts the interval between median birth weeks is not a demographic generation time, and in the null models it drops to 5 weeks (4 in some runs), below the age of first reproduction, which inflates their Γ (1.4–1.6). With the standard demographic definition, the mean age of the mothers at birth ($T_{\text{gen}} = 14.3$ weeks in C1), $\Gamma \approx 0.7$ in C1 and 0.60–0.85 in the other arms where it was computed; Γ is of order one under either definition (Appendix D.7). Γ does not set the depth of the boom. With five times less crowding damage ($\gamma = 0.02$) $\Gamma = 1.6$ (1.8 on the founders), and the boom is still one to two generations deep: the mean generation of the births is 1.42 against 1.35, R_{net} of the first generation is 0.52, and O17p passes in 17 of 20 runs. Pups born at $s_{\text{nb}} \approx 0.4$ ($s_0 = 0.25$) deepen the boom more at about the same Γ (1.3): the mean generation is 1.51 and $R_{\text{net}} = 0.62$ (20 runs per point; Appendix D.7). The depth is set in the first step, from the founders to their offspring: pups are born below the deterioration threshold, conception and pup survival scale with the parents’ state, and pups learn from neighbourhoods made mostly of other pups and of founders that are already deteriorating. In Universe 25 the deterioration took the whole phase C, about 35 weeks, with $T_{\text{gen}} \approx 10$ weeks, so $\Gamma \approx 3.5$ (our estimate). With a different operational definition, that is an order-of-magnitude comparison, not a test.
- *Extinction time.* Without mortality before senescence, $T_{\text{ext}} = \max_i(b_i + \ell_i) \simeq t_{\text{last}} + \ell_{\text{max}}$: the time of the last birth plus the longest residual lifetime. In C1 $\ell_{\text{max}} \approx 110$ weeks with an interquartile range of a few weeks, of the order of the inverse slope of the logistic hazard, $1/k \approx 7$ weeks. The spread of T_{ext} is therefore that of t_{last} , the extreme of a sparse tail of births whose rate decays over $\tau_B = (t_{B99} - t_{B90})/\ln 10 \approx 7$ weeks (t_{Bq} is the week by which a fraction q of all births has occurred). At fixed density the median of T_{ext} grows with the logarithm of the system size, from 162 weeks at $L = 15$ to 184 at $L = 60$, and its interquartile range over the median is 0.15 for $L \leq 20$ and about 0.08 for $L \geq 30$ (Appendix D.9). A small dispersion of T_{ext} is a property of how abruptly natality stops and of the size of the system, not a signature of the rules. In the same way $(T_{\text{ext}} - t_{\text{pk}})/t_{\text{pk}} \approx 2.1$ (Universe 25: ≥ 1.84) depends on where t_{pk} falls on the plateau: it is 4.0 if the peak time is the start of the plateau,

t'_{pk} , and the plateau itself shrinks from 51 to 17 weeks with a background mortality of 0.003 per week. Similarly $t_{\text{pk}}/a_{\text{max}} \approx 0.45$ (0.28 with t'_{pk}), against 0.7 in Universe 25.

Expected regimes and their test. Before the production sweeps, the mean field (Section 6) and the calibration suggested four regimes. The production planes test them (Section 8). The planes report the fraction of runs that pass a composite classifier (O17p). That fraction is not an order parameter, and its boundaries mix the mechanism with the thresholds of the definitions. We therefore also report continuous order parameters: ρ_{pk} , the empty fraction f_0 , the class order parameter $\Psi = \min(\Phi_{\text{BO}}^{\text{pers}}, \Phi_{\text{CR}})$ (O13p is $\Psi \geq 0.1$), the net reproductive number of the first generation and τ_B (Section 8.5; the widths of the transitions could not be resolved with 20–30 runs per point).

1. *Recurrent growth* (A2), when recovery is individual and reversible. *Confirmed*: without the window P_{ext} falls from 1 at $r = 0.02$ to 0 for $r \geq 0.2$.
2. *Unbounded growth* (A1), when crowding damage is weak or the population is homogenised. *Confirmed* for $\gamma = 0$.
3. *A single collapse without recovery*, the Calhoun-like regime. Extinction alone does not identify it, since the long-lived demography without new rules also goes extinct. What identifies it is the structure of the collapse, which requires R1 and R2. In C0 the classes are set by (α, κ) , inside the band of Fig. 1(c). With refuges the band disappears (Q5, refuted). No recovery appears in larger systems either: at fixed density, A2 is cleared in all 150 exploratory runs with L from 20 to 84 (30 per size, with the single-pass refuge admission), up to about 21 000 animals (Appendix D.9).
4. *Trivial early collapse*: founders in one cell, where local nuclei deteriorate before the colony spreads ($\text{Da}_{\text{fill}} \approx 10$), or very strong aggregation.

Persistence of the isolated class is a separate axis: it requires $\phi_R \gtrsim 0.1$ and $\pi \gtrsim 100$, and its cohort structure depends on the basis b of the retreat.

5 Calibration of the candidates

The rules of Section 4 were not fitted to a population curve. They were selected, and their parameters set, by maximising the number of qualitative patterns of Table 1 reproduced across stochastic replicates, with the smallest set of rules. This section summarises that process. The development ensembles, the local sensitivity analysis, the diagnosis of the deteriorated classes and the founder protocol are in Appendix D.

Calibration is not validation. The rules and parameters were chosen with the same evaluator and the same patterns that are later reported as reproduced. The production ensembles are statistically independent, but no pattern was held out. Several modelling choices also mirror the operational definitions of the patterns:

- the refuge capacity $c_R = 2$ equals the isolation threshold $m \leq 2$ of O13;
- the persistence requirement of O13 was adopted after C0 had been found to produce only transient isolation, and the refuges were then designed to satisfy it;
- m_c is both the damage threshold and the threshold that defines the crowded class;

- the fraction of empty cells at the peak was used informally to discard candidates that left most of the lattice empty, and it is part of a criterion of the primary evaluator: the primary O4 requires $f_0 \geq 0.10$, so O4 tests a quantity that was an informal target of the calibration.

Reproducing O13 with refuges therefore shows that a persistent isolated class is *possible* under local rules; it is not a prediction. Several further patterns hold by construction and are consistency checks of the design, not results:

- the failure of phase-D transplants (O12, Q1): under R1 an adult cannot recover, and conception scales as $s_f s_m$, so animals with $s \simeq 0$ cannot found a colony whatever their partners;
- the persistence of the isolated class (O13p), through $c_R = 2$;
- the absence of social mortality (A4) and the senescent decline (O8), through $\mu_s = 0$;
- the coincidence of the end of natality with the peak (O5) when the peak time is $\arg \max N$, because without mortality before senescence t_{pk} falls on the last birth before the first death (Section 4.1); measured from the start of the plateau, as in the primary evaluator, O5 is not automatic but generic: it passes in four of the null models (Table 6);
- the life-table law of phase D (Q4), which follows from the demography.

The tests that do not depend on the calibration targets are the relative weight of window and learning rate (Q6), the endogenous collapse without crowding (Q7), both with the restrictions of Section 8.4, and the comparison with null models, in which a pattern that also passes carries no evidence (Section 8). Agreements of magnitudes normalised by t_{pk} , such as $(T_{ext} - t_{pk})/t_{pk} \approx 2.1$ (Calhoun: 2.2 projected), are not tests: they depend on where t_{pk} falls on the plateau (Section 4.4).

Procedure. Each run is scored by the evaluator of Section 3. MARGINAL counts as not passing, and ensemble fractions carry 95% Wilson intervals. The calibration proceeded in five stages, with independent seeds and a horizon of $T = 600$ weeks:

1. screening of rule families (47 points $\times$ 10 replicates);
2. ablations of the selected rules (10 points $\times$ 40);
3. one-at-a-time sensitivity (17 points $\times$ 30);
4. development ensembles of the candidates, with 60–120 replicates and transplants;
5. tests of the founder protocol.

It used about 45 min on four cores. The transplant protocol and the stages are detailed in Appendix D.1.

5.1 The candidates

Table 4 lists the essential patterns of the four candidates of Table 3 in the production ensembles, with the exact hard capacity and the primary evaluator (the start of the plateau t'_{pk} as peak time, O5 measured with the end of effective natality t_{B99} , the primary O4, O10 and O11, and the crowded class $m > 8$). With the dispersed initial condition ($N_0 = 400$ at random positions), every applicable essential pattern passes in at least 80% of the runs of every candidate:

Table 4: Essential patterns of the four candidates in the production ensembles ($n = 200$ each; exact hard capacity; primary evaluator; C0, C1' and C1 from the reference ensemble, C2 from its contrast ensemble; Section 8.1, where the controls are also given). Fractions of passing runs. The 95% Wilson half-width is at most 0.07, and at most 0.02 for fractions of 0 or 1. O17 uses the instantaneous O13; O17p, the primary outcome, also requires a persistent isolated class (O13p). O12: founding transplants of phase-D animals (taken at $1.75t'_{\text{pk}}$) over trials (60 per candidate); with $s \simeq 0$ it is a consistency check of R1. $\rho_{\text{pk}} = N_{\text{pk}}/K_{\gamma}$, with $K_{\gamma} = m_c|G|$ the damage-threshold capacity (not a physical capacity; Universe 25: $2200/3840 = 0.57$), and f_0 , the median fraction of empty cells at the peak (Universe 25: about 20% of nest boxes); both are comparable with Calhoun's only in sign.

Candidate	O4	O5	O6	O7	O10	O11	O12	O13	O17	O17p [95% CI]	ρ_{pk} / f_0
C0 (R1, R2, AV)	1.00	1.00	1.00	1.00	1.00	1.00	0/60	0.92	0.92	0.00 [0.00, 0.02]	0.41 / 0.18
C1' (R1, R2, R7)	1.00	0.855	1.00	0.995	1.00	0.975	0/60	1.00	0.835	0.835 [0.78, 0.88]	0.33 / 0.52
C1 (R1, R2, AV, R7)	1.00	0.995	1.00	1.00	1.00	1.00	0/60	1.00	0.995	0.995 [0.97, 1.00]	0.36 / 0.16
C2 (C1, maternal b)	0.90	0.815	0.98	0.98	1.00	0.995	0/60	1.00	0.725	0.705 [0.64, 0.76]	0.42 / 0.11

- a low peak with free space (O4);
- natality ending at high N (O5);
- no rebound (O6);
- extinction (O7);
- deterioration of the adults before the peak (O10);
- failure of the crowded-born cohorts (O11);
- failure of phase-D transplants (O12);
- two deteriorated classes (O13).

Only the candidates with refugees make the isolated class persistent (O13p) and thus pass the primary outcome O17p. O1–O3 apply only to founder colonies (Appendix D.6). Not all of these patterns carry the same weight. O12 is a consistency check of R1 (see above). Against the null models of Table 6 only O13 and O13p discriminate, against three of the four (not against full inheritance without social learning). O11 discriminates against all four only when scored at six weeks: at the end of the plasticity window it also passes in 0.81 of the long-lived runs and in 0.755 of the runs without rules, so the failure of the late cohorts is mostly crowding damage. O4, O5, O6, O7 and O10 pass in at least two null models and are generic. In particular the low peak with free space (O4) also passes in C1 without its four rules ($\rho_{\text{pk}} = 0.45$, $f_0 = 0.74$): it comes from the attraction $\alpha = 4$ with crowding damage, not from the rules. The development ensembles of Appendix D.2 agree qualitatively.

5.2 Necessity of the rules

Whether a rule is necessary depends on the others. Single ablations are in Table 29 for C0 and in Section 8.3 for C1 and C1'.

Without refuges (C0). R1, R2 and AV are each necessary:

- without R2, O11 and O13 are never reproduced (O17 0/40). Viable pups come from the healthiest mothers and inherit their competence;
- without AV, deteriorated animals collapse into a few pools, 76% of the lattice is empty and the isolated class disappears;
- without R1, adults recover when the colony thins, and O17 falls to 0.40.

With refuges. R7 supplies the isolated class, and AV is no longer necessary for the primary outcome. The rules necessary for O17p are R1, R2 and R7: the single ablations of C1' give O17p of 0 in every case (40 runs each). AV is not: C1', which is C1 without AV, passes O17p in 0.835 [0.78, 0.88], against 0.995 [0.97, 1.00] for C1. This necessity is conditional on the thresholds of the classes and of the ensemble (Section 8.3). What AV controls is the use of space, and it *reduces* the free space rather than supplying it: without AV the deteriorated animals gather in a few pools and half of the lattice is empty at the peak ($f_0 = 0.52$), with AV $f_0 = 0.16$ (Universe 25: about 20% of nest boxes). It also controls the senescent decline, with O8 at 0.21 against 0.625. Partial inheritance is not necessary either: C1 with $w = 0$ and $r = 0.15$ passes O17p in all 120 runs. The essential part of R2 is social learning.

Crowding is the trigger. Without crowding damage ($\gamma = 0$) no colony collapses, and the mean adult social state stays near 0.6. With the calibrated socialisation, the failure of socialisation under R2 needs crowding to start (Q7): crowding deteriorates the founders, the only competent teachers, within about one generation time ($\Gamma \approx 1.1$; Section 4.4 and Appendix D.7).

5.3 What the calibration does not achieve

1. *Founder colonies (O1–O3).* No candidate reproduces latency, exponential growth and deceleration robustly from four pairs. The best point of the founder plane reproduces O1–O3 jointly with O17p in 0.43 [0.27, 0.61] of the runs, and only with $a_{\max} = 200$, outside the preset (Appendix D.6).
2. *Initial condition.* The low peak ($N_{\text{pk}}/K_\gamma \approx 0.4$) holds for the dispersed initial condition. With C0, the peak density falls with the seeding density, from 0.40 at $N_0 = 400$ to 0.12 at $N_0 = 50$. With initial ages spread uniformly over 4–52 or 4–80 weeks the classes, the peak density, O11 and the generational structure are unchanged, but O17p falls to 0.93 and 0.63, through O5 alone. O5 then becomes MARGINAL, not FAIL (73 MARGINAL and one FAIL among the 74 runs that do not pass with 4–80 weeks). Natality still ends at the population maximum ($N(t_{B99}) \geq 0.97 N_{\text{pk}}$ in every run); with older founders the first deaths come earlier and balance the last births, so the plateau starts earlier and the last 1% of the births falls more than $0.5 t'_{\text{pk}}$ after its start. The patterns that discriminate against the null models (O13, O13p) pass in every run, and so does O11. O5 is generic and carries no evidence for the rules with any initial condition (Section 8.5; Appendix D.7).
3. *A boom of one to two generations.* The colony does not go through a cascade over many generations. About 70% of all births are offspring of the founders and 96% belong to the first two generations, the net reproductive number falls from 3.8 (founders) to 0.4 (first generation), and the phases B and C of Calhoun appear compressed: growth with competent adults lasts 13

weeks and two doublings, against 30 weeks and about five in Universe 25 (Appendix D.7; reference ensemble). It also holds, in 20 replicates of each production arm, with a non-synchronous initial condition, a nest period, background mortality and the asynchronous update (mean generation of the births 1.22–1.43), and with five times less crowding damage. It is not caused by fast deterioration: deterioration takes of the order of one generation time ($\Gamma \approx 1.1$; 0.7 with the mean age of the mothers at birth as generation time), and slowing it barely deepens the boom (Section 4.4).

4. *Late cohort (O13b)*. Births are front-loaded: 90% occur before $0.45 t_{pk}$. Hence no retreat mechanism can make half of the isolated class be born after $0.8 t_{pk}$. Basing the retreat on maternal socialisation (C2) makes the isolated class younger than the crowded one, but only reaches MARGINAL, and at the cost of the senescent decline and of O17p (0.705; Appendix D.5).
5. *Fertility (O15, O15b) and sex bias (O9)*. Conception ($\propto s_f s_m$) and pup survival ($\propto s_f$) collapse together, and the model has no sex-specific mortality.
6. *Update scheme*. The candidates were calibrated with synchronous moves, with which refugees with a preference require a hard capacity, because a soft one produces stampedes. With a random sequential update a soft capacity also gives a persistent isolated class, but a higher peak density (O17p 1.00 and 0.995 with hard and soft capacity, $\rho_{pk} = 0.43$ against 0.36; Section 8.5). The hard capacity is therefore a remedy for the collisions of the synchronous update, not a mechanism of its own.
7. *A plateau, not a peak*. After natality dies, the young adult population barely changes until senescence, because there is no mortality before senescence. $t_{pk} = \arg \max N$ thus lies on a plateau, and it coincides with the last birth in 69% of the runs, which makes O5 nearly automatic when it is measured from $\arg \max N$. Every criterion normalised by t_{pk} inherits this ambiguity; the primary evaluator uses the start of the plateau, t'_{pk} , and measures the end of natality by t_{B99} instead of the last, extreme birth. With a background mortality of 0.003 and 0.008 per week the plateau shrinks from 51 to 17 and 10 weeks, and the primary outcome falls to 0.55 and 0.06, through O5 alone. With 0.003 the failing runs are almost all MARGINAL (83 of 90): natality still ends at $N \approx 0.99 N_{pk}$, but t'_{pk} comes earlier and the clause on $(t_{B99} - t'_{pk})/t'_{pk}$ is not met. With 0.008 O5 fails outright in 70 of 200 runs. In both arms O11, O13 and O13p pass in every run, so what background mortality removes is the coincidence of the end of natality with the start of the plateau, a generic pattern, and not the classes (O13, O13p) that discriminate against the null models (Section 8.5).
8. *Mobility*. Mixing is slow compared with deterioration ($Da \approx 1.7 \times 10^2$, against $Da \ll 1$ in Universe 25; Section 4.4). With more movement sub-steps the coexistence of classes is lost (Appendix D.6).

6 Mean-field analysis

The macroscopic behaviour of the agent-based model is governed by two slow variables: the population size $N(t)$ and the mean social state $S(t) = N^{-1} \sum_i s_i$. A hierarchy of mean-field descriptions explains the regimes of the core model without the rules, extracts the mechanism and the thresholds of the irreversibility rules R1 and R2, and gives the quantitative predictions tested in Section 8.4. The analysis itself is in Appendices B and C.1; here we state its scope and its predictions, with their outcome.

6.1 Scope

Most of the mean-field analysis describes the *short-lived control* and the long-lived control (the core model without the four rules: $a_{\max} = 60$ or 120 , $\alpha = 1.4$, $r = 0.01$, $\gamma = 0.1$; Section 4.3), not the calibrated candidates. In that model reproduction is limited by mate encounter, which gives a built-in Allee effect, and deterioration is driven by the Poisson tail of the local occupancy, which makes the social state a switch between recovery and loss at a population N^* ; a two-variable mean field then has a superstable limit cycle, the oscillations of the spatial model are sustained by *healthy pockets* of uncrowded individuals, and extinction is a mixture of an early deterministic-like mode and a rare late mode, without secular drift of the peaks. Its correspondence with C1 is limited: the switch point ($N^* \approx 6200$ with Poisson occupancy, 3900 – 4600 with negative-binomial occupancy of dispersion $\xi = 1$ – 2 , 5300 for $\xi = 5$) lies close to the damage-threshold capacity $K_\gamma = 7200$ and far above the peak of C1 (≈ 2600), which is set by aggregation and refuges; under R1 adults do not recover, so the limit cycle does not exist in C1; and the first-moment closure fails where it matters (at $N = 1600$ the effective weekly loss of social state is 0.75 , against 0.001 predicted with Poisson occupancy). What carries over to C1 is the demographic law of phase D, Eq. (7), which uses only the mortality; the capacity bound $\Phi_{\text{BO}}^{\text{pers}} \leq \phi_R$ of Section 4.4; and two heuristic statements about R2 with the parameters of C1, the elasticity θ of Q6 and the generational map of Q7. Under R1 recovery becomes a *lineage* process, whose bottleneck is the fecundity $s_f^2 s_m$ of deteriorated parents (Table 11); under R2 the recovery time grows with the social-learning weight λ and is a lower bound for the irreversibility threshold (Eq. (11)). A kinetic mean field with persistent cell disorder confirms both rules in the long-lived scenario for the two rates it was run with, $r \in \{0.01, 0.03\}$: R1 gives certain extinction for every window $a_w \leq 24$, and R2 alone needs a weight λ above a threshold that grows with r (defining the threshold as the first value of the grid $\{0, 0.3, 0.5, 0.7, 0.9, 1\}$ at which at least half of the 10 seeds go extinct: $\lambda_c \leq 0.3$ at $r = 0.01$, with $8/10$ seeds already at 0.3 , and $\lambda_c \in (0.7, 0.9]$ at $r = 0.03$). These are bounds and orders of magnitude, not predictions for C1.

6.2 Predictions for the calibrated candidate and their outcome

Before the production sweeps, the analysis was turned into quantitative predictions for C1 (Q1–Q7, Section 7). Q2 (necessity of each rule) comes from the calibration; the other six have a mean-field origin, and only Q3–Q4, Q6 and Q7 rest on pieces that carry over to C1. The outcomes below are those of the production runs with the primary evaluator; where the preregistered evaluator gives a different verdict it is stated (Section 8.4).

Q1, transplant. Under R1, phase-D animals ($s \approx 0$) cannot found a colony, because their recovery must pass through reproduction (Table 11); with individual recovery at $r = 0.1$ the time to viability, about 3 weeks, is shorter than the fertile time left, so they can, up to an age ceiling. *Outcome:* the null hypothesis is rejected ($0/60$ against $51/60$ at $1.75 t'_{\text{pk}}$) and the predicted rate in the enclosure without R1 and R2 (≥ 0.25) is reached (0.85); at the later sampling time of the preregistered protocol the same animals give $12/60$, because of the age ceiling. Since the transplanted animals have $s \approx 0$ and fecundity scales as $s_f^2 s_m$, the $0/60$ follows from R1 by construction: Q1 is a consistency check of R1, as Q4 is of the demography. Transplants of intermediate animals confirm the bottleneck (8% found a colony under R1 against 95% without it) but not the collapse of the lineage: their offspring settle at the parents' level.

Q3, Q4, the phase-D law. Once births stop, extinction is the death of the last survivor of a

closed, ageing population. With n animals born at the last birth event,

$$\mathbb{E}[T_{\text{ext}} - t_{\text{last}}] \simeq \sum_{a \geq 1} \left[1 - (1 - S(a))^n \right], \quad S(a) = \prod_{b=1}^a [1 - h(b)], \quad h(b) = [1 + e^{-k(b-a_{\text{max}})}]^{-1}, \quad (7)$$

where $S(a)$ is the probability that an animal born at t_{last} is still alive $a - 1$ weeks later (the product starts at $b = 1$ because the first mortality draw happens at age one, in the week of birth). The expectation grows one-to-one with a_{max} and only logarithmically with n . *Outcome:* confirmed (slope 0.996 ± 0.014); the logarithmic dependence on n appears in the lattice-size stage. Since this follows from the demography alone, it is a consistency check rather than a test of the rules.

Q5, preferred occupancy. A deteriorated individual weighs a cell with n acquaintances by $(1 + \alpha n)e^{-\kappa n}$, maximal at $n^* = 1/\kappa - 1/\alpha$ (Eq. (6)); pools above m_c and isolated animals were predicted to coexist only if $n^* \approx m_c$. The prediction was derived and calibrated on C0, which has no refuges, and the preregistration should have anticipated that R7 decouples the isolated class from this balance. *Outcome:* refuted in C1.

Q6, window against rate. Under R2 with $\lambda = 1$ a juvenile ends the window with $s(a_w) = w s_f + r \int_0^{a_w} \bar{s}_V(a) da$; the elasticity of the learned part is one with respect to r and $\theta = \bar{s}_V(a_w)/\langle \bar{s}_V \rangle_{[0, a_w]} < 1$ with respect to a_w , if the competence of the neighbours decreases with age. *Outcome:* the prediction $0 < \theta < 1$ is not supported with the primary evaluator: the Calhoun-like boundary of the plane is set by the learning rate and the window has little or no effect ($\hat{\theta} = -0.15$ $[-0.20, -0.05]$ in a single-index model that an additive model beats, $\Delta\text{QAIC} = 28$); with the preregistered evaluator the single-index model is the preferred one and gives $\hat{\theta} = 0.30$ $[0.15, 0.50]$, as in the runs of the preregistered design. The difference is the criterion O11: measured on the matured late cohorts, the boundary is set by their socialisation at fast learning, where a longer window fails less often, so the weak dependence on a_w runs against the predicted sign without amounting to a reversal. The elasticity argument describes the learned competence at the end of the window, not which cohorts end up below the threshold.

Q7, generational map. Without crowding, the mean competence at maturity obeys

$$S_{g+1} = w \tilde{S}_g + \varphi \Lambda S_g, \quad \Lambda = r a_w, \quad \tilde{S}_g = \frac{\mathbb{E}[s^3]}{\mathbb{E}[s^2]} \equiv \sigma S_g, \quad (8)$$

where $\sigma \geq 1$ is the selection factor of the birth flux and φ the competence of a pup's neighbours relative to S . Competence dies out if $\sigma w + \varphi \Lambda < 1$, and since the reproduction number $R_0(S) \approx 70 S^3$ of the long-lived demography falls below one for $S < 0.24$, the colony then goes extinct *without* crowding; at the C1 values ($w = 0.2$, $\Lambda = 1.2$) the map is supercritical. *Outcome:* the test confirms an endogenous collapse without crowding in the subcritical corner, which is not Calhoun-like because the colonies first overshoot the capacity. With a_w varied, the transition in $\Lambda = r a_w$ falls in the same grid bracket for $a_w = 12$ and 24 and one grid step lower for $a_w = 6$, so the exponent of a_w relative to r lies in $(0, 1)$ between 6 and 12 weeks and is compatible with 1 between 12 and 24 ; the grid resolves neither the preregistered equivalence margin $[2/3, 3/2]$ nor a single exponent (51 of 54 points have $P_{\text{ext}} \in \{0, 1\}$), so the product form holds only approximately and short windows are less effective than it predicts. The linearity of the boundary in (w, Λ) is imposed by the logistic predictor, and the fitted $\hat{\varphi} \approx 1.1$ – 1.2 , above the preregistered $\varphi \leq 1$, has no valid interval.

The status of the exploratory mean-field statements is listed in Appendix C.1.

7 Statistical design

Unit of analysis and primary outcome. The unit of analysis is the independent stochastic replicate; seeds are derived as `SeedSequence([base_seed, point, replicate])`, and common random numbers are used across design points in the screening and variance-based designs. Each run is scored by the automatic evaluator of Section 3. The primary per-run outcome is *O17p*: the run is Calhoun-like, meaning that every applicable essential pattern passes and no anti-pattern is triggered, *and* its isolated class of O13 is persistent. With the default initial condition ($N_0 = 400$), O1–O3 are not applicable and are excluded from O17p; they are studied separately with the founder protocol. Every other essential pattern must pass: a pattern that cannot be evaluated in a run (NA) counts as a failure of O17p, so that no pattern is passed vacuously. The primary evaluator, which is not preregistered, differs from the preregistered one: it takes the peak time at the start of the plateau (t'_{pk} , first week within 2% of the maximum), the end of natality at 99% of the births (t_{B99}), O4 with free space at the peak, O10 on the adults, O11 on the cohorts born in the second half of the natality and the crowded class at $m > 8$; every run is also scored with the preregistered evaluator and with each option changed separately (Appendix C.6). Some patterns are defined for an ensemble and scored per run: the per-run version of O7 is “extinct before the horizon T ”, its ensemble version is $P_{\text{ext}}(T) \geq 0.9$ estimated by Kaplan–Meier, and the fraction of runs that pass O7 is the binomial estimate of the latter.

Presets and stages. All designs are expressed relative to the reference candidate C1 (Table 3). The controls are C0 (C1 without R7), C1 with $\alpha = 0$, and the short-lived and long-lived controls (the core model with the four rules switched off, $\alpha = 1.4$, and $a_{\text{max}} = 60$ or 120; Section 4.3); the confirmatory contrast is C2; the null models (Section 8.2) are run from the initial condition of C1. Rule ablations switch a rule off by setting its parameters to their neutral values (for R2, full inheritance and individual recovery at the core rate $r = 0.01$). The design has six kinds of stages: production ensembles of the candidates, controls and null models, with one reference ensemble for C1, C1', C0 and the $\alpha = 0$ control; a 2^4 factorial ablation with implementation variants; transplant experiments (paired cross-transplant, mixed transplants with healthy partners, sampling at a common absolute time, transplants of adults of intermediate social state); phase planes in the socialisation, space and refuge parameters, a seeding-density plane and a founder plane; robustness variants (background mortality, nest period, asynchronous update, pups born above the deterioration threshold, non-synchronous initial condition, lattice size at fixed density); and global sensitivity analysis in three steps: Morris elementary effects over 15 factors (20 trajectories, 10 runs per point), two independent 15-factor Latin hypercubes (200 points, 10 runs each, plus 40 runs at the points near the boundary), and Sobol' indices with the Saltelli design and Jansen estimators [28] for 11 factors (the 8 retained by the screening plus w , r and a_w), with $N = 256$ and 3 runs per point, extended to 6 [32]. Morris and Sobol' indices are descriptive, not tests; they are reported with bootstrap intervals and with an *empirical noise floor*, the same estimators applied to the split-half contrast of the replicate means at each design point (Appendix C.9).

Confirmatory hypotheses. Seven hypotheses were fixed before the production runs, each with a quantitative prediction derived from the mean-field analysis or the calibration. The plan, with designs and predictions, was fixed on 1 October 2026, 22:02 UTC, before the launch of the production runs, and is reproduced in `docs/preregistered_plan.md` of the code repository; the primary

evaluator, the stages not in that plan and their decision rules are not preregistered, but were fixed (`docs/production_plan.md`, 2 October 2026, 05:34 UTC) after exploratory runs with the primary evaluator (smoke tests on the data of the preregistered design and on two replicates per point) and before the production runs. One option of the primary evaluator, the birth reference t_{B99} of O5, was chosen after those exploratory runs had shown that the alternative fails O5 in most C1 runs; it is the only option on which a confirmatory decision (Q2) depends, and both are reported (Table 27). They form one family with Holm correction at $\alpha = 0.05$: Q1, structural irreversibility (the same phase-D animals transplanted to a C1 enclosure and to one without R1 and R2; exact one-sided McNemar test); Q2, each rule is necessary (intersection–union test over the four single-rule ablations, one-sided Fisher tests, $n = 120$ per arm); Q3, near-deterministic collapse versus the short-lived control (Fisher test on P_{ext} and the coefficient of variation of T_{ext}); Q4, the phase-D law, $T_{\text{ext}} - t_{\text{last}}$ one-to-one with a_{max} (TOST with margins $[0.8, 1.25]$); Q5, the coexistence band of O13 centred at $n^* \approx m_c$ (Firth regression on $\log n^*$); Q6, the plasticity window weighs less than the learning rate (profile likelihood of θ in $\log r + \theta \log a_w$, $\theta < 1$; quasi-binomial p); Q7, endogenous collapse without crowding damage in the subcritical corner (Fisher). A second confirmatory family compares C1 and C2 on every pattern and on six continuous magnitudes ($n = 200$ per arm; two-sided Fisher and Mann–Whitney tests, log-rank for T_{ext} , Holm correction over 29 rows, which hold 27 distinct tests because O6, O7 and A2 are decided by the same four C2 colonies; nine rows are constant in both arms and enter with $p = 1$, so the correction is conservative). One confirmatory test is not preregistered but was fixed before the production runs: the equivalence of the exponent of a_w with that of r at $\gamma = 0$ (TOST with margins $[2/3, 3/2]$), with a likelihood ratio for $\theta = 1$; its design turned out to lack the resolution for that test (the logistic model is separated), so the stage is reported non-parametrically (Section 8.4). Rejection of a null hypothesis and fulfilment of the quantitative prediction are reported separately. A Holm-adjusted p rejects if it is at most the level, and adjusted values within 0.01 of the level are given with four digits. Q2 is an intersection–union test: it rejects only if every component test rejects at the full level, so the four component tests are not corrected among themselves. Exploratory contrasts are grouped into families and controlled with Benjamini–Hochberg at 10%. The rules for the stages not in the preregistered plan were also fixed before the runs: a criterion is *generic* if it passes in at least two of the four null models, a robustness variant *preserves* the result if O17p ≥ 0.8 , and the feasible fraction of the hypercubes is reported with the empirical-Bayes estimator of Appendix C.9.

Intervals, survival and budget. Unless stated otherwise, intervals are 95% intervals: Wilson for proportions; Newcombe for differences of two proportions; percentile bootstrap, resampling runs or design points, for the coefficient of variation and the spread of extinction times, for the Calhoun-like fraction of the hypercubes and for sensitivity indices; profile likelihood for the exponents $\hat{\theta}$, quasi-binomial when stated; Greenwood log–log for Kaplan–Meier curves and medians. Extinction times are right-censored at the horizon T , and means over extinct runs only are never reported: we use Kaplan–Meier estimates, $P_{\text{ext}}(T)$ with Wilson intervals, log-rank comparisons, Weibull fits and a cure model with an escaping fraction where a fraction of the colonies survives [24]; the spread of the extinction time is the interquartile range over the median of the Kaplan–Meier estimate. The 95% Wilson half-width at $p = 0.5$ is ± 0.17 , ± 0.12 , ± 0.09 and ± 0.07 for $n = 30, 60, 120$ and 200 ; production, contrast, null-model and robustness ensembles have 200 runs (100 for the short-lived and long-lived controls, 60 for the background-mortality curve and the lattice sizes), single ablations 120, multiple ablations 40, transplant arms 60 replicates, phase-plane points 30 (100 at the 40 points nearest to a boundary), Morris and hypercube points 10 [21]. No power analysis was done after the runs: observed power is a monotone function of the p -value and cannot explain a non-rejection [18], so we report effect sizes with their intervals instead. The production program comprises 51828

simulations and 960 transplant replicates in 26 stages, about 4.4 hours of logged wall time on four cores, 1.8 of them for the Sobol’ design (Table 26); a run of **C1** or one of its variants costs 0.5–2.6 s of CPU time. Table 27 (Appendix C.14) lists every deviation from the preregistered plan, including the redefinition of the primary evaluator; analyses not fixed before the production runs are labelled post hoc wherever they are reported.

8 Results

All results come from the production design of Section 7: the preregistered stages, with the exact refuge capacity and the primary evaluator (t'_{pk} , t_{B99} , free space in O4, O10 on adults, O11 on the second half of the births, crowded class at $m > 8$; Table 27), plus seven stages not in the preregistered plan; 51 828 simulations and 960 transplant replicates. Every run was also scored with the preregistered evaluator and with each option changed separately (Appendix C.6). Unless stated otherwise, the numbers for **C1**, **C1'**, **C0** and the $\alpha = 0$ control come from one reference ensemble ($n = 200$, $T = 600$), those of **C2** from its production ensemble ($n = 200$, $T = 700$), and the null models, among them the long-lived control (the core model with the four rules switched off, $\alpha = 1.4$ and $a_{\text{max}} = 120$ weeks; Section 4.3), start from the initial condition of **C1** (Appendix C.13). f denotes the fraction of runs that pass a pattern; intervals are 95% Wilson intervals unless another kind is named. Pass fractions are rounded to two decimals in the tables; in the text, O17p of the $n = 200$ ensembles is quoted with three ($199/200 = 0.995$, $167/200 = 0.835$, $141/200 = 0.705$), because the third decimal is exact there and separates 0.995 from 1.00.

8.1 Production ensembles

C1 reproduces every applicable essential pattern. Each applicable essential pattern passes in at least 99% of the 200 runs, and the primary outcome O17p in $f = 0.995$ [0.97, 1.00]; the independent ensemble at $T = 700$ gives 0.98 [0.95, 0.99], the preregistered evaluator 0.98 and the single-pass refuge admission, run as a variant, 0.995: neither the admission rule nor the seeds change the primary outcome. The time course (Fig. 2) is a single boom, a plateau and an extinction by old age. The mean social state of the adults (age ≥ 6 weeks, which includes the maturing pups of the first cohort) falls below 0.5 at week 13 [13, 14], while the population is still growing. Part of this fall is demographic dilution by pups born at $s = 0$: measured on the animals past the plasticity window ($a \geq a_w = 12$), or on the founders alone, the crossing comes at week 16, so that deterioration takes of the order of one generation, $\Gamma = t_{\text{det}}/T_{\text{gen}} \approx 1.1$ ($t_{\text{det}} \approx 10$ weeks, $T_{\text{gen}} \approx 9$; 1.2 on the founders; Section 4). The socially functional growth phase is short either way: O10b passes in 0.02 of the runs with the adult mean of the primary evaluator and in 0.28 (marginal in the rest) when measured on the animals past the window, a phase B of 13 weeks that holds two thirds of the births, against 30 weeks in Universe 25. The population comes within 2% of its maximum at $t'_{\text{pk}} = 34$ [32, 36] weeks, 99% of the births have occurred by week 40, the last surviving birth is at week 59, the plateau lasts 51 [49, 53] weeks, and the colony then ages at one week per week and dies out at week 169 [167, 171] (Kaplan–Meier median). The peak uses $\rho_{\text{pk}} = 0.36$ of the damage-threshold capacity $K_\gamma = m_c|G|$ and leaves 16% of the cells empty (Universe 25: 0.57 and about 20%; an agreement in sign only, and f_0 was used in the calibration); 17% of the adults form a persistent isolated class (0.6% in **C0**). The desirable patterns are only partly reproduced: O8 passes in 0.62 [0.56, 0.69] of the runs, O9 in 0.45 and O13b in none (Table 12). Without mortality before senescence the maximum $t_{\text{pk}} = \arg \max N$, at week 55 [50, 60], falls on the last birth before it in 69% of the runs, 21 weeks after the plateau begins; every criterion normalised by the peak time inherits this ambiguity, which is why the primary

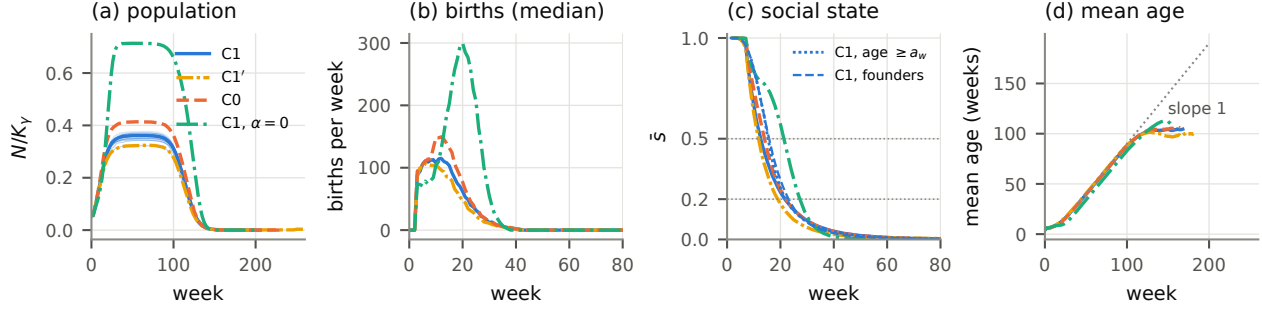


Figure 2: Time course of the reference ensemble ($n = 200$ each). **(a)** Population relative to the damage-threshold capacity $K_\gamma = m_c|G|$: median and 10–90% band for C1 (thin lines are 12 individual runs), and medians for C1', C0 and the $\alpha = 0$ control. **(b)** Weekly births (median). **(c)** Mean social state $\bar{s}$ of the adults (age ≥ 6 weeks; solid and dashed coloured lines as in a), median over living colonies, shown while at least half of the colonies are alive, on a linear scale; for C1 also the mean over the animals past the plasticity window (age $\geq a_w$, dotted) and over the founders (dashed), which are not diluted by the newborn pups. The dotted grey lines mark the 0.5 level of O10 and the 0.2 deterioration threshold. **(d)** Mean age, with a reference line of slope one: after natality stops, the colony simply ages.

evaluator uses t'_{pk} and t_{B99} . The decline lasts $(T_{ext} - t_{pk})/t_{pk} = 2.07$ peak times with $\arg \max N$, close to Calhoun's 1.84–2.2, but 3.97 with t'_{pk} : this agreement is not a robust result.

C1', controls and C2. C1' (R1 + R2 + R7, without AV) passes O17p in 0.835 [0.78, 0.88] of the runs (0.93 with the preregistered evaluator). What fails is O5: in 15% of its runs the 99th percentile of the births comes when the population has already fallen below $0.7 N_{pk}$ or later than $1.5 t'_{pk}$, because its tail of births is longer ($\tau_B = 8.7$ against 6.9 weeks). Without AV the deteriorated animals gather in pools that leave half of the lattice empty at the peak ($f_0 = 0.52$ against 0.16) and the senescent decline O8 passes in 0.21 instead of 0.62. C0 passes every instantaneous essential pattern (O17 in 0.92) but its isolated class is never persistent (O17p = 0): persistence is what refuges add. Without attraction ($\alpha = 0$) the colony fills 71% of K_γ with 3% of empty cells and fails the primary O4 and O14. The short-lived control, the core model with the four rules switched off, $\alpha = 1.4$ and a short life ($a_{max} = 60$, $a_{rep} = 50$ weeks), fails O11 and O13 in every run, and 53% of its colonies survive the first collapse and oscillate; the long-lived control, identical except for $a_{max} = 120$ and $a_{rep} = 80$, goes extinct in 0.98 of the runs, with a peak at 0.85 of K_γ and no failure of the late cohorts scored at six weeks of age (scored at the end of the plasticity window they fail in 0.81 of the runs from the initial condition of C1; Section 8.2). Near-certain extinction is therefore not specific to the new rules; what they add is the structure of the collapse. C2, which ties the retreat to maternal socialisation, makes its isolated class a later-born cohort (O13b at least marginal in 200/200 runs against 0/200 in the C1 ensemble of the same stage and 2/200 in the reference ensemble) but pays with its demography: O8 fails in every run, O5 falls to 0.815 and O4 to 0.90, and O17p is 0.705 [0.64, 0.76] against 0.98 in the independent C1 ensemble of the same stage ($T = 700$; Holm $p = 4 \times 10^{-14}$ over the 29 rows of the C1–C2 family, 27 distinct tests; 0.88 with the preregistered evaluator), whereas the calibration had predicted no difference (Appendix C.12).

Table 5: Production ensembles: fraction of runs that pass each essential pattern, primary evaluator. **C1**, **C1'**, **C0** and the $\alpha = 0$ control: the reference ensemble ($n = 200$, $T = 600$); **C2**: $n = 200$, $T = 700$; short-lived control ($\alpha = 1.4$, $a_{\max} = 60$) and long-lived control ($\alpha = 1.4$, $a_{\max} = 120$), both without the four rules and from the dispersed initial condition: $n = 100$, $T = 1000$. Patterns are evaluated over the applicable runs; O17 and O17p over all runs. The 95% Wilson half-width is at most 0.07 for $n = 200$ and 0.10 for $n = 100$, and at most 0.02 and 0.04 for fractions of 0 or 1. No anti-pattern is triggered in the candidates; A2 (recovery of natality) is triggered in 0.02 of the **C2** runs, 0.55 of the short-lived runs and 0.03 of the long-lived runs. Magnitudes are ensemble means, except the Kaplan–Meier median and the spread of the extinction time (interquartile range over the median of the Kaplan–Meier estimate). Transplants: Table 23.

ID	Pattern	C1	C1'	C0	$\alpha=0$	C2	short-lived	long-lived
O4	peak below capacity, free space	1.00	1.00	1.00	0.00	0.90	0.87	0.08
O5	natality ends at high N	0.99	0.85	1.00	1.00	0.81	0.25	0.90
O6	no rebound	1.00	1.00	1.00	1.00	0.98	0.45	0.97
O7	extinction	1.00	0.99	1.00	1.00	0.98	0.47	0.98
O10	adult deterioration before peak	1.00	1.00	1.00	1.00	1.00	1.00	1.00
O11	late cohorts fail	1.00	0.97	1.00	1.00	0.99	0.00	0.00
O13	two deteriorated classes	1.00	1.00	0.92	1.00	1.00	0.00	0.00
O13p	persistent isolated class	1.00	1.00	0.00	0.00	0.94	0.00	0.00
O17	Calhoun-like run (O17)	0.99	0.83	0.92	0.00	0.72	0.00	0.00
O17p	primary outcome	0.99	0.83	0.00	0.00	0.70	0.00	0.00
	95% Wilson interval	0.97–1.00	0.78–0.88	0.00–0.02	0.00–0.02	0.64–0.76	0.00–0.04	0.00–0.04
	$\rho_{\text{pk}} = N_{\text{pk}}/K_{\gamma}$	0.36	0.33	0.42	0.71	0.42	0.77	0.85
	empty cells at peak, f_0	0.16	0.52	0.19	0.03	0.11	0.30	0.59
	persistent isolated, $\Phi_{\text{BO}}^{\text{pers}}$	0.17	0.16	0.01	0.09	0.13	0.00	0.00
	median T_{ext} (weeks, KM)	169	182	166	155	218	–	186
	spread of T_{ext} (KM IQR/median)	0.09	0.12	0.08	0.05	0.14	–	0.16

8.2 Which patterns carry evidence: null models

A criterion is informative only if it can fail. Table 6 gives the pass rate of every essential pattern in four null models run from the initial condition of **C1** ($n = 200$): no crowding damage ($\gamma = 0$), full inheritance without social learning ($-\text{R2}^*$: $w = 1$, individual recovery at $r = 0.01$), the long-lived demography without new rules, and **C1** without its four rules; and in the two partial ablations of the reference ensemble ($\alpha = 0$ and **C0**); Table 17 (appendix) adds the desirable patterns. NA counts as a failure. By the rule fixed before the runs, a criterion is *generic* if it passes ($f \geq 0.8$) in at least two of the four null models, and it *discriminates* against a model if the upper Wilson bound of its pass rate there is below 0.8 and the Newcombe interval of the difference with **C1** excludes zero. O5, O6, O7 and O10 are generic (they pass in three or four null models), and so is the primary O4, which still passes in $-\text{R2}^*$ and without rules (the free space at the peak needs no rule: $f_0 = 0.74$ without rules, 0.59 in the long-lived control; and f_0 , part of the primary O4, was an informal target of the calibration): these five criteria carry no evidence for the rules; they verify that the model has a boom, an end of natality and an extinction. Scored at six weeks, as in the primary evaluator, the primary O11 discriminates against every null model, but not at the end of the plasticity window (below); O13 and O13p discriminate against three of the four, but not against $-\text{R2}^*$, where both deteriorated classes form by full inheritance without social learning (O13p 1.00 there), and O13p also discriminates against both partial ablations. O17p fails in every null model and in both partial ablations; which criteria carry that evidence depends on the age at which O11 is scored (below).

Table 6: Discriminating power of the essential criteria (primary evaluator; fraction of all runs that pass, NA counting as failure). Columns: the reference ensemble (C1, C1', $\alpha = 0$, C0), the four null models with the initial condition of C1 ($\gamma = 0$; -R2*: $w = 1$ without social learning; long-lived control; no rules: C1 without its four rules), $n = 200$ each, and the short-lived control ($n = 100$). “Generic”: passes in at least two of the four null models. “Discriminates”: upper Wilson bound below 0.8 and Newcombe interval of the difference with C1 excluding zero (long: long-lived control; none: no rules; short: short-lived control). O10b (functional growth phase) is listed because its failure in C1 is discussed in the text. O11 is scored at six weeks of age, as in the primary evaluator. Scored at the end of the plasticity window ($a_w = 12$; same runs, `results_v2/o11_aw`) it passes in C1 0.995, C1' 0.935, $\gamma=0$ 0.00 (NA), $\alpha=0$ 1.00, C0 0.99, -R2* 0.21, long 0.81 [0.75, 0.86] and none 0.755 [0.69, 0.81] (short-lived control not re-scored): it is then not generic (it passes in one null model) and discriminates only against $\gamma=0$ and -R2*. The desirable patterns are in Table 17.

	C1	C1'	$\gamma=0$	$\alpha=0$	C0	-R2*	long	none	short	generic	discriminates against
O4	1.00	1.00	0.00	0.00	1.00	0.99	0.17	1.00	0.87	yes	$\gamma=0$, $\alpha=0$, long
O5	0.99	0.85	1.00	1.00	1.00	0.97	0.89	0.91	0.25	yes	short
O6	1.00	1.00	0.00	1.00	1.00	1.00	0.99	1.00	0.45	yes	$\gamma=0$, short
O7	1.00	0.99	0.00	1.00	1.00	1.00	0.99	1.00	0.47	yes	$\gamma=0$, short
O10	1.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00	yes	$\gamma=0$
O11	1.00	0.97	0.00	1.00	1.00	0.00	0.00	0.00	0.00	no	$\gamma=0$, -R2, long, none, short
O13	1.00	1.00	0.00	1.00	0.92	1.00	0.00	0.00	0.00	no	$\gamma=0$, long, none, short
O13p	1.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	no	$\gamma=0$, $\alpha=0$, C0, long, none, short
O17p	0.99	0.83	0.00	0.00	0.00	0.00	0.00	0.00	0.00	no	$\gamma=0$, $\alpha=0$, C0, -R2, long, none, short
O10b	0.02	0.01	1.00	1.00	0.18	0.05	0.20	0.12	1.00	no	none

With the preregistered evaluator O10 passed in every null model and O11 passed vacuously without crowding damage, which is why both were redefined (Table 18). O11 is in part definitional, because pups are born at $s = 0$ and are scored at six weeks of age, inside the plasticity window. Re-scoring the same runs with the late cohorts measured at the age $a_w = 12$ instead (same thresholds; `results_v2/o11_aw`), C1 and C1' barely change (O11 0.995 and 0.935, O17p 0.990 and 0.815), but O11 then also passes in 0.81 of the long-lived runs and in 0.755 of the runs without rules, whose late cohorts fall below 0.2 through accumulated crowding damage, so it no longer discriminates against those two null models; against -R2* it passes in 0.21 (O17p 0.205), and without crowding damage it stays NA. Their late cohorts even end lower than those of C1 (median $\bar{s}$ 0.074 in the long-lived runs and 0.055 without rules, against 0.130). The dependence on s_{base} almost disappears at that age (below). At six weeks, then, O11 separates C1 from the null models largely through the birth state of the pups ($w = 0.2$, $s_{\text{base}} = 0$; with $s_{\text{base}} = 0.25$ it falls to 0.08), not through a failure that the rules add to the crowding damage, and the evidence of O17p against the long-lived demography and the model without rules rests on O13 and O13p; against -R2* it rests on O11 at six weeks and is weaker at a_w . O10b, the functional growth phase, passes only where crowding is absent or weak and fails in C1 with the adult mean (0.28 when measured on the animals past the window).

8.3 Rule ablations and the minimal candidate

Fig. 8 (appendix) removes the rules of C1 singly, in pairs, in triples and all together; each single ablation fails in a specific way. *Without R7* the colony is Calhoun-like in the instantaneous sense (O17 in 0.94), but the isolated class is never persistent (O17p = 0). *Without R2* (full inheritance and individual recovery) the late cohorts do not fail at six weeks (O11 and O17p in 0.00 [0, 0.03];

0.17 with the preregistered O11, which passed when the late cohorts were empty). *Without R1* adults recover when the colony thins: natality resumes, only 23% of the colonies go extinct by week 600, and O17p is 0.21 [0.15, 0.29]. *Without AV* O17p falls to 0.79 [0.71, 0.85], through O5. Every double, triple or quadruple ablation has O17p = 0, and without crowding damage ($\gamma = 0$) every colony grows to the cap of 15 000 individuals. Inheritance is not needed: with $w = 0$ and a faster learning rate $r = 0.15$, O17p is 1.00, so social learning is the essential part of R2. R7 depends on its implementation and on the update scheme: without a preference ($\pi = 0$) the refugees stay empty and O17p is 0; with a soft capacity under synchronous moves O17p is also 0, because simultaneous moves send many deteriorated animals into the same refuge, whereas under a random sequential update a soft capacity gives the same result as the hard one (0.995 against 1.00): the hard capacity is a remedy for the collisions of the synchronous update, not a mechanism of its own (Appendix C.4).

Q2 and the minimal candidate C1'. The preregistered intersection–union test is the largest of the four one-sided Fisher p -values, and it comes from AV. With the primary evaluator $p_{\text{IUT}} = 1.1 \times 10^{-7}$, $p_{\text{Holm}} = 2.2 \times 10^{-7}$, so the null hypothesis that some rule can be removed without loss is rejected; with the preregistered evaluator the same runs give $f(-\text{AV}) = 0.90$ and $p_{\text{Holm}} = 0.067$, not rejected, which replicates the borderline result of the runs of the preregistered design ($p_{\text{Holm}} = 0.0503$). The rejection is therefore a consequence of the primary O5 alone, and specifically of its birth reference t_{B99} : that option was adopted after exploratory runs with the primary evaluator had shown that the alternative, the last sustained birth combined with t'_{pk} , fails O5 in half of the C1 runs (0.48 in the production runs, Table 16); it is a choice informed by the result (Table 27), and O5 is a generic criterion (Section 8.2). The rejection of Q2 thus shows an effect of AV on a generic criterion, significant but smaller than predicted: with either evaluator the prediction $f(\text{C1} - r) \leq 0.5$ fails for AV (0.79), and the necessity of AV, as defined in the plan, is refuted. The difference with C1 is 0.20 [0.13, 0.28] in the factorial ($n = 120$ per arm) and 0.16 [0.11, 0.22] in the reference ensemble (Newcombe intervals). AV is a contributing rule: it makes the end of natality sharper (O5, through a shorter tail of births) and supplies the senescent decline (O8 0.62 against 0.21), while it *reduces* the empty space at the peak ($f_0 = 0.16$ against 0.52), which is present without any rule. In C1', R1, R2 and R7 remain necessary (O17p of 0 without each, $n = 40$; Table 13). C1' is the minimal set *conditional* on the evaluation thresholds. Re-scoring the stored class fractions (Fig. 10), O17p of C1' is 0.83 for class thresholds $\theta \leq 0.125$ and 0.33 at $\theta = 0.15$ (0.31 in the $-\text{AV}$ arm of the factorial, $n = 120$); that of C2 falls from 0.70 to 0.01 at 0.125; C1 holds 0.96 at 0.15. On the grid of class and ensemble thresholds (θ, τ) , AV is “necessary” exactly when $\tau \geq 0.8$ at $\theta \leq 0.125$ (0.79 lies below 0.8, with an interval that contains it) or when $\theta \geq 0.15$; R1, R2 and R7 are necessary everywhere. The necessity of R7 is necessity for O13p, a criterion designed together with R7 after C0 had been seen; for O11, O6 and O7 removing R7 changes nothing (O17 in 0.94).

8.4 Confirmatory tests Q1–Q7

Table 7 summarises the confirmatory family with the primary evaluator; Table 14 repeats it with the preregistered evaluator. With the primary evaluator six of the seven null hypotheses are rejected after Holm adjustment and the quantitative predictions are met for Q1, Q3, Q4 and Q7 and not for Q2, Q5 and Q6; with the preregistered evaluator five are rejected and four predictions hold (Q3, Q4, Q6 and Q7). The two counts differ in three places: Q2 is rejected only with the primary O5 (below), the prediction of Q1 holds only at the earlier transplant time $1.75 t'_{\text{pk}}$, and the prediction of Q6 holds only with the preregistered O11. Q5 is refuted in the opposite direction with both. Of the predictions that hold, Q1 is a consistency check of R1 and Q4 of the demography, and the contrast of Q3 with the short-lived control is confounded by the demography. A rejected null hypothesis is

Table 7: Preregistered confirmatory family with the primary evaluator (hypotheses and tests in Section 7; Holm correction at $\alpha = 0.05$ over seven tests; the p of Q6 is the quasi-binomial one). “Rejected” refers to the null hypothesis after Holm adjustment; “Met” to the quantitative prediction fixed before the runs. X is the cross-transplant arm. The same family with the preregistered evaluator is in Table 14.

	Prediction	Estimate	p	p_{Holm}	Rejected	Met
Q1	$P_T(D \mid C1) \leq 0.05$, $P_T(D \mid X) \geq 0.25$	0/60 vs. 51/60 ($P_T = 0.85$ [0.74, 0.92]); discordant pairs 51:0	4.4×10^{-16}	1.3×10^{-15}	yes	yes
Q2	$f(C1) \geq 0.9$, $f(C1 - r) \leq 0.5 \forall r$	0.99 vs. -R1 0.21, -R2 0.00, -AV 0.79, -R7 0.00	1.1×10^{-7}	2.2×10^{-7}	yes	no
Q3	$P_{\text{ext}} \geq 0.99$, $CV \leq 0.10$; short-lived ≤ 0.7	200/200 vs. 47/100; CV 0.066 [0.059, 0.071]	$< 10^{-16}$	$< 10^{-16}$	yes	yes
Q4	slope $\in [0.8, 1.25]$, $ \bar{r} \leq 10$ wk	slope 0.996 ± 0.014 ; $\bar{r} = -2.0 \pm 0.3$ wk	$< 10^{-16}$	$< 10^{-16}$	yes	yes
Q5	$b_2 < 0$, CI of $n_{\text{opt}}^* \subset [4, 12]$	$b_2 = +185.3 \pm 64.3$ (convex); no interior maximum	1.00	1.00	no	no
Q6	$0 < \theta < 1$	$\hat{\theta} = -0.15$ [-0.20, -0.05] (quasi-binomial, $\phi = 3.1$)	$< 10^{-16}$	$< 10^{-16}$	yes	no
Q7	corner ≥ 0.8 , $C1 \leq 0.2$	120/120 vs. 0/30 (30/30 explode)	$< 10^{-16}$	$< 10^{-16}$	yes	yes

not the same as a Calhoun-like outcome, as Q7 shows.

Q1 and the transplants. Four males and four females of each C1 colony, taken at $1.75 t'_{\text{pk}}$ (week 60; ages 37–64), never found a colony in a C1 enclosure (0/60), but do in 51/60 trials in an enclosure without R1 and R2 with $r = 0.1$ (arm X ; exact McNemar $p = 4 \times 10^{-16}$); the prediction (≥ 0.25 in X) holds. The same animals have $s \approx 0$ (median 0.000, maximum 0.36) and fecundity scales as $s_f^2 s_m$, so under R1 they cannot found a colony with any partner (0/120 with healthy partners): a consistency check of R1, not a finding about the mechanism. The rate in arm X is a matter of age: at $1.75 t_{\text{pk}} = \text{week } 97$ (preregistered protocol; ages 59–80) the same colonies give 12/60, and at week 95 in every arm, the common time that removes the confound between rule and sampling time, 5/60 (Appendix C.11). An additional arm samples adults past the plasticity window with $0.2 \leq s \leq 0.5$ (median 0.36, weeks 8–11): under R1 they found a colony in 5/60 trials, against 57/60 for the same animals in arm X and 50/60 without R1, and in the colonies they do found the adult descendants settle at the social state of their parents (0.28–0.45 in C1, 0.33–0.37 in X). By the rule fixed in advance they recover partly: the irreversibility of R1 + R2 is strict only for $s \approx 0$, and for intermediate animals R1 acts through fecundity, not through the lineage.

Q3 and Q4. C1 went extinct in 200 of 200 runs by $T = 600$, against 47 of 100 runs of the short-lived control by $T = 1000$ (one-sided Fisher $p < 10^{-16}$), with a coefficient of variation of T_{ext} of 0.066 [0.059, 0.071], as predicted. The contrast that isolates the rules is the long-lived demography alone, which goes extinct in 0.98–1.00 of the runs ($p = 0.11$): extinction does not discriminate. The Kaplan–Meier interquartile range over the median of T_{ext} is 0.089 [0.071, 0.105] in C1, 0.160 [0.144, 0.180] in the long-lived control with the same initial condition and 0.121 [0.106, 0.148] in C1', whose difference from the control, -0.040 [-0.065 , -0.006] (percentile bootstrap over runs), is *not* within the preregistered equivalence margin of ± 0.05 , which the runs of the preregistered design had met. A narrow spread is not a signature of the rules (0.045 without attraction, 0.084 in C0): it is set by how abruptly natality stops and by the lattice size (below). The time from the last birth to

extinction grows one-to-one with $a_{\max}$ (slope 0.996 ± 0.014 over 300 runs, TOST $p < 10^{-16}$), as the demography requires (Eq. (7); Appendix C.5).

Q5 and Q6. The coexistence band predicted at $n^* = 1/\kappa - 1/\alpha \approx m_c$ (Eq. (6)) is refuted: O13 passes in at least 95% of the runs at 46 of the 49 points of the (α, κ) plane with $n^* > 0.5$, and the fitted quadratic in $\log n^*$ is convex; with refuges the isolated class no longer depends on the balance between attraction and aversion. The prediction of Q6, $0 < \theta < 1$ for the Calhoun-like boundary in $\log r + \theta \log a_w$ on the plane (r, a_w) , is not supported with the primary evaluator. The boundary is nearly vertical in r : O17p is at least 0.67 for $r \leq 0.10$ and at most 0.60 for $r \geq 0.2$ at every window, so the window has little or no effect, and the single-index fit gives $\hat{\theta} = -0.15$ $[-0.20, -0.05]$ (profile likelihood, quasi-binomial correction $\phi = 3.1$) in a model that an additive one beats ($\Delta\text{QAIC} = 28$); with the preregistered evaluator the single-index model is the preferred one ($\Delta\text{QAIC} = 11$ in its favour) and gives $\hat{\theta} = 0.30$ $[0.15, 0.50]$, as in the runs of the preregistered design (0.35). The small negative value should not be read as a reversal of the predicted dependence: it comes from the primary O11, which at fast learning ($r \geq 0.2$) fails more often the shorter the window (O11 passes in 0.10 of the runs at $r = 0.2$ with $a_w = 4$ and in 0.87 with $a_w = 24$), and the change of criterion also changes which model fits the plane. The elasticity argument of Section 6 does not describe this plane.

Q7: endogenous collapse without crowding, and the dependence on $\Lambda = r a_w$. With crowding damage switched off ($\gamma = 0$), the four subcritical points ($w \leq 0.1$, $r \leq 0.03$) went extinct in 120/120 runs, while all 30 C1 colonies exploded (one-sided Fisher $p < 10^{-16}$); the colonies of the corner first overshoot the damage-threshold capacity ($\rho_{\text{pk}} = 0.91\text{--}1.60$), so none is Calhoun-like. With $a_w \in \{6, 12, 24\}$ varied at $\gamma = 0$ (54 points, 20 runs each; Fig. 9, Table 15), 51 of the 54 points have $P_{\text{ext}} = 0$ or 1, so the logistic model in $\log r + \theta \log a_w$ fixed before the production runs is separated: its profile intervals and likelihood ratio measure the geometry of the grid, not sampling error, and the planned equivalence test of the two exponents (TOST, margin $[2/3, 3/2]$) cannot be carried out as inference (Appendix C.5). What the design does establish is non-parametric. The transition in $\Lambda = r a_w$ is bracketed by two adjacent grid points, a factor 1.25–1.5 apart; at every inheritance weight the curves for $a_w = 12$ and 24 lie in the same bracket and the curve for $a_w = 6$ one grid step below ($\Lambda_{1/2} \in (0.60, 0.84)$ against $(0.84, 1.2)$ at $w = 0$; log-linear interpolation 0.71, 0.99 and 1.00). The exponent implied between windows is therefore in $(0, 1)$ between 6 and 12 weeks (interpolated 0.5–0.7 over the three w) and compatible with 1 between 12 and 24 weeks (interpolated 0.88–1.00), where the grid bounds it only within about $(0.5, 1.5)$. The firm result is qualitative: at equal product $r a_w$ the six-week window is less effective, so a single exponent does not describe the three windows and the effective learning grows less than linearly with the window at short windows. The boundary fitted as $\sigma w + \varphi \Lambda = 1$ gives $\hat{\varphi} \approx 1.1\text{--}1.2$ in both planes, above the preregistered $\varphi \leq 1$, but it is an extrapolation of the same separated predictor and carries no valid interval.

8.5 Robustness

Demographic and update variants, and the initial condition. Fig. 11 and Table 19 report C1 and C1' under the robustness variants ($n = 200$ each), scored with both evaluators and also with O17p restricted to the criteria that discriminate against the null models (O11 scored at six weeks, O13, O13p and the anti-patterns); by the rule fixed before the runs, a variant preserves the result if O17p ≥ 0.8 . With a constant background hazard μ_{bg} at all ages the plateau shrinks (51 weeks at 0, 35 at 0.001, 17 at 0.003 and 10 at 0.008 per week) and O17p falls to 0.83, 0.63, 0.55 $[0.48, 0.62]$ and 0.06 in C1 (0.67, 0.38, 0.28 and 0.04 in C1'; 0.45 at 0.003 with the preregistered evaluator). The only

criterion that fails is O5, and it fails through its timing clause, not through its level: at $\mu_{bg} = 0.003$ O5 is marginal in 83 of the 200 runs and fails in 7, because the start of the plateau t'_{pk} moves from week 34 to 29 while t_{B99} moves from 40 to 43, so $(t_{B99} - t'_{pk})/t'_{pk}$ crosses the 0.5 threshold of O5 while natality still ends with the population near its maximum; at $\mu_{bg} = 0.008$ O5 fails outright in 70 runs (marginal in 118). The classes, O11 and the low peak are unchanged, and the restricted outcome is 1.00 at every μ_{bg} . Since O5 is generic (Section 8.2), what the background mortality removes is the generic part of the primary outcome, not the part that carries evidence for the rules; the primary outcome as defined nevertheless holds only without, or with very weak, background mortality, which is stated as a limitation (Section 9.1). With initial ages drawn uniformly from 4–52 weeks instead of all equal, O17p is 0.93 [0.89, 0.96]; from 4–80 weeks, which includes post-reproductive founders, 0.63 [0.56, 0.69] (0.68 and 0.31 in C1'), again through the timing clause of O5 (73 marginal and 1 failed of 200; plateau 28 and 22 weeks, t'_{pk} at weeks 30 and 29) and with the restricted outcome at 1.00. The timing of the end of natality relative to the start of the plateau is therefore a property of the synchronous initial condition and of the absence of background mortality, not of the rules; the end of natality near the maximum itself persists. The other variants preserve the result: a nest period of three weeks gives 0.835 [0.78, 0.88], borderline because the interval contains 0.8 (0.60 in C1'; O11 0.90, $\rho_{pk} = 0.31$); a random sequential update gives 1.00 with the hard and 0.995 with the soft capacity, with $\rho_{pk} = 0.43$; and the exact refuge capacity gives 0.995 against 0.995 with the single-pass admission. Pups born above the deterioration threshold ($s_{base} = 0.25$ or 0.5 instead of 0) give O17p of 0.08 and 0.00 (0.64 and 0.36 with the preregistered O11), entirely through O11 scored at six weeks of age: the failure of the late cohorts at that age is partly definitional, tied to $s_{base} = 0$. Scored at the age a_w , the same runs give O11 of 0.975 and 0.785 and O17p of 0.965 and 0.76 (0.755 and 0.77 in C1'): by the end of the window the late cohorts are below 0.2 whatever their birth state, through crowding damage and learning from deteriorated neighbours.

Lattice size, seeding density and phase planes. Over $L = 15$ –60 at fixed density $N_0/L^2 = 0.44$ ($n = 60$; Table 20), the peak density (0.35–0.37) and the persistent fraction (0.17–0.18) do not change, the empty fraction decreases slightly (0.18 to 0.15), and O17p rises from 0.82 [0.70, 0.89] at $L = 15$ to 1.00 at $L \geq 45$; the losses at $L \leq 20$ come from O5 (5 of 60 runs at each size: a relatively longer tail of births with $N_0 = 100$ –178), from O11 (1 of 60 at each) and, at $L = 15$ only, from the anti-pattern A2, a recovery of natality in 6 of 60 runs, and the interval at $L = 15$ contains 0.8. The spread of the extinction time falls from 0.15 at $L = 15$ to 0.08 at $L \geq 30$ together with the relative spread of the last birth (0.58 to 0.18), while $T_{ext} - t_{last}$ stays at 110–112 weeks: the spread of T_{ext} is that of the end of natality, as Eq. (7) implies, and its value is a property of the system size. Over seeding densities from 0.02 to 3.6 animals per cell the peak density is monotone in N_0/L^2 and O17p reaches 0.8 only for $0.2 \lesssim N_0/L^2 \lesssim 0.9$; with Calhoun’s four founding pairs no run is Calhoun-like, and the best point of the founder plane reproduces O1–O3 with O17p in 0.43 [0.27, 0.61] of the runs (Appendix C.10). Around C1 the primary outcome holds over part of the explored planes (Appendix C.8): with crowding damage every point of the (r, w) plane goes extinct and O17p is at least 0.83 for $w \leq 0.3$ and $r \leq 0.10$, falling at $w \geq 0.5$ and $r \geq 0.2$ through O11; every finite window $a_w \leq 24$ gives certain extinction; in the (α, κ) plane O17p is at least 0.8 at 34 of the 48 points with $\alpha \geq 2$, failing where the free space of the primary O4 is lost; and a persistent isolated class requires a strong preference and a total refuge capacity $f_{RCR} \geq 0.4$. These are point estimates with 30 runs per cell (Wilson half-width up to ± 0.17); re-running the 40 points nearest to a boundary with 100 runs changed them by 0.07 on average.

Global: Latin hypercubes and variance decomposition. Over two independent 15-factor Latin hypercubes (400 points, 10 runs each, plus 40 runs at the 18 points near the boundary), 2.0%

[0.7, 3.6] of the points are Calhoun-like with the primary evaluator ($k/n \geq 0.8$; bootstrap interval over points); the mean of O17p over the box, an unbiased estimate of the volume weighted by the probability of being Calhoun-like, is 4.0% [2.6, 5.9], and the empirical-Bayes fraction of points with $p \geq 0.8$, which shrinks each point towards a logistic emulator, is 1.8% [0.7, 3.2], with 8 points still uncertain (Table 21, Fig. 14). With the preregistered evaluator the same runs give 7.2%, 11.0% and 6.9% [4.8, 9.2], the values of the runs of the preregistered design; the drop comes mostly from O5 with t'_{pk} and from the free-space condition of O4. Every such fraction is relative to the box and to the sampling scales of Table 3. Part of the small fraction is definitional and part is mechanism: with $c_R = 3$ refuge residents exceed the isolation threshold and none of the 100 such points is Calhoun-like; restricting successively to $c_R \leq 2$, $m_c \leq 10$, $f_R c_R \geq 0.3$ and $\pi \geq 100$ (chosen after the runs) raises the fraction to 2.7%, 3.1%, 5.6% and 10.8% [2.7, 21.6] (4 of 37 points; 46% with the preregistered evaluator), and even then 33 of the 37 points fail, through O11 (16), the persistence of the class (14), O5 (9) and O4 (9); with class thresholds of 5%, 10% and 15% the fraction is 5.5%, 2.3% and 0.5%. The Sobol' decomposition uses 11 factors (the 8 selected by the Morris screening plus w , r and a_w ; Saltelli design, $N = 256$, 6 runs per point), so its indices are conditional on four factors held at their C1 values (f_R , π , $a_{\max}$, δ). O17p is zero at 78% of its points; its total indices are led by c_R (0.49 ± 0.16), w (0.43), κ (0.41), r (0.32) and m_c (0.31), which cannot be ordered, and its first-order indices add up to only 0.15, so it is almost entirely interactions; the continuous outputs are nearly additive (ρ_{pk} : r , κ , η , γ ; f_0 : κ , m_c ; Φ_{BO}^{pers} : c_R , m_c). Appendix C.9 gives the full decomposition.

9 Discussion

The envelope is generic; the rules add the classes. The main lesson of the production runs is a negative one. A long-lived demography alone, without any of the new rules, drives nearly every colony to extinction (98/100 in the long-lived control, 199/200 in the corresponding null model run from the initial condition of C1). Once natality stops, the colony simply ages out, and the phase-D law Q4 is a consistency check of that demography. Extinction is therefore *not* a result of the rules, and neither is the rest of the envelope: with crowding damage and the calibrated attraction $\alpha = 4$, the candidate without its four rules has a low peak with free space, an end of natality near the maximum and no rebound (Table 6), and, scored at the end of the plasticity window, late-born cohorts that fail to acquire social competence. We distinguish what the rules encode, what they add, and what the demography and the initial condition set.

Consistency checks of the design. Several patterns follow from the rules by construction; they verify the implementation, but they are not findings.

- *Transplants (O12, O12b, Q1).* Under R1 an adult cannot recover, and conception and pup survival scale with the social state of the parents. Phase-D animals have $s \approx 0$ (median 0.000 in every arm with R1; Table 24), so they cannot found a colony, with healthy partners or without them (0/60 and 0/120 in C1). R1 was motivated by Marsden's transplants [5], and the cross-transplant arm only shows that removing the ratchet restores recovery. The non-trivial part is upstream: that the colony drives essentially every adult to $s \approx 0$, which O10 and O11 measure and which crowding damage alone largely achieves. Adults of intermediate social state ($0.2 \leq s \leq 0.5$, which exist only around weeks 8–20) found a colony in 5 of 60 trials under R1, against 57 of 60 without R1 and R2: the ratchet also blocks them, through their fecundity, but where they do found a colony their offspring settle at the parents' level ($\bar{s} \approx 0.3$ –0.4) and the lineage does not collapse (Appendix C.11).

- *Persistence of the isolated class (O13p)*. R7 was designed after the first candidate produced only a transient isolated class, and its capacity $c_R = 2$ equals the isolation threshold of O13.
- *Senescent decline and no social deaths (O8, A4)*. Social mortality is off in every preset.

Table 6 gives the pass rate of each essential pattern in the null models; the patterns above are marked $\ddagger$ in Table 1, whose caption names the rule that implies each of them.

What the rules add. Of the eight essential criteria of Table 6, five (O4, O5, O6, O7 and O10) pass in at least two of the four null models and carry no evidence for the rules: they verify that the model has a boom, an end of natality and an extinction. In particular the low peak with free space is generic: the candidate without its four rules passes the primary O4 in 200 of 200 runs ($\rho_{pk} = N_{pk}/K_\gamma = 0.45$, $f_0 = 0.74$), and the long-lived control fails it only through its high peak ($\rho_{pk} = 0.85$), not through a lack of free space ($f_0 = 0.59$). $K_\gamma = m_c|G|$ is the population at which every cell would sit at the damage threshold, not a physical capacity. What the rules add, and the null models lack, is two patterns:

- *Coexisting crowded and isolated deteriorated classes (O13)*. They are absent in the long-lived control, without crowding damage and without the rules, and present with crowd aversion or refuges alone (1.00 in C1', 0.92 in C0); full inheritance without social learning also has them. This is a calibrated property, obtained in a nearly sessile regime (below), and it is lost with more movement sub-steps.
- *The persistence of the isolated class (O13p)*, which discriminates against three of the four null models but holds by design ($c_R = 2$ equals the isolation threshold).

The failure of the cohorts born during the boom (O11) is not among them. Scored at sexual maturity (six weeks), as in the primary evaluator, O11 passes in 1.00 of the C1 runs against 0.00 in each null model and needs R1 and R2 (0.71 of the applicable runs and 0.00 without each), but that separation was partly definitional: pups are born with $s \leq w s_f \leq 0.2$, the threshold that O11 uses, and at six weeks they are still inside the plasticity window, so with pups born above the threshold ($s_{base} = 0.25$ or 0.5) O11 failed and O17p dropped to 0.08 and 0.00. Scored at the end of the window ($a_w = 12$ weeks) on the same runs, C1 still passes (O11 0.995, O17p 0.990), the dependence on s_{base} disappears (O17p 0.965 and 0.76), and O11 also passes in 0.81 of the long-lived runs and in 0.755 of the runs without rules, whose late cohorts fall below 0.2 through accumulated crowding damage. Only against full inheritance without social learning does it still separate in part (0.21). The failure of the late cohorts is therefore mostly the crowding damage of the envelope, not a contribution of the rules. Against the long-lived control and the model without rules, the evidence that the primary outcome O17p carries for the rules rests on O13, with O13p as a design property; against full inheritance without social learning it rests on O11 at six weeks and is weaker at the end of the window. Crowd aversion is not needed for either pattern (C1' passes them), and it does not supply the free space at the peak: it reduces it ($f_0 = 0.16$ with AV, 0.52 without).

The central limitation: demography and initial condition. The model has no mortality before senescence, and its default initial condition is a synchronous cohort of 400 healthy four-week-old animals. Both matter for the primary outcome. Without background mortality nobody dies until old age, so the population sits on a plateau of about 51 weeks and t_{pk} is an arbitrary point of it; the ratio $(T_{ext} - t_{pk})/t_{pk}$ is 2.1, close to Calhoun's 1.84–2.2, with $t_{pk} = \arg \max N$, but 4.0 from the start of the plateau, so we do not count it as agreement. Calhoun recorded adult deaths during phases B and C. With a background hazard of 0.003 per week at all ages the plateau shrinks to

17 weeks and O17p falls from 0.995 to 0.55; with 0.008 per week, to 0.06. With initial ages spread over 4–52 weeks O17p is 0.93, but over 4–80 weeks it is 0.63 (Section 8.5). In every case the only criterion that fails is O5, and it fails by its timing clause, not by its level clause. Natality still ends with the population near its maximum (at the 99th percentile of the births the population is about $0.99 N_{\text{pk}}$), but the deaths bring the start of the plateau forward (t'_{pk} moves from 34 to 29 weeks at 0.003 per week, medians over 200 runs) while the end of natality hardly moves (40 to 43 weeks), so $(t_{B99} - t'_{\text{pk}})/t'_{\text{pk}}$ crosses 0.5 and O5 becomes MARGINAL. This is the case in 83 of the 90 failing runs at 0.003 per week and in 73 of the 74 failing runs with ages over 4–80 weeks; the remaining ones, and the 70 runs that fail O5 at 0.008 per week, exceed the timing clause by more than a factor two. O5 is generic, and the patterns that the rules add (O13 and O13p), as well as O11, pass in every run of these arms. The dependence on the demography and on the initial condition is therefore a dependence of the timing of the end of natality relative to the start of the plateau, a quantity that also rests on the post hoc choice of t'_{pk} . It is not evidence against what the rules add, nor for it, since O5 discriminates against no null model; but it does mean that O17p = 0.995 overstates how robustly the full conjunction of patterns is reproduced. A nest period of three weeks (0.835, borderline) and a random sequential update (1.00) preserve the primary outcome.

The dynamics is a boom and its senescence, a compressed sequence of phases. The time course of C1 is essentially one transient (Tables 32 and 35; reference ensemble, 200 runs). The founders deteriorate by crowding during their own adult life, from $s = 0.90$ at the end of the plasticity window to 0.20 at 26 weeks of age. Their offspring, about 70% of all births, are born below the deterioration threshold and fail to acquire competence, because competent adults have become scarce. The net reproductive number falls from 3.8 in the founders to 0.40 in their daughters, so natality stops within about one generation, and the third and later generations add 3.2% of the births. The null models without the rules have a *deeper* boom: with the same attraction and crowding damage, C1 without its four rules has a mean generation of the births of 1.59 against 1.34 and a net reproductive number of 0.66 in the first generation against 0.40, and the long-lived control 1.95 and 1.2 (30 runs each; Table 33). The rules, R2 in particular ($-R2^*$ gives 1.62), shorten the generational depth rather than create a long cascade. The shallowness of the boom is set by how pups are born and socialised: they are born below the deterioration threshold ($w = 0.2$, $s_{\text{base}} = 0$), conception and pup survival scale with the parents' social state, and social learning (R2) draws on adults who are already deteriorating, so the net reproductive number drops in the first generation. A ratio of time scales gives a heuristic comparison with Universe 25, not the cause. Measured on animals past the plasticity window, so that newborns do not dilute it, $\Gamma = t_{\text{det}}/T_{\text{gen}}$ is ≈ 1.1 (1.2 on the founders; about 0.7 if T_{gen} is taken as the mean age of the mothers at birth, 14.3 weeks, so it is of order one under either definition); in Universe 25 the “death of societal organization” took the whole of phase C, about 35 weeks, with a generation time of about 10 weeks, so $\Gamma \approx 3.5$, an order of magnitude with a different operational definition. The model has a growth phase with socially competent adults, but it is about half as long as Calhoun’s phase B (13 against 30 weeks), although it holds two thirds of the births, and O10b, which asks for it, passes in 0.28 of the runs when the adult mean is replaced by that of the animals past the window (0.02 with the adult mean, which the newborns dilute). The phases appear in Calhoun’s order, but B and C are compressed and an artificial plateau precedes D. Within this transient, crowding damage is the trigger (without it colonies explode), social learning (R2) the amplifier, the plasticity window (R1) the ratchet and the refuges (R7) make the isolated class persistent. The ratchet matters because of the refuges: they are pockets of low density in which, without R1, deteriorated animals recover and breed again (natality resumes and O17p falls to 0.21), whereas without refuges the collapse stays irreversible in most runs even without R1. R1 is what lets a persistent isolated class coexist with an irreversible

collapse (Section 4.2). Only where socialisation is subcritical does the learning cascade propagate over several generations and kill a colony without crowding (Q7), and those colonies first overshoot the crowding threshold, so this is not Calhoun’s collapse.

The timing of extinction is extreme-value statistics. A post hoc analysis of the runs of the preregistered design suggested that crowd aversion makes the extinction time more reproducible. It is not a signature of the mechanism. Without social mortality $T_{\text{ext}} \approx t_{\text{last}} + \ell_{\text{max}}$, the last birth plus the longest residual lifetime (Appendix D.9). The second term, about 110 weeks, fluctuates only on the scale of the inverse slope of the senescent hazard, as expected for the maximum of many lifetimes [15, 7], so the spread of T_{ext} is that of the last birth (correlation 0.92), set by how abruptly natality stops. Without attraction the spread is 0.04, smaller than in C1, and it falls with the size of the lattice (0.14 at $L = 20$, 0.08 at $L = 30$ –60; Table 20).

What “minimal” means here. “Minimal” does not mean few parameters: the core model has more than a dozen, and the added rules about ten (R1: a_w ; R2: λ , w , s_{base} ; AV: κ , p ; R7: f_R , c_R , π , q , b and the hard capacity), all fixed by qualitative calibration against the evaluated patterns. It means that every added rule is local, has a neutral value that recovers the core model and is motivated by an observation in Calhoun’s report; that rules found unnecessary were removed (long-range jumps, state-dependent attraction, territorial exclusion, social mortality); and that each rule of the minimal set is necessary by ablation *at the evaluation thresholds used*: a class threshold of 10%, isolation at $m \leq 2$, persistence of four weeks and an ensemble threshold of 0.8. The necessity of R7 is necessity for O13p, a criterion added together with R7; for O6, O7 and O11 removing R7 changes nothing, whereas R1 and R2 remain needed (O11 in 0.71 of the applicable runs and 0.00 without each). The preregistered necessity test Q2 is rejected with the primary criteria ($p_{\text{Holm}} = 2.2 \times 10^{-7}$) but not with the preregistered criteria ($p_{\text{Holm}} = 0.067$); the rejection comes only from the post hoc choice of t_{B99} in O5, a generic criterion (Table 27). Removing AV lowers the primary outcome by 0.20 [0.13, 0.28] in the factorial. The predicted size of the effect, however, fails: without AV the primary outcome was to fall to at most 0.5, and it is 0.79 [0.71, 0.85]. AV therefore contributes but is not necessary in the preregistered sense; we do not use the observed power of the test, which cannot explain a decision [18]. At these thresholds the minimal set for the primary outcome is C1’ (R1, R2 and R7): 0.835 [0.78, 0.88] of 200 runs, borderline because the lower bound is below 0.8, against 0.995 for C1 (not preregistered). It falls to 0.33 at a class threshold of 15%, and that of C2 from 0.70 to 0.01 at 12.5% (Fig. 10), so minimality is not a property of the rules alone. Adding AV makes O5 more reliable and the decline senescent (O8 in 0.62 against 0.21), and reduces the free space at the peak ($f_0 = 0.16$ against 0.52); partial inheritance is not necessary ($w = 0$ with $r = 0.15$ gives 1.00). Making the list of required patterns and their thresholds explicit, and testable run by run, is in our view the main methodological contribution of this work.

9.1 Limitations and circularity

Circularity between the patterns and the calibration. The same patterns and evaluator were used to *choose* the rules and to *report* them; no pattern was held out. The refuge capacity $c_R = 2$ coincides with the isolation threshold of O13 (with $c_R = 3$, O17p is zero at every point of the sensitivity designs); the persistence criterion O13p was added after C0, and R7 was designed to satisfy it; m_c is both the damage threshold and, in the preregistered evaluator, the boundary of the crowded class (with a fixed threshold $m > 8$ the Calhoun-like fraction of the hypercubes is 2.0%, against 2.5% with $m > m_c$). The empty fraction at the peak was used informally to discard candidates and is a condition of the primary O4; O12 was used to choose $\alpha = 4$ (Section 5). The

fulfilment of the patterns by C1 therefore shows that they are *possible* under local rules, not that they were predicted. Among the preregistered tests, Q1 and Q4 are consistency checks, the contrast of Q3 is confounded by the demography, the prediction of Q6 is not supported (with the primary O11 the window has little or no effect on the boundary), and Q7 establishes an endogenous collapse without crowding, not the form of its boundary. All deviations from the plan are listed in Table 27.

Robustness. Against stochasticity robustness is high: O17p is 0.995 [0.97, 1.00] over 200 runs, independent ensembles replicate it, eight refuge layouts gave no difference, and exploratory runs on lattices up to $L = 84$ at fixed density (30 runs per size, single-pass refuge admission) give the same peak density and persistent class, with no healthy pocket reseeding the colony. Locally it holds over most of the phase planes, whose boundaries are estimated from 30 runs per point. Globally it is limited and relative to the sampled box: over two 15-factor Latin hypercubes the mean primary outcome is 0.040 [0.026, 0.059] and 2.0% [0.7, 3.6] of the points are Calhoun-like (7.2% with the preregistered criteria). Part of this is definitional ($c_R = 3$, large m_c) and part is mechanism (a total refuge capacity $f_{RCR} \geq 0.3$ and a strong preference $\pi \geq 100$); even with the four restrictions together, chosen after the runs, 33 of 37 points fail. The Sobol’ indices are conditional on the factors fixed after screening. Demographically, the primary outcome depends on the timing of the end of natality, as discussed above, while the patterns that the rules add do not.

Scales and the comparison with Universe 25. With at most one Moore step per week, the Damköhler number $t_{\text{mix}}/t_{\text{det}}$ is about 1.7×10^2 in C1 (about 10 for founder colonies), against $\lesssim 4 \times 10^{-3}$ in Universe 25, whose mice crossed the enclosure every day (Table 36). The spatial structure of the model is that of animals almost sessile on the scale of the deterioration, and with more movement sub-steps the coexistence of classes is lost. The model lacks the separation $t_{\text{mix}} \ll T_{\text{gen}} \ll t_{\text{det}}$ of Universe 25. We map a cell to a nest box, m_c to its comfortable occupancy (“fifteen mice could comfortably rest in a single nest box” [5]), K_γ to Calhoun’s 3840 places and the refuges to the little-preferred high nests; $c_R = 2$ is the isolation threshold, not the capacity of a nest. With this mapping f_0 and ρ_{pk} have the same definitions as Calhoun’s 20% of empty nest boxes and 0.57, but $|G|$ and m_c were not calibrated and ρ_{pk} depends on the seeding density, so they agree only in sign. R7 was calibrated with synchronous moves; the exact hard capacity leaves the primary outcome unchanged, and with a random sequential update a soft capacity suffices ($\rho_{\text{pk}} = 0.43$).

What is not reproduced. The founder phase (O1–O3): from four founding pairs a colony spreads as a slow front, and no point of an exploratory founder plane is Calhoun-like (O17p at most 0.37). The beautiful ones (O13b): the persistent isolated class of C1 is made of refugees of the early boom, most of whom entered the refuges as pups, not of a late-born cohort; the variant C2 produces a later-born class at a demographic cost. Total infant mortality (O15) and the sex bias of the remnant (O9) are not reproduced, because conception and pup survival decline together and there is no sex-specific mortality. Details, the refuted coexistence band (Q5) and the comparison with [16] are in Appendix E. That model reproduces the founder phase and the timing of the curve, which ours does not; ours shows that local rules, without population-level switches, produce coexisting classes of deteriorated animals, one of them persistent, on top of an envelope (a low peak with free space, an end of natality and the failure of the late-born cohorts) that crowding damage and the demography already give. Both agree that replicate variability is large and that Calhoun’s trajectory is one admissible outcome among many. We make no claim about human populations.

Future work. A founder protocol with realistic mobility and an explicit adaptation phase; patterns held out from the calibration; gestation, lactation and background mortality, so that births spread up to the peak, which is the condition for a late-born isolated cohort and for natality to end at high population without a synchronous start; decoupling conception from pup survival and adding sex-specific risks; and a closer comparison with maternal-effect theories of population cycles [12, 19].

Code and data availability

All code, simulation outputs and analysis scripts are available in the repository <https://github.com/jfbonder/universe25-abm>; the version used in this paper (release v1.0) is archived in Zenodo, doi:10.5281/zenodo.23111283. The model is implemented in the Python package `u25`; the candidates of this paper are the presets `calhoun_C0`, `calhoun_C1p` ($C1'$), `calhoun_C1` and `calhoun_C2`. Every ensemble reported here is generated by `scripts/production/run_2b_v2.sh` with fixed base seeds recorded in each `results_v2/f2b_*/meta.json`; the confirmatory family and the remaining stages are analysed in `results_v2/f2b_final/`, and every number quoted in Results is dumped, with its ensemble, to `results_v2/results_numbers.json`, including $O11$ at age a_w (re-run with the same seeds in `results_v2/o11_aw/`, `make o11-aw`) and Γ (re-run in `results/fisico_v3/` by `scripts/analysis/gamma_v3.py`). All figures and tables are regenerated by `make figs`, as described in the repository README. The preregistered statistical plan is reproduced in `docs/preregistered_plan.m`.

Declaration of generative AI use

The author designed the project and provided a preliminary implementation of the agent-based model, together with a draft describing its objectives and interaction rules, which were taken as the starting point. Claude (Anthropic), used through Claude Code in a multi-agent setting (agents with the roles of mathematician, physicist, programmer and literature reviewer, coordinated by a principal agent), contributed to the reimplementing and extension of the code, the simulations, the statistical analysis, the literature review and the writing of the manuscript. An independent system of Claude agents carried out an internal review in three rounds. The author supervised the work, reviewed all of its content and takes full responsibility for it.

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A Model description following the ODD protocol

We follow the second update of the ODD protocol [13]. Parameter names refer to the `Params` class of the `u25` package.

A.1 Purpose and patterns

The purpose is to identify a minimal set of *local* rules that, with unlimited resources and without thresholds on the total population, generates the qualitative sequence reported for Universe 25 [5]:

1. latency;

2. exponential growth;
3. deceleration with social deterioration;
4. cessation of surviving births near the peak;
5. decline without recovery to extinction;
6. differentiated classes of deteriorated animals.

The model is evaluated against the patterns of Table 1 and, for each pattern, against null models (Section 8.2). With the default initial condition only items 3–6 are testable, and the model reproduces them as a boom of one or two generations followed by extinction by senescence; of these items, the crowding damage and the demography already give the end of natality and the decline, and the rules add the classes of item 6 (Section 9).

A.2 Entities, state variables and scales

Agents (mice). Each agent i carries:

- an identifier;
- sex $\sigma_i \in \{M, F\}$, fixed at birth;
- a cell (x_i, y_i) on an $L \times L$ lattice;
- an age a_i in weeks;
- a social state $s_i \in [0, 1]$, where 1 means full social competence;
- optionally, a refractory counter and the maternal social state at birth h_i (used by C2).

Pairs. Each pair carries a symmetric affinity $F_{ij} \in [0, 1]$, stored sparsely: only pairs with $F_{ij} \geq \varepsilon$ are kept.

Cells. A cell has its occupancy n_c , which is derived. Ordinary cells have no occupancy limit. A fixed random fraction f_R of cells are *refuges* with capacity c_R . The layout is drawn once with its own random generator (`refuge_seed` = 0) and is the same in every run of every stage, so all intervals are conditional on it; eight alternative layouts gave no detectable difference.

Scales.

- One time step is one week.
- The lattice has $L = 30$ and hard walls.
- A cell is an abstract meeting place, not a metric distance.
- Mapping: a cell corresponds to a resting place (a nest box of Universe 25 [5]), the damage threshold m_c to its comfortable occupancy, and $K_\gamma = L^2 m_c = 7200$, the population at which every cell would sit at the damage threshold, to Calhoun’s 3840 places. K_γ is a reference for the peak density, not a physical capacity, because cells have no occupancy limit, and $|G|$ and m_c were not calibrated: model densities are compared with Calhoun’s only in sign. The refuges correspond to the little-preferred high nests; their capacity $c_R = 2$ is the isolation threshold of O13, not the capacity of a nest (Table 36).

- Mobility is at most one Moore step per week. Crossing the lattice of $L = 30$ cells takes at least 29 weeks in a straight line and about 1.7×10^3 weeks by the observed diffusion, whereas real mice crossed the enclosure daily. The ratio of the mixing time to the deterioration time (a Damköhler number) is about 1.7×10^2 in C1 and about 10 for founder colonies, against $\lesssim 4 \times 10^{-3}$ in Universe 25 (Section 4.4).

A.3 Process overview and scheduling

Each week the following steps are executed in order:

1. movement. This is simultaneous, with weights evaluated on the pre-move snapshot, repeated `move_substeps` times. Refuge capacity is enforced after each sub-step (see Movement below). An asynchronous variant, in which agents move one at a time in random order on the current occupancy, is available (`async_move`) and reported as a robustness check;
2. affinity update;
3. occupancy m_i after movement, including i itself;
4. social-state update;
5. reproduction;
6. ageing;
7. mortality;
8. recording of observables.

Updates within each submodel are synchronous, except movement in the asynchronous variant.

A.4 Design concepts

Basic principles. The model rests on three ideas:

- social stress from crowding reduces reproduction and offspring survival [6, 30];
- maternal transmission of individual quality [12, 19];
- the behavioural sink: attraction to familiar individuals produces aggregation despite free space [2]. In Universe 25 this appears as “the learned need for proximity to others . . . gains dominance over the primary need” [5].

Emergence. The following are not imposed by any rule:

- the population trajectory;
- spatial aggregation;
- the effective deterioration threshold;
- the classes of crowded and isolated deteriorated animals.

The following are imposed by design and are therefore consistency checks, not emergent properties (Section 9): the irreversibility of adults, which is R1, and with it the failure of transplanted late-phase animals; the persistence of the isolated class in refuges of capacity $c_R = 2$, equal to the isolation threshold; and the absence of social deaths and the senescent decline, since social mortality is off.

Adaptation, objectives, learning, prediction. Agents do not optimise anything. “Social learning” (R2) is a contagion rule.

Sensing. Agents sense only their Moore neighbourhood: their affinities with the occupants of each neighbouring cell, the occupancy of those cells, the mean social state of neighbours, and whether a cell is a refuge and is full. No rule uses the total population size.

Interaction. Interactions are co-presence (affinity), mating within the cell (or the Moore neighbourhood if `mate_radius` = 1), social learning, and crowding.

Stochasticity. The following are random:

- destination choice;
- mate choice, conception, litter size, neonatal survival and sex;
- death;
- the order of entry into full refuges;
- the non-refuge cell to which a pup, a founder or a transplanted animal is moved when a refuge is full;
- in the asynchronous variant, the order in which agents move;
- in the non-synchronous variant, the initial ages.

The refuge layout is fixed (see Cells). The state $N = 0$ is absorbing.

Observation. Per-step time series, per-run summaries, and the automatic pattern evaluator (Section 3).

A.5 Initialization

- *Default:* 200 males and 200 females aged 4 weeks with $s = 1$, placed uniformly at random, with no affinities. This is a synchronous cohort; a non-synchronous initial condition is reported as a robustness check: integer ages drawn uniformly between 4 and 52 weeks, or between 4 and 80 weeks (`initial_age_dist="uniform"`, `initial_age_hi`).
- *Founders* (presets `founders`, `founders_longevo`): 4 males and 4 females aged 7 weeks with $s = 1$ in the central cell. This mimics Calhoun’s four pairs of 48-day-old mice [5].
- *Transplant:* 4 males and 4 females of reproductive age are taken from a donor run in phase B or D, keep their s , and are placed in the central cell of an empty lattice.

A.6 Input data

None. The environment is constant: no resource limitation, no predation, no emigration.

A.7 Submodels

Movement. Agent i chooses a destination y in its Moore neighbourhood $V(i)$, which includes its own cell, with weight

$$w_i(y) = [1 + \alpha A_i(y)] e^{-\kappa(1-s_i)^{p_{ny}}} [1 + \pi(1 - b_i)^q \mathbf{1}_{\mathcal{R}}(y)], \quad A_i(y) = \sum_{j \in y, j \neq i} F_{ij}.$$

The terms are:

- n_y is the occupancy of y excluding i ;
- $\mathbf{1}_{\mathcal{R}}(y) = 1$ if y is a non-full refuge ($n_y < c_R$) and 0 otherwise, $\mathcal{R}$ being the set of non-full refuges;
- κ is `crowd_aversion` and p is `crowd_power` (rule AV);
- π is `refuge_pref` and q is `refuge_pref_power` (rule R7);
- $b_i = s_i$ in C1 and $b_i = h_i$ in C2 (`refuge_pref_basis`).

Full refuges cannot be entered. With hard capacity, the agents already inside have priority, the entering agents are drawn in random order, and rejected agents return to their previous cell. Admission (`refuge_hard="strict"`) is iterated to a fixed point: a rejected agent returning to a refuge it had left counts as a resident, and whoever took its place is rejected in turn. Pups that would put a refuge above capacity are placed in a uniformly chosen non-refuge cell of the nearest Chebyshev ring around the mother's cell that contains one (her Moore neighbourhood, unless all its cells are refuges, which happens mostly at corners of the lattice), and founders and transplanted animals are placed in the same way. A test checks that no refuge ever exceeds its capacity in real runs of C1, C1' and C2. A single-pass variant, `refuge_hard="legacy"`, is reported as a robustness check: it does not count a rejected agent returning to a refuge it had left, nor pups born in a refuge, so refuges can exceed their capacity (up to 11 occupants with $c_R = 2$). If all the weights of an agent are zero, it stays in its cell; such events are counted (`n_zero_weight_tot`).

Affinity. If i and j are at Chebyshev distance ≤ 1 after movement, $F_{ij} \leftarrow (1 - \eta)F_{ij} + \eta$. Otherwise $F_{ij} \leftarrow (1 - \delta)F_{ij}$. Pairs with $F_{ij} < \varepsilon$ are pruned.

Social state.

$$s_i \leftarrow \Pi_{[0,1]} \left(s_i + r \mathbf{1}_{a_i \leq a_w} [(1 - \lambda) + \lambda \bar{s}_{V(i)}] - \gamma (m_i - m_c)_+ \right),$$

where:

- r is the recovery (learning) rate (`rec`);
- $\bar{s}_{V(i)}$ is the mean social state of the *other* agents in $V(i)$, and is 0 if there are none;
- a_w is the plasticity window (rule R1: `plast_age_max = a_w`, `plast_outside_rec = 0`);
- λ is the social-learning weight (rule R2: `social_rec_lambda`).

The neutral values $a_w = \infty$ and $\lambda = 0$ (with $w = 1$ and $r = 0.01$) recover the core model.

Reproduction.

1. Each reproductive, non-refractory female ($a_{\min} \leq a \leq a_{\text{rep}}$) that shares her cell with at least one reproductive male chooses one uniformly at random.
2. She conceives with probability $p_0 s_f s_m$.
3. The litter size L is drawn from the empirical distribution (mean 6.55), and $\text{Binomial}(L, q_0 s_f)$ pups survive, with $q_0 = 0.95$.

4. Pups are born in the mother’s cell at age 0, with sex ratio $1/2$, and social state $s = w s_f + (1 - w) s_{\text{base}}$. With $w = 0.2$ and $s_{\text{base}} = 0$, every pup is born with $s \leq 0.2$. Pups are independent agents from birth: they move and add to local crowding. A variant with a nest period, in which pups stay in the maternal cell and do not add to crowding until weaning, is reported as a robustness check (`nest_period` = 3 weeks). Nest-bound pups count towards refuge capacity and form affinities.

Mortality. After ageing, each agent dies with probability $1/(1 + e^{-k(a - a_{\text{max}})})$. Two optional social terms exist, an additive social hazard and the factor μ_s of Eq. (4) (`social_mortality_factor`, applied with `social_mortality_form="shift"`), but both are zero in all presets: in Universe 25 adults died of old age and the beautiful ones were “physically healthy” [5]. There is no mortality before senescence in the presets; a background hazard μ_{bg} (up to 0.01 per week) is reported as a robustness check (`mu_bg`). It applies at all ages, pups included, combines with the senescent hazard as $1 - (1 - p_{\text{age}})(1 - \mu_{\text{bg}})$, is independent of the social state and is not counted as social mortality.

A.8 Parameters

Table 8: Parameters of the short-lived control (SL), values in C1, and justification (complements Table 3, which also gives the long-lived control). [E] empirical, [O] order of magnitude, [M] modelling choice, [C] calibrated qualitatively.

Parameter	SL	C1	Justification
<code>grid_size</code> L	30	30	[M] abstract space; Calhoun had 16 cells, 256 nest boxes
N_0	200+200	200+200	[M]; founders 4+4 as in [5]
<code>initial_age</code>	4	4	[M]; 7 for founders (48 days)
$\eta, \delta, \varepsilon$	0.2, 0.01, 10^{-6}	same	[M] bond formation ~ 5 weeks, memory ~ 100 weeks
α	1.4	4	[C] attraction to familiar individuals (behavioural sink)
m_c	8	8	[M/O] “Fifteen mice could comfortably rest in a single nest box”; groups of “over 10 individuals” [5]
r (<code>rec</code>)	0.01	0.1	[M/C] recovery/learning rate
γ	0.1	0.1	[M]
p_0	0.3	0.3	[O] weekly conception probability given an encounter
q_0	0.95	0.95	[M]
a_{min}	6	6	[O] sexual maturity ≈ 6 weeks
a_{rep}	50	80	[E] menopause at “about 560 days” ≈ 80 weeks [5]
litter sizes	3–10, mean 6.55	same	[O] mean 6.12 in [16]’s source
a_{max}, k	60, 0.15	120, 0.15	[E] “large numbers” lived to 800 days; last survivors > 987 days [5]
a_w (R1)	∞	12	[C] juvenile plasticity window
λ (R2)	0	1	[C] social learning
w, s_{base}	1, 1	0.2, 0	[C] partial maternal inheritance
κ, p (AV)	0, 1	0.15, 4	[C] crowd aversion of deteriorated animals

Parameter	SL	C1	Justification
refuge fraction, capacity	0, –	0.3, 2 (hard)	[C] R7; layout <code>refuge_seed = 0</code>
π , q , basis (R7)	0	100, 4, current	[C]; C2: 1000, 1, maternal
<code>max_pop</code>	60000	15000	numerical safeguard

Values checked against the four presets `calhoun_C0`, `calhoun_C1p`, `calhoun_C1` and `calhoun_C2` of `u25/params.py` (Table 3). `C1'` (`calhoun_C1p`) is C1 with $\kappa = 0$. In all four presets `move_substeps = 1` and `mate_radius = 0`.

B Mean-field derivations and numerical details

B.1 Demography

With $h(a) = [1 + e^{-k(a-a_{\max})}]^{-1}$, $k = 0.15$, $a_{\max} = 60$, the survival function $\mathcal{S}(a) = \prod_{0 \leq a' < a} [1 - h(a')]$ (the probability of reaching age a) gives $\mathcal{S}(40) = 0.74$, $\mathcal{S}(50) = 0.27$, $\mathcal{S}(60) = 0.006$, a life expectancy $\text{LE} = \sum_{a \geq 0} \mathcal{S}(a) = 44.8$ weeks and $\sum_{a=6}^{50} \mathcal{S}(a) = 38.0$ expected fertile weeks per newborn. The stationary fraction of reproductive individuals is $\phi = 0.85$, so $\phi_F = \phi_M = 0.42$; $\mu = 1/\text{LE} = 0.0223$ per week. The litter law has $\bar{L} = 6.55$, $\mathbb{E}L^2 = 45.5$. The maximal per-capita birth rate is $b_{\max} = \phi_F p_0 \bar{L} q_0 = 0.79$ per week. With the long-lived demography of the candidates ($a_{\max} = 120$, fertility up to 80 weeks) $R_0 = 70$ and $s_v = R_0^{-1/3} = 0.24$. The fractions ϕ_M and ϕ_F are those of the stationary age distribution; Eq. (10) applies them during growth, when the population is younger, as an approximation.

B.2 Occupancy statistics

For Poisson occupancy $D(\nu) = \sum_{k \geq m_c} (k - m_c + 1) e^{-\nu} \nu^k / k!$; the balance $D(\nu^*) = r/\gamma = 0.1$ gives $\nu^* = 4.13$. For a negative binomial of mean ν and size ξ (variance $\nu + \nu^2/\xi$, index of dispersion $1 + \nu/\xi$) the same balance gives $N^* = 2840, 2225, 1718$ for $\xi = 5, 2, 1$. In the fast ABM with $\alpha = 0$ the effective ξ fitted from the index of dispersion is 1.4 during growth (many fresh litters) and 3.9 during decline. In the short-lived control ($\alpha = 1.4$) the index of dispersion is 29 at the peak and 21 in the quasi-stationary regime; Lloyd's mean crowding $m^* = \sum_c n_c^2 / \sum_c n_c - 1$ is 34 and 21 respectively, while 43% of the individuals sit in cells with $m \leq m_c$: the distribution of experienced crowding is strongly bimodal.

B.3 Two-variable model: linearisation and first integral

At the equilibrium of (9),

$$J = \begin{pmatrix} N^* b'(N^*) S^{*3} & 3N^* b(N^*) S^{*2} \\ -\gamma D'(\nu^*)/|G| & 0 \end{pmatrix}, \quad \text{tr } J = \mu \frac{\phi_M \nu^* e^{-\phi_M \nu^*}}{1 - e^{-\phi_M \nu^*}} > 0.$$

For Poisson occupancy $N^* = 3717$, $S^* = 0.324$, eigenvalues $0.00411 \pm 0.1034i$; integrating the unclamped system from $1.1N^*$ the successive maxima grow as $e^{0.00409t}$. With b frozen at its saturation value the maxima are constant to 10^{-5} , verifying the first integral. For the clamped system, Table 9 lists the limit cycle.

occupancy	N^*	S^*	first peak	later peaks	trough	period (weeks)
Poisson	3717	0.324	11185	5841	1679	96
neg. binomial $\xi = 5$	2840	0.337	10024	5043	1182	107
neg. binomial $\xi = 2$	2225	0.352	8805	4482	854	119
neg. binomial $\xi = 1$	1718	0.370	7496	4052	598	133
ABM $\alpha = 0$ (40 runs)	—	—	6954 ± 413	5528	median 700	113.5 ± 10

Table 9: Limit cycle of the clamped two-variable mean field from $(N_0, S_0) = (80, 1)$, compared with the fast ABM started from the same 80 animals.

B.4 Age-structured mean field

Weekly cohorts n_a, s_a ($a \leq 130$): $s_a \leftarrow \Pi_{[0,1]}(s_a + r - \gamma D(\nu))$; births

$$B = \sum_a \frac{1}{2} n_a \mathbf{1}_{6 \leq a \leq 50} (1 - e^{-n_M/|G|}) p_0 s_a \bar{s}_M \bar{L} q_0 s_a,$$

with $\bar{s}_M$ the fecundity-weighted male state, newborn state $s_{\text{new}} = \sum_a B_a s_a / B$; ageing and logistic survival. A class of healthy-pocket individuals, of fraction ρ , experiences a reduced density $f_P \nu$ (healthy pockets are uncrowded cells of the core model without the rules, unrelated to the refuges of R7). With $f_P = 0$ the bifurcation to sustained oscillation is at $\rho_c = 0.0216$ (Poisson) and 0.0286 ($\xi = 2$); just above ρ_c the trough is $N_{\min} \approx 77 \approx N_A$ and the period is ≈ 170 weeks. The well-mixed ABM (uniform relocation each week) gives $N_{\max} = 8860 \pm 80$, $t_{\text{ext}} = 160 \pm 12$ weeks over 40 seeds, a regression test with tight tolerances for any reimplementations.

B.5 Kinetic mean field with persistent cell disorder

Individuals carry age, sex and s ; cells carry fixed weights $w_c \sim \Gamma(\xi_c, 1/\xi_c)$, so occupancy is Poisson–Gamma with index of dispersion $1 + \nu/\xi_c$. Each week an individual relocates with probability $1/\tau_F$ to a cell drawn with probability $\propto w_c$; pups are born in their mother’s cell; R1 and R2 act as in the ABM (R2 uses the mean s of the other occupants of the cell). Calibrated to the quasi-stationary regime of the short-lived control ($N \approx 1735$, $S \approx 0.23$): $\xi_c = 0.5$, $\tau_F = 8$ gives $N_{\text{QS}} \approx 1450$, $S \approx 0.29$, 56% with $s < 0.2$; $\xi_c = 1$, $\tau_F = 5$ gives $N_{\text{QS}} \approx 2330$. Frozen disorder does not reproduce the initial overshoot ($N_{\max} \approx 5500$) because clusters in the ABM form on the affinity time scale $1/\eta$, after phase B; that regime is covered by the homogeneous models above. Each run costs 0.5–1.5 s.

B.6 Lineage criterion for R1

A lineage that starts from s_0 accumulates $\prod_g R_0(s_g)$ expected descendants before it reaches viability s_v , with $R_0(s) = R_0 s^3$ when the partner is equally deteriorated (conception $\propto s_f s_m$, neonatal survival $\propto s_f$) and $R_0 s^2$ in the first generation when the partner is healthy. Between generations, competence grows additively without R2 (full inheritance and individual recovery during the window, $s_{g+1} = s_g + \Delta$, $\Delta = r a_w$) and multiplicatively with R2 ($\lambda = 1$, inheritance w , learning from neighbours of competence φS : $s_{g+1} = (w + \varphi \Lambda) s_g$). This is the expected value of a branching process without crowding and without learning from neighbours outside the lineage, so it is a heuristic bound, not a prediction (scripts/analysis/mf_lineage_v2.py).

The erosion of healthy adults under R1 is $\gamma D(\nu_{\text{loc}})$ per week (permanent): with $\xi = 2$, a local density of 2, 2.5 and 3 per cell removes 0.19, 0.46 and 0.89 over 44 fertile weeks, against an individual regeneration without R1 of $r \cdot 44 = 0.44$ –1.3.

scenario	r	$a_w = 6$	12	24	48	∞	
$\xi_c = 0.5, \tau_F = 8$, short life	0.01	0.3	0.2	0.0	0.0	0.0	
	0.03	0.1	0.0	0.0	0.0	0.0	
$\xi_c = 1, \tau_F = 5$, short life	0.01	1.0	0.7	0.4	0.0	0.0	
	0.03	0.6	0.7	0.0	0.0	0.0	
$\xi_c = 0.5, \tau_F = 8$, long-lived	0.01	1.0	1.0	1.0	1.0	0.0	
	0.03	1.0	1.0	1.0	0.0	0.0	
		$\lambda = 0$	0.3	0.5	0.7	0.9	1.0
$\xi_c = 0.5, \tau_F = 8$, short life	0.01	0.0	0.0	0.0	0.0	0.0	0.3
	0.03	0.0	0.0	0.0	0.0	0.0	0.0
$\xi_c = 1, \tau_F = 5$, short life	0.01	0.0	0.1	0.1	0.2	0.8	0.8
	0.03	0.0	0.0	0.0	0.0	0.0	0.6
$\xi_c = 0.5, \tau_F = 8$, long-lived	0.01	0.0	0.8	1.0	1.0	1.0	1.0
	0.03	0.0	0.0	0.0	0.0	1.0	1.0
$\xi_c = 1, \tau_F = 5$, long-lived	0.03	0.0	0.0	0.0	1.0	1.0	1.0

Table 10: Extinction probability (10 seeds, 1500 weeks; 10/10 has a Wilson interval of $[0.72, 1]$) in the kinetic mean field under R1 (top, columns a_w) and R2 (bottom, columns λ), for the two recovery rates tested. The long-lived scenario has $a_{\max} = 120$ and fertility up to 80 weeks; with $\xi_c = 1, \tau_F = 5$ the long-lived control itself goes extinct for $r = 0.01$.

B.7 Recovery time under R2 and critical λ

From (11) with $s_v = 0.31$ (weeks):

r, S_0	$\lambda = 0$	0.3	0.5	0.7	0.9	1.0
0.01, 0.02	29	39	50	71	130	274
0.01, 0.10	21	28	35	48	77	113
0.03, 0.02	10	13	17	24	43	91
0.03, 0.10	7	9	12	16	26	38

Solving $T_{\text{rec}}(\lambda_c) = \tau_{\text{disp}}$ with $S_0 = 0.02$: $\lambda_c = 0.04, 0.50$ for $r = 0.01$ and $\tau_{\text{disp}} = 30, 50$; $\lambda_c = 0.61, 0.80, 0.93$ for $r = 0.03$ and $\tau_{\text{disp}} = 20, 30, 50$. These values are consistent with the kinetic thresholds of Table 10 for the long-lived demography: below 0.3 at $r = 0.01$ and in $(0.7, 0.9)$ at $r = 0.03$, or $(0.5, 0.7)$ with faster mixing.

B.8 Hysteresis and transplant

Hysteresis area $A_H = \int |b_{\downarrow} - b_{\uparrow}| d \log N / \int b_{\uparrow} d \log N$ of the per-capita birth rate over the first cycle (deterministic age-structured mean field): short-lived control 0.932 ($r = 0.01$), 0.916 ($r = 0.03$), long-lived control 0.969; R1 ($a_w = 12$) and R2 ($\lambda = 1$) 1.000. Transplant of four pairs ($s = 0.02$, ages 6–30, local neighbourhood of 25 cells, 50 replicas): short-lived control $P_T = 0.00$ ($r = 0.01$), 0.36 ($r = 0.03$), 0.64 and 1.00 long-lived; R1 ($a_w = 12$) 0.00 in all cases; R2 $\lambda = 1$: 0.00; R2 $\lambda = 0.7$: 0.02 (short life), 0.78 (long-lived); healthy controls 0.94.

B.9 Statistical methods

Extinction times are right-censored at the horizon; we report Kaplan–Meier estimates with Greenwood log-log bands, Wilson intervals for $P_{\text{ext}}(T)$, Nelson–Aalen cumulative hazards, piecewise-constant hazards with likelihood ratio tests, and maximum-likelihood exponential, Weibull and

demography, map	partner	$s_0 = 0.02$	0.05	0.10	0.20
short-lived ($R_0 = 35$, $s_v = 0.31$), $\Delta = 0.12$	deteriorated	$2 \cdot 10^{-5}$	$6 \cdot 10^{-4}$	0.013	0.28
	healthy	$8 \cdot 10^{-4}$	0.013	0.13	1.4
long-lived ($R_0 = 70$, $s_v = 0.24$), $\Delta = 0.12$	deteriorated	$1 \cdot 10^{-4}$	$3 \cdot 10^{-3}$	0.052	0.56
	healthy	$5 \cdot 10^{-3}$	0.060	0.52	2.8
long-lived, $\Delta = 0.36$	deteriorated	$6 \cdot 10^{-4}$	$9 \cdot 10^{-3}$	0.070	0.56
	healthy	0.028	0.17	0.70	2.8
long-lived, R2 with $w + \varphi\Lambda = 1.4$	deteriorated	$2 \cdot 10^{-14}$	$1 \cdot 10^{-6}$	0.007	0.56
	healthy	$9 \cdot 10^{-13}$	$3 \cdot 10^{-5}$	0.071	2.8
long-lived, R2 with $w + \varphi\Lambda = 0.8$	either	0	0	0	0

Table 11: Expected number of descendants accumulated by a lineage before it reaches viability from s_0 , without further crowding. The C1 values are $w = 0.2$ and $\Lambda = ra_w = 1.2$; $\varphi = 1$ gives the supercritical factor 1.4 and $\varphi = 0.5$ a subcritical 0.8, for which the lineage never reaches s_v . For $s_0 \leq 0.05$ the expected number is at most 10^{-2} in every row with a deteriorated partner and at most 0.17 with a healthy one; for $s_0 = 0.1$ it is 0.01–0.07 with a deteriorated partner and up to 0.7 with a healthy one.

lognormal fits on the conditional tail with parametric-bootstrap goodness of fit. Peak drift is measured as the slope of $\log N_{\text{peak}}$ versus cycle index (peaks with prominence $\geq 10\%$ of N_{max} , first peak excluded), with Mann–Kendall tests per run, a Wilcoxon test of the per-run slopes and a within-run regression with cluster-robust errors. Global sensitivity uses Morris elementary effects and Sobol indices (Saltelli sampling) on replicate-averaged metrics, with bootstrap confidence intervals and Holm or Benjamini–Hochberg corrections.

C Supplementary results

C.1 Mean-field analysis of the core model: limit cycle, healthy pockets and drift

This subsection collects the mean-field analysis of the *core model without the four rules*, that is, of the short-lived control ($\alpha = 1.4$, $a_{\text{max}} = 60$) and of its variants in α , summarised in Section 6.1; none of it applies to the calibrated candidates beyond what is stated there.

Occupancy. The effective weekly loss of social state at $N = 1600$ in the short-lived control is 0.75, against a Poisson prediction of 0.001. With negative-binomial occupancy of dispersion ξ , the switch point N^* drops from 3700 ($\xi = \infty$) to 2200 ($\xi = 2$) and 1700 ($\xi = 1$) (Appendix B.2).

Two-variable mean field: a limit cycle, not Lotka–Volterra. With a first-moment closure, $\mathbb{E}[s_f^2 s_m] \approx S^3$, the slow variables N and S obey

$$\dot{N} = N[b(N)S^3 - \mu], \quad \dot{S} = r - \gamma D(N/|G|), \quad S \in [0, 1]. \quad (9)$$

The interior equilibrium is $D(N^*/|G|) = r/\gamma$, $S^* = (\mu/b(N^*))^{1/3} \approx 0.32\text{--}0.37$. Three facts fix the nature of the oscillations. (i) If b were constant, (9) would be separable, with first integral $H = bS^4/4 - \mu S - \int^N [r - \gamma D(n/|G|)] dn/n$: a continuum of neutral closed orbits around a centre, as in the Lotka–Volterra system, although the vector field is not bilinear. (ii) The Allee factor gives $\text{tr } J = N^*b'(N^*)S^{*3} > 0$: the equilibrium is a weakly unstable focus, with eigenvalues $0.0041 \pm 0.103i$ per week (linear period 61 weeks, amplitude e -folding time about 240 weeks). (iii) The clamping

$S \in [0, 1]$ is dissipative: an orbit that reaches $S = 1$ while $N < N^*$ stays there until $N = N^*$ and then follows *the* orbit through $(N^*, 1)$, whatever its past. Hence every trajectory that reaches $S = 1$ joins in finite time a single periodic orbit. We call this cycle *superstable* because the clamp makes its Floquet multiplier zero: nearby orbits do not approach it geometrically but merge with it. This is a sketch, not a proof that every trajectory gets there: it excludes the equilibrium, and it would require showing that the growing spiral reaches $S = 1$ before N crosses the Allee threshold; all the trajectories we integrated did (Fig. 3). The period of the cycle, 96–133 weeks depending on ξ (Table 9), brackets the 113 ± 10 weeks of the agent-based model with $\alpha = 0$; this is a range obtained by varying ξ , not a point prediction. The first peak is a transient overshoot of the homogeneous initial condition $S_0 = 1$ (ratio 1.25 in the agent-based model, 1.9 in the ODE).

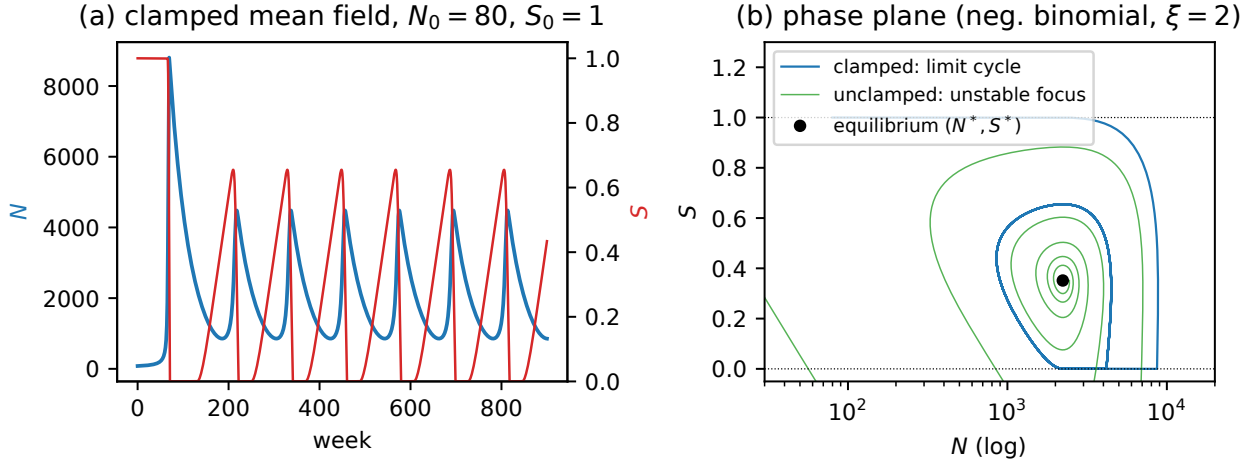


Figure 3: Two-variable mean field (9) with negative-binomial occupancy ($\xi = 2$). (a) $N(t)$ and $S(t)$ from $N_0 = 80$ and $S_0 = 1$: one overshoot followed by a periodic limit cycle. (b) Phase plane: the clamped system (blue) converges to the maximal orbit; without clamping (green) the equilibrium is an unstable focus whose amplitude grows until it hits $S = 0$ or $S = 1$.

Homogeneous collapse versus healthy pockets. The two-variable model lacks demographic inertia. An age-structured mean field and a well-mixed variant of the agent-based model (Poisson occupancy) both predict a single hump followed by deterministic extinction (Fig. 4): at the peak the whole young cohort reaches $s \approx 0$ at once, recovering to viability $s_v = R_0^{-1/3} \approx 0.31$ takes $s_v/r \approx 30$ weeks, and by then the cohort has left its fertile window. The well-mixed agent-based model gives $N_{\max} = 8860 \pm 80$, $t_{\text{ext}} = 160 \pm 12$ weeks and 40/40 extinctions; the age-structured mean field gives $t_{\text{ext}} = 203$ weeks. The oscillations of the spatial model are produced instead by heterogeneity. A minority of individuals living in uncrowded cells, which we call *healthy pockets* (10–20% with $s > 0.2$ at the peak), keeps $s \approx 1$ and repopulates the trough. Their effective fecundity exceeds S^3 by a factor 10–20 at the peak, so the first-moment closure fails exactly where it matters. In the age-structured mean field a fixed fraction ρ of pocket individuals turns extinction into sustained oscillations above a sharp threshold $\rho_c \approx 2.2\%$ (Poisson occupancy) to 2.9% ($\xi = 2$), at which the trough equals the Allee threshold, $N_{\min} \approx N_A$. That threshold assumes that pocket individuals experience no crowding at all ($f_P = 0$), whereas the pockets of the agent-based model are only less crowded, so the comparison is qualitative. The three regimes found when scanning the aggregation strength α in the core model without rules are: (I) homogeneous overshoot for small α with long-lived mixing; (II) a source–sink

coexistence sustained by healthy pockets; (III) collapse by aggregation, when clusters exceed m_c at low N and no healthy pocket survives. Healthy pockets are an emergent property of the core model and are unrelated to the refuges of rule R7.

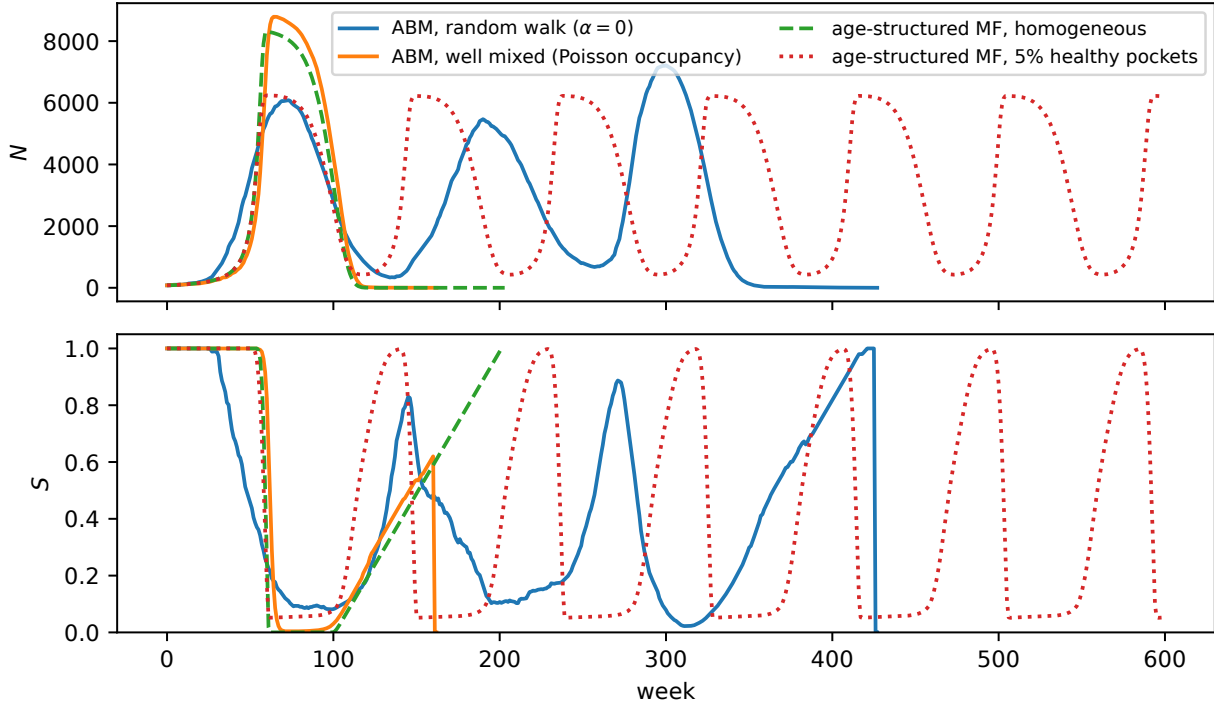


Figure 4: Population and mean social state for the spatial agent-based model with random motion, the well-mixed agent-based model (Poisson occupancy), the homogeneous age-structured mean field and the same mean field with a 5% fraction of healthy-pocket individuals. Homogeneous descriptions collapse once; healthy pockets produce the oscillatory regime.

Extinction and the absence of secular drift. Because the attractor is a cycle, extinction occurs at the troughs, when $N_{\min}$ falls below the Allee threshold of an aged population. The short-lived control is bimodal. In the exploratory ensemble ($\alpha = 1.4$, $n = 100$), 49% [39, 59] of the colonies die in the first collapse ($t \leq 184$ weeks), and the survivors have a hazard of 7×10^{-5} per week [2.6, 15] $\times 10^{-5}$ (Fig. 5a); a piecewise-constant hazard is strongly preferred to a constant one (likelihood ratio 148, $p < 10^{-16}$), while the tail alone is compatible with an exponential (Weibull shape 0.67, $p = 0.28$). The independent production ensemble ($n = 100$, $T = 1000$) gives an early fraction of 0.44 [0.35, 0.54] with the split at 1.5 times the median event time, 0.42–0.45 for splits between 1.25 and 2.5 times the median, and a tail hazard of $(0.5\text{--}1.1) \times 10^{-4}$ per week; 53 of 100 colonies were still alive at T . Extinction is a mixture of a deterministic-like early mode and a rare late mode, not a heavy-tailed law. The inheritance rule $s_{\text{pup}} = s_{\text{mother}}$ was conjectured to produce a secular downward drift of successive peaks. The mean field says otherwise: mothers are weighted by $s_f^2 s_m$ in the birth flux, so newborns carry $\bar{s}_{\text{birth}} = \mathbb{E}[s^3]/\mathbb{E}[s^2] \geq S$; inheritance acts as *selection*, and the clamp at $S = 1$ erases memory in every deep trough. The slope of $\log N_{\text{peak}}$ per cycle (first peak excluded) is $+0.027 \pm 0.019$ for $\alpha = 0$ (40 runs, 2000 weeks) and $+0.005 [-0.001, +0.012]$ for $\alpha = 1.4$ (50 runs with at least four peaks; within-run regression $+0.001 \pm 0.002$); no run shows a

significant negative Mann–Kendall trend (Fig. 5c), and the preregistered production ensemble gave $+0.007 [-0.014, +0.027]$ (Wilcoxon $p = 0.41$). Collapse in the short-lived control is thus *contingent*: it requires the recovery number $\Pi_R = r \tau_{\text{disp}}/s_v$ (τ_{disp} the fertile time left after crowding) to be below one and the fraction of healthy pockets to be below ρ_c .

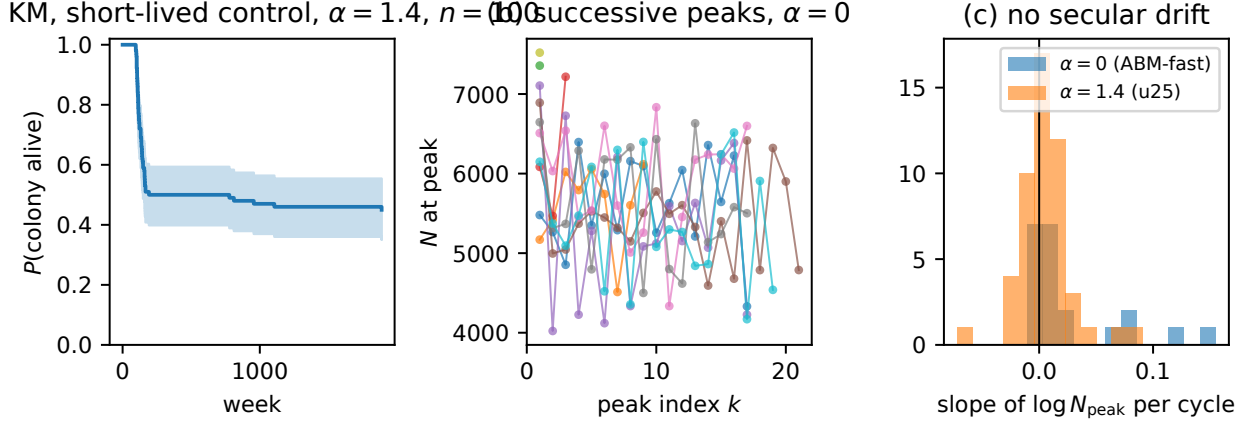


Figure 5: (a) Kaplan–Meier survival of the colony for the short-lived control ($\alpha = 1.4$, 100 replicates, right-censored at 2000 weeks) with Greenwood band. (b) Successive peak populations for ten runs with $\alpha = 0$. (c) Per-run slope of $\log N_{\text{peak}}$ versus cycle index: no secular drift.

Demography and local occupancy. Let $\nu = N/|G|$ be the mean number of mice per cell ($|G| = 900$). With the parameters of the short-lived control the logistic hazard gives a life expectancy of 44.8 weeks, with death almost deterministically “of old age”. A female that always finds a mate and keeps $s = 1$ produces $R_0 = \frac{1}{2} p_0 \bar{L} q_0 \tau_{\text{fert}} \approx 35$ daughters over her reproductive window (70 with the long-lived demography of the candidates), so reproduction is limited by *mate encounter*, not by fecundity: a reproductive female conceives only if a reproductive male shares her cell. Under random mixing the per-capita birth rate

$$b(N) = \phi_F p_0 \bar{L} q_0 (1 - e^{-\phi_M N/|G|}) \quad (10)$$

increases with N : the model has a built-in *Allee effect*, with a deterministic extinction threshold $N_A \approx 61$ for $s \equiv 1$. Here $\phi_M \approx \phi_F \approx 0.42$ are the fractions of reproductive males and females in the stationary age distribution; during growth the age structure is younger, so Eq. (10) is an approximation there. The latency of small founding groups and the failure of very small ones follow from (10). Social deterioration is driven by the local excess occupancy $(m_i - m_c)_+$. With independent positions the expected weekly loss is $\gamma D(\nu)$, $D(\nu) = \mathbb{E}[(1 + X - m_c)_+]$ with $X \sim \text{Poisson}(\nu)$. Being a Poisson tail expectation, D is extremely convex ($D = 0.0014$ at $\nu = 2$, 0.10 at $\nu^* = 4.13$, 1.04 at $\nu = 7$), so the social state behaves as a *switch*: in the core model ($r/\gamma = 0.1$) recovery dominates up to $N^* = |G|\nu^* \approx 3700$, and deterioration of order one per week sets in just above it. Pups are born in their mother’s cell, so the occupancy is clustered (index of dispersion 3–6 during growth for $\alpha = 0$, up to 29 at the peak for $\alpha = 1.4$). A negative-binomial occupancy with dispersion ξ captures this and lowers N^* to 2200 for $\xi = 2$; affinity-driven motion lowers ξ further, which is why the peak decreases monotonically with α .

Irreversibility: rules R1 and R2. In the core model, recovery is an individual process that only needs time: an adult with $s_0 \approx 0$ returns to s_v in $(s_v - s_0)/r$ weeks, within its fertile window. Under R1 an adult is a ratchet ($\dot{s} \leq 0$), and recovery becomes a *lineage* process: only pups gain, and producing a pup requires reproduction, whose fecundity scales as $s_f^2 s_m$. Table 11 gives the expected number of descendants that a lineage accumulates before it reaches viability. It is the expected value of a branching process without learning from neighbours outside the lineage, so it is a heuristic bound, not a prediction. With the long-lived demography of the candidates ($R_0 \approx 70$, $s_v = R_0^{-1/3} \approx 0.24$) the expected number is at most 10^{-2} for $s_0 \leq 0.05$, whatever the gain per generation and the partner. It is 0.01–0.07 for $s_0 = 0.1$ when the partner is equally deteriorated, and up to 0.7 when the partner is healthy, because fecundity then scales as s^2 in the first generation. A lineage of $s_0 \approx 0.2$ can recover. The bottleneck at small s_0 is the fecundity of deteriorated parents, not the gain per generation. In addition, every crowding hit on a healthy adult is permanent, so healthy pockets erode instead of regenerating. The transplants of Section 8.4 confirm the bottleneck (animals with $s \approx 0$ never found a colony; animals with $s \approx 0.36$ found one in 8% of the trials under R1 against 95% without it) but not the second part of the argument: the offspring of intermediate founders settle at their parents' level instead of collapsing. Under R2 recovery is $r[(1 - \lambda) + \lambda \bar{s}_V]$, with $\bar{s}_V$ the mean state of the neighbours. In the homogeneous limit $S = 0$ is absorbing for $\lambda = 1$, and for $\lambda < 1$

$$T_{\text{rec}}(\lambda) = \frac{1}{r\lambda} \ln \frac{(1 - \lambda) + \lambda s_v}{(1 - \lambda) + \lambda S_0}, \quad (11)$$

so the colony is irreversible when T_{rec} exceeds the fertile time left after crowding. Unlike R1, R2 depends on r , and in the spatial model a healthy cell is a focus from which s spreads, so (11) gives lower bounds for the threshold. A kinetic mean field with persistent cell disorder (Appendix B.5, Table 10, Fig. 6) quantifies both rules for $r \in \{0.01, 0.03\}$, the only rates it was run with, and 10 seeds per cell (10 extinctions out of 10 have a Wilson interval of $[0.72, 1]$): with the long-lived demography, R1 gave extinction in 10/10 seeds for every $a_w \leq 24$ weeks at both rates, so within this range its effect does not depend on r , and the agent-based plane (r, a_w) later extended this to $r \in [0.02, 0.3]$; R2 alone extinguishes the long-lived colony when λ exceeds a threshold that grows with r ; the threshold λ_c is defined throughout as the first value of the grid $\{0, 0.3, 0.5, 0.7, 0.9, 1\}$ at which at least half of the 10 seeds go extinct, so it is $\lambda_c \leq 0.3$ at $r = 0.01$ (8/10 seeds at $\lambda = 0.3$; 10/10 from 0.5 on), $\lambda_c \in (0.7, 0.9]$ at $r = 0.03$, and $\lambda_c \in (0.5, 0.7]$ at $r = 0.03$ with faster mixing; with the short life span R2 needs $\lambda \geq 0.9$ and fast mixing, and even then extinction is not certain (at most 8/10 seeds).

Thresholds of R1 and R2 in the kinetic mean field. In the kinetic mean field (Appendix B.5), calibrated to the quasi-stationary regime of the short-lived control, R1 alone with the short life span and persistent healthy pockets reduces the rebound but does not extinguish the colony ($P_{\text{ext}} \leq 0.3$); faster mixing makes it lethal for $a_w \leq 12$, with weak dependence on r . In the long-lived scenario ($a_{\text{max}} = 120$, fertility to 80 weeks) R1 gives $P_{\text{ext}} = 1$ and zero rebound for $a_w \leq 24$ at both $r = 0.01$ and 0.03 (10/10 seeds per cell, Wilson interval $[0.72, 1]$), with the critical window $a_w^c \in (24, 48)$ at $r = 0.03$ and above 48 weeks at $r = 0.01$ (same definition: first grid value with at least half of the seeds extinct). R2 requires $\lambda \gtrsim 0.9$ with the short life span, and in the long-lived scenario $\lambda_c \leq 0.3$ ($r = 0.01$; $P_{\text{ext}} = 0.8$ already at $\lambda = 0.3$) or $\lambda_c \in (0.7, 0.9]$ ($r = 0.03$), in agreement with Eq. (11); with faster mixing $\lambda_c \in (0.5, 0.7]$ at $r = 0.03$ (Fig. 6). For $\lambda = 1$, Eq. (11) gives $T_{\text{rec}} = \ln(s_v/S_0)/r = 274\text{--}413$ weeks for the measured $S_{\text{min}} \approx 0.01\text{--}0.02$ and $r = 0.01$.

Status of the exploratory mean-field statements.

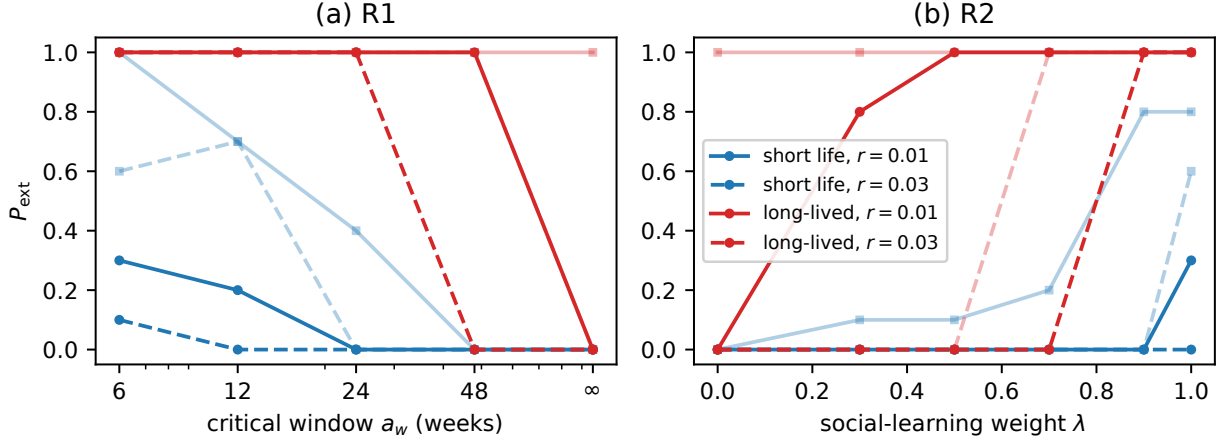


Figure 6: Extinction probability in the kinetic mean field (10 seeds, 1500 weeks) as a function of (a) the critical window a_w of R1 and (b) the social-learning weight λ of R2, for the short ($a_{\text{max}} = 60$) and long ($a_{\text{max}} = 120$) life spans and two recovery rates. Circles: persistent healthy pockets ($\xi_c = 0.5$, $\tau_F = 8$); faint squares: faster mixing ($\xi_c = 1$, $\tau_F = 5$).

- The kinetic mean field predicted that R1 with the long-lived demography gives $P_{\text{ext}} = 1$ and no rebound for every window $a_w \leq 24$ at the two rates it was run with ($r = 0.01$ and 0.03). The agent-based plane (r, a_w) extends this to $r \in [0.02, 0.3]$: $P_{\text{ext}} = 1$ at 48 of 48 points. Without the window, extinction occurs only for $r \leq 0.03$, and recurrent growth (A2) dominates for $r \geq 0.07$ (Appendix C.8).
- That R1 blocks the recovery of transplanted adults is confirmed. That R2 with $\lambda = 1$ alone blocks it is not: without R1, phase-D transplants recover in 0/60 trials at week 86 but in 10/60 at week 126, and animals of intermediate state found colonies in 50/60 trials (Appendix C.11), as Eq. (11) allows for $r = 0.1$ ($T_{\text{rec}} \approx 27$ weeks).
- The ageing decline, $(T_{\text{ext}} - t_{\text{pk}})/t_{\text{pk}} \approx 2$ with a small spread, is confirmed (2.07 with $t_{\text{pk}} = \arg \max N$), but the ratio depends on how the peak time is defined on the plateau (Appendix C.2).
- Not tested in the production sweeps: the thresholds in λ ($\lambda = 1$ in all candidates), the effect of long-range mixing (the jump rule was discarded) and critical slowing down (the candidates do not oscillate).

C.2 Production ensembles: details

Table 12 completes Table 5 with the desirable patterns, the timings of the boom and further magnitudes.

The plateau and the definition of the peak time. In the 200 runs of C1 (medians, with interquartile ranges in brackets), the population first comes within 2% of its maximum at week 34 [32, 36] and leaves that band at week 86 [85, 86], so the plateau lasts 51 [49, 53] weeks. The maximum itself, $t_{\text{pk}} = \arg \max N$, is reached at week 55 [50, 60], 21 [16, 25] weeks after the plateau begins, and coincides with the last surviving birth in 69% of the runs. The 99th percentile of the

births falls at $1.17 t'_{\text{pk}}$ [1.12, 1.24], and the decline ratio is $(T_{\text{ext}} - t'_{\text{pk}})/t'_{\text{pk}} = 3.97$ [3.72, 4.29] against 2.07 [1.85, 2.33] with t_{pk} . The primary evaluator uses t'_{pk} and t_{B99} ; Appendix C.6 gives every pattern with both definitions.

The escaped run. One C1' run in 200 was still alive at $T = 600$: it fails O7 and is censored in the survival analysis. No C1 run escaped in the reference ensemble; in the ensemble at $T = 700$ all 200 died.

C.3 Survival of colonies

Fig. 7 shows Kaplan–Meier estimates of colony survival; runs still alive at the horizon enter as right-censored observations.

- C1 and C0 die out within a narrow window. Their median extinction times are 169 [167, 171] and 166 [165, 167] weeks, and the Weibull shapes of the collapse are 14.7 and 11.9.
- C1' dies at 182 [180, 185] weeks, with a longer tail (Weibull shape 3.3 without a cure fraction, 4.4 with $\hat{\pi} = 0.005$, the single censored colony).
- C2 dies later, at 218 [217, 222] weeks, and with more spread (KM interquartile range over median 0.14). Four of its 200 colonies persist to $T = 700$ and recovered natality (A2).
- The short-lived control is a mixture: 47 of 100 colonies died by $T = 1000$, 44% [35, 54] in the first collapse, the others in a quasi-stationary oscillation with a tail hazard of 0.7×10^{-4} per week (Weibull shape 0.60 over all times; cure fraction $\hat{\pi} = 0.53$).
- In the long-lived control 98 of 100 colonies die, at 186 [181, 190] weeks; with the initial condition of C1 (null-model ensemble, $n = 200$), 199 of 200 die, with a KM spread of 0.160.

The survival curves of C1 and the short-lived control differ (log-rank $\chi^2 = 53$, $p = 3 \times 10^{-13}$). A likelihood-ratio test of the cure fraction is degenerate for C1 (no censored colony) and is not reported; because no C1 colony dies before about 140 weeks, a Weibull without a location parameter mixes location and shape, so the shape 14.7 should not be read as a hazard exponent. The Kaplan–Meier quantiles are the safer summary. Among the single ablations, removing R1 is the only one that changes survival qualitatively (27 of 120 colonies go extinct by week 600); removing AV delays extinction by about 18 weeks (median 185 against 167); removing R2 or R7 leaves the survival curve unchanged (170 and 167).

C.4 Ablations: factorial model and the minimal candidate

A Firth logistic model of the full 2^4 factorial (exploratory, Benjamini–Hochberg at 10%) gives, on the logit scale: positive main effects for R7 and R2 (4.9 ± 1.2 each), none for R1 (-0.5 ± 1.1 , $p = 0.63$) or AV (-0.5 ± 1.1 , $p = 0.68$), positive interactions R1:R7 and R1:R2 (6.6 ± 1.9) and weaker ones R2:AV and AV:R7 (3.9 ± 1.8 , $p = 0.036$). R1 acts only together with R2 or R7, because without them O17p is zero whether R1 is present or not. Unlike in C0 (Section 5.2), stronger attraction does not rescue the ablation of R1: with $\alpha = 6$, O17p is 0.04. Table 13 compares C1' with C1 in the reference ensemble and in the factorial, and gives its single ablations (the double ablations of C1 that remove AV, $n = 40$).

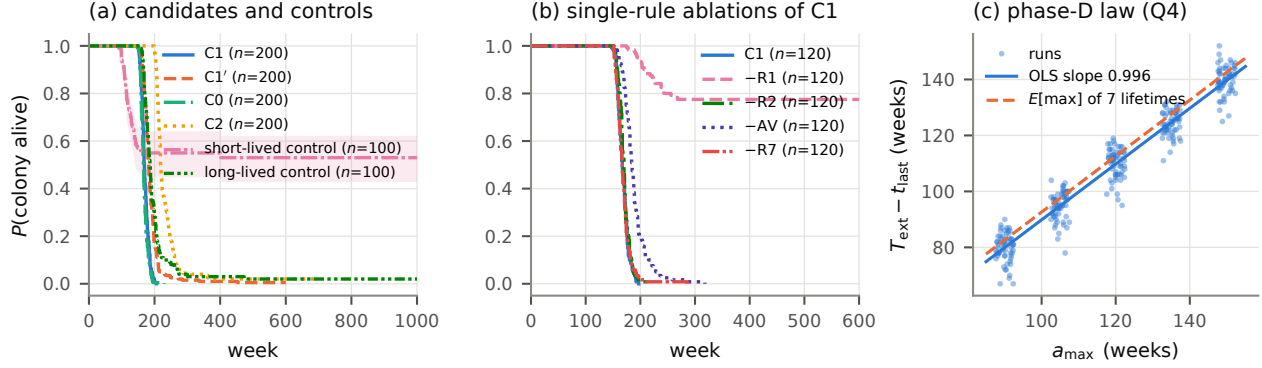


Figure 7: Survival of colonies. (a) Kaplan–Meier curves of the candidates and of the short-lived and long-lived controls (reference ensemble for C1, C1' and C0). Shaded bands are log–log Greenwood intervals for C1 and the short-lived control. Censoring is at $T = 600$ (C1, C1', C0), $T = 700$ (C2) and $T = 1000$ (short-lived and long-lived controls). (b) Single-rule ablations of C1 ($n = 120$, $T = 600$). (c) Phase-D law (Q4): time from the last birth to extinction against a_{max} ($n = 300$, jittered). The solid line is the OLS fit. The dashed line is the expected maximum of n_{last} logistic lifetimes, with $n_{\text{last}} = 7$, the median number of births in the last ten weeks.

C.5 Confirmatory tests: further details and the preregistered evaluator

Table 14 gives the confirmatory family with the primary evaluator and with the preregistered evaluator; the latter replicates the preregistered analysis on the production runs.

- *Q1*. The transplanted C1 animals are 37–64 weeks old at $1.75 t'_{\text{pk}}$ (week 60), 59–80 at $1.75 t_{\text{pk}}$ (week 97) and 65–80 at week 95; the fertile window closes at 80 weeks. In arm X the founding rates are 51/60, 12/60 and 5/60 (McNemar $p = 4 \times 10^{-16}$, 2×10^{-4} and 0.03).
- *Q4*. The law explains why O8 depends on longevity: O8 passes in 0.15, 0.30, 0.67, 0.83 and 0.92 of the runs for $a_{\text{max}} = 90, 105, 120, 135$ and 150, while every essential pattern is unchanged (O17p 0.93–1.00).
- *Q5*. O13 fails only for the strongest aversion with the strongest attraction ($\kappa = 0.3$, $\alpha = 8$; $n^* \approx 3.2$), where the crowded fraction falls to the 10% threshold. The one-sided test of $b_2 < 0$ gives $p = 1$ (Pearson $\phi = 1.0$).
- *Q6*. Profile-likelihood intervals: $[-0.15, -0.10]$ binomial and $[-0.20, -0.05]$ quasi-binomial ($\phi = 3.1$) with the primary evaluator; $[0.20, 0.40]$ and $[0.15, 0.50]$ ($\phi = 1.8$) with the preregistered evaluator. The grid of the profile is $\theta \in [-1, 3]$ in steps of 0.05. The quasi-binomial p is the one that enters the Holm sequence. The additive model in standardised $\log r$ and $\log a_w$ with quadratic terms beats the single-index model by $\Delta\text{QAIC} = 27.8$ over the 48 points and by 27.7 without the point that fails through the persistent class (dispersion of the additive model 1.9). With the preregistered evaluator the comparison goes the other way: the single-index model is preferred by $\Delta\text{QAIC} = 11$, so the change of criterion changes the adequate model as well as the sign of $\hat{\theta}$. With the primary evaluator the O17p boundary of the plane is nearly vertical in r (at least 0.67 for $r \leq 0.10$ and at most 0.60 for $r \geq 0.2$ at every window), which is why the prediction is reported as “not supported” rather than as a reversal.
- *Q7*. The logistic fit of the preregistered plane places the extinction boundary at $\gamma = 0$ at $w + 0.24 \Lambda = 0.215$, that is $\hat{\sigma} = 4.65$ and $\hat{\varphi} = 1.11$ in the form $\sigma w + \varphi \Lambda = 1$ (Fig. 12b). The

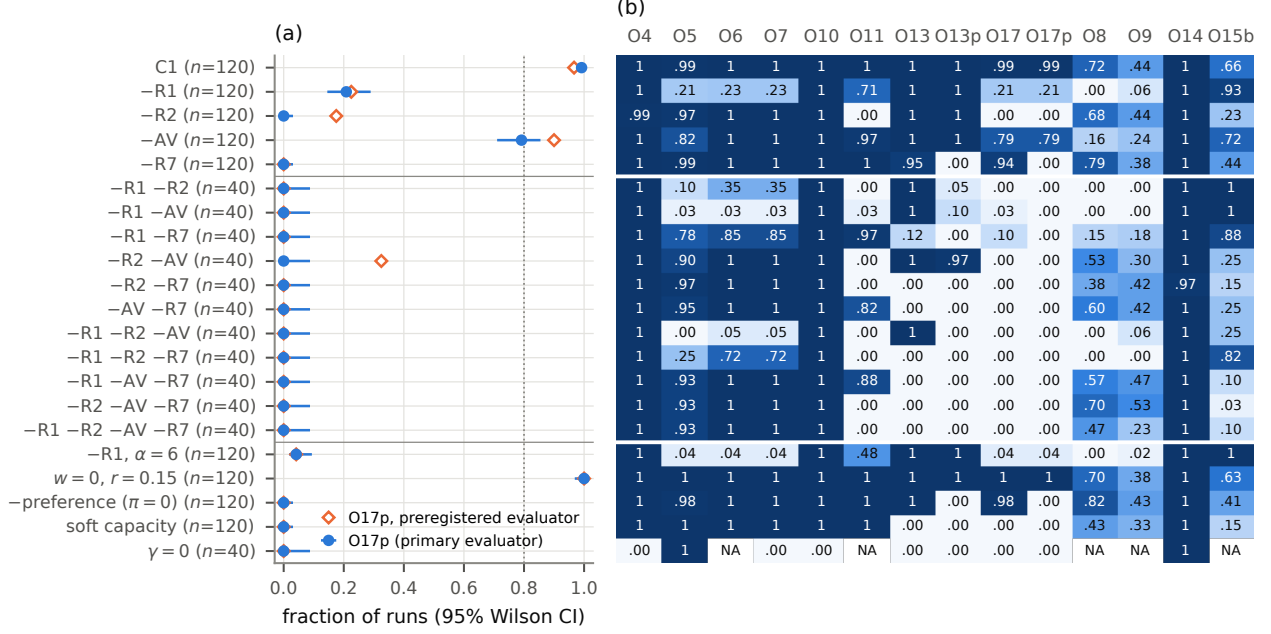


Figure 8: Ablations of C1. (a) Primary outcome O17p with the primary evaluator (filled, 95% Wilson interval) and with the preregistered evaluator (open diamonds). Rows are C1 and its single ablations ($n = 120$), double to quadruple ablations ($n = 40$), and implementation variants. The dotted line marks the 0.8 threshold for an essential pattern. (b) Fraction of passing runs for each pattern (primary evaluator). NA marks a pattern that is not applicable in any run; with $\gamma = 0$ the colonies never go extinct.

grid brackets the boundary coarsely: at $w = 0$ colonies go extinct for $r \leq 0.05$ (0.9 at $r = 0.07$) and survive for $r \geq 0.10$; at $w = 0.1$ the change happens between $r = 0.03$ and 0.05 ; at $w \geq 0.2$ no colony goes extinct. In the subcritical corner the median peak is 6 500–11 500 individuals at weeks 77–96 (ρ_{pk} between 0.85 and 1.66 over the 120 runs), and O4 fails in every run.

Q7 with the window varied. Fig. 9 shows P_{ext} at $\gamma = 0$ against $\Lambda = r a_w$ for $a_w \in \{6, 12, 24\}$ and $w \in \{0, 0.05, 0.1\}$: six values of Λ per window (0.24, 0.36, 0.48, 0.6, 0.84, 1.2; consecutive ratios 1.25–1.5), 54 points, 20 runs each. The analysis fixed before the production runs is a logistic model in $z = \log r + \theta \log a_w$ and w , with a profile-likelihood interval for θ , a likelihood ratio for $\theta = 1$ and an equivalence test of the two exponents (TOST, margin $[2/3, 3/2]$). The data do not support that analysis: 51 of the 54 points have $P_{ext} = 0$ or 1 (the other three are 19/20, 14/20 and 5/20), the Pearson dispersion is 0.05 and the maximum likelihood coefficients diverge, that is, the model is separated. In a separated logistic model the profile $\ell(\theta)$ is flat wherever θ keeps the ordering of the points in z that separates zeros from ones and drops where it breaks it, so the profile interval ($\hat{\theta} = 0.725$ [0.70, 0.75], likelihood ratio 219 for $\theta = 1$) reflects the geometry of the grid rather than sampling error; the fitted transition widths are 10^{-8} – 10^{-19} in $\log \Lambda$, below the grid spacing at every a_w , and the parametric bootstrap of the fitted boundary ($\hat{\sigma} = 4.3$, $\hat{\varphi} = 1.17$) resamples essentially the same data. None of these quantities is reported as inference; they are kept, labelled, in `results_numbers.json`. Table 15 gives instead a non-parametric summary per window: the last grid point with $P_{ext} \geq 1/2$ and the first with $P_{ext} < 1/2$, which bracket $\Lambda_{1/2}$ to a factor 1.25–1.5; a log-linear interpolation between them; and, between consecutive windows, the exponent

$\theta_{\text{loc}} = 1 - \log[\Lambda_{1/2}(a_2)/\Lambda_{1/2}(a_1)]/\log(a_2/a_1)$ implied by the brackets (its grid bounds) and by the interpolated values. At every w the brackets of $a_w = 12$ and 24 coincide and that of $a_w = 6$ is one grid step below, so $\theta_{\text{loc}} \in (0, 1)$ between 6 and 12 weeks (interpolated 0.51, 0.60 and 0.70 for $w = 0, 0.05$ and 0.1) while between 12 and 24 weeks the grid only bounds it within about $(0.5, 1.5)$ (interpolated 0.99, 1.00 and 0.88). The equivalence margin $[2/3, 3/2]$ is thus resolved in neither interval, and the one firm statement is qualitative: at a given product ra_w a short window with fast learning is less effective than a long window with slow learning, and a single exponent does not describe the three windows. Resolving θ would need a grid of ratio about 1.05 around each $\Lambda_{1/2}$ with more runs per point; it was not run.

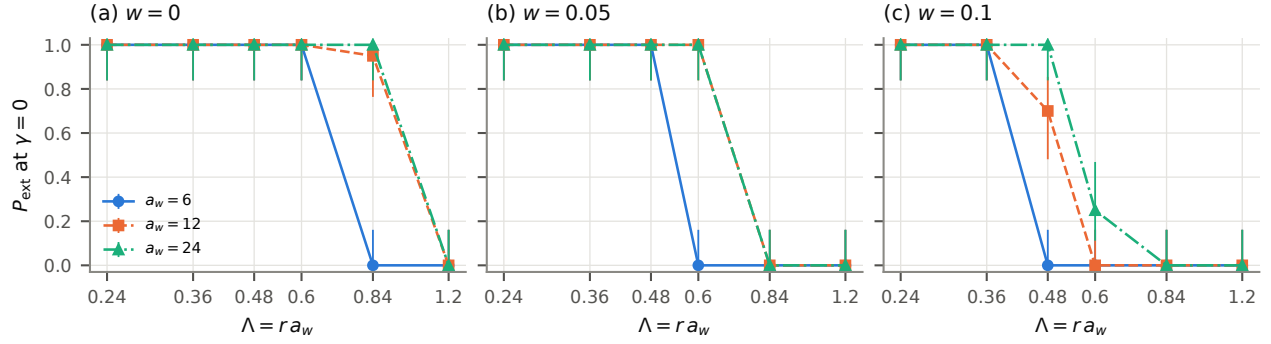


Figure 9: Extinction probability without crowding damage ($\gamma = 0$) against $\Lambda = ra_w$ for three plasticity windows, at three inheritance weights (95% Wilson intervals, 20 runs per point).

C.6 Re-scoring with the preregistered evaluator and with each option changed separately

Every run of the production program was scored with the primary evaluator, with the full preregistered evaluator and with each option changed separately: peak time at $\arg \max N$ or at the end of the plateau instead of its start; crowded class at $m > m_c$ instead of $m > 8$; O5 on the last sustained birth instead of t_{B99} ; the preregistered O4 (no free-space condition). Table 16 gives O17p for the main arms, and Table 18 the discriminating power of the redefined criteria with the preregistered evaluator.

- The peak-time definition matters only through O5. With t'_{pk} and the last sustained birth, O5 fails in half of the C1 runs (O17p 0.48): the thin tail of the last births lies more than $0.5 t'_{\text{pk}}$ after the start of the plateau. Taking the end of effective natality at 99% of the births removes this artefact (0.995). With $\arg \max N$ the result is insensitive to the birth reference (1.00). The choice of t_{B99} was made after exploratory runs with the primary evaluator (smoke tests on the data of the preregistered design and on two replicates per point) had shown this failure of the alternative, so it is informed by the result, and it is the one option on which the rejection of Q2 depends (Table 27).
- The fixed crowded-class threshold ($m > 8$) changes nothing at the C1 values, where $m_c = 8$; it lowers C2 from 0.78 to 0.70 together with the primary O4.
- The primary O4 (free space) fails only without attraction and in parts of the (α, κ) plane and of the hypercubes.

- The primary O10 and O11 remove the vacuous passes of the preregistered criteria: the preregistered O10 passed in every null model, and the preregistered O11 in 1.00 of the $\gamma = 0$ runs and in 0.09–0.21 of the other null models. The primary O11 also turns the preregistered value of $-R2$ (0.17) into 0 and that of $s_{\text{base}} = 0.25$ (0.64) into 0.08.
- For the demographic variants the drop in O17p is the same with both evaluators ($\mu_{\text{bg}} = 0.003$: 0.55 against 0.45); with $\arg \max N$ as peak time it is masked (0.88), because t_{pk} then moves to the end of natality.

Threshold of the deteriorated classes. O13 requires both the crowded and the isolated deteriorated classes to hold at least $\theta = 10\%$ of the adults, and O13p the same of the persistent isolated class. The stored class fractions at t'_{pk} , $t'_{\text{pk}} + 10$ and $t'_{\text{pk}} + 20$ allow both to be re-scored for other θ without new simulations; the other patterns of O17p are left unchanged. Fig. 10 gives O17p against θ for the candidates, two variants and the single ablations, on the grid $\theta \in \{0.05, \dots, 0.20\}$ against the ensemble thresholds $\tau \in \{0.5, \dots, 0.9\}$. C1 holds 0.99 up to $\theta = 0.125$ and 0.96 at 0.15, and collapses at 0.175; C1' holds 0.83 up to 0.125 and falls to 0.33 at 0.15 (0.31 in the $-AV$ arm of the factorial, $n = 120$); C2 falls from 0.70 to 0.01 at 0.125; the asynchronous variant and the nest variant of C1 hold up to 0.125 and 0.15. R1, R2 and R7 are necessary at every (θ, τ) of the grid; AV is necessary for $\tau \geq 0.8$ when $\theta \leq 0.125$ (0.79 against 0.8, with a Wilson interval $[0.71, 0.85]$ that contains the threshold) and for every τ when $\theta \geq 0.15$. In the hypercubes, the Calhoun-like fraction is 5.5%, 2.3% and 0.5% at $\theta = 0.05, 0.10$ and 0.15 . The conclusions about refuges are therefore conditional on the 10% threshold, through the capacity bound $\Phi_{\text{BO}}^{\text{pers}} \leq \phi_R$.

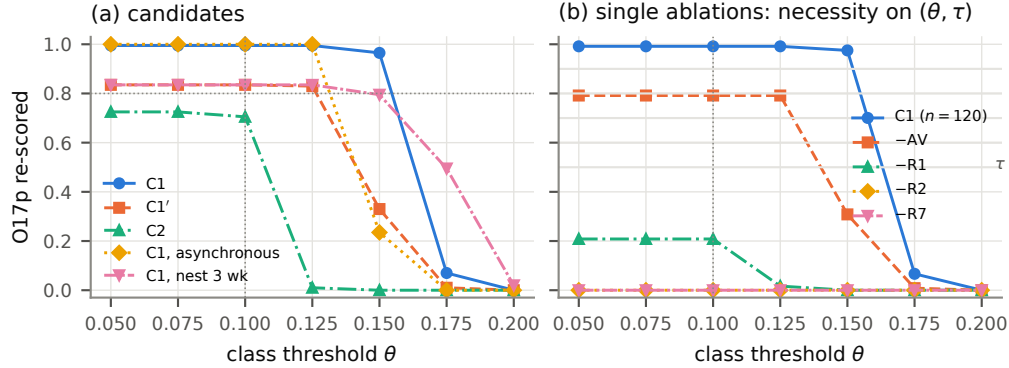


Figure 10: O17p re-scored with the class threshold θ (the production value is 0.10, dotted). (a) Candidates and two variants of C1. (b) C1 and its single ablations ($n = 120$); the horizontal lines are the ensemble thresholds $\tau = 0.5$ –0.9 of the necessity grid.

C.7 Robustness variants and system size: tables

Table 19 lists the robustness variants of Section 8.5 and Table 20 the lattice sizes at fixed density. In every variant that loses the primary outcome the loss comes from a single criterion: O5 for the background mortality and the non-synchronous initial conditions (and, marginally, for C1'), O11 for the nest period and for pups born above the deterioration threshold. O5 is lost through its timing clause $(t_{B99} - t'_{\text{pk}})/t'_{\text{pk}} \leq 0.5$, not through its level clause $N(t_{B99}) \geq 0.7 N_{\text{pk}}$: at $\mu_{\text{bg}} = 0.003$ the O5 verdicts of C1 are 110 PASS, 83 MARGINAL and 7 FAIL (median t'_{pk} 29 weeks against 34 in the reference ensemble, median t_{B99} 43 against 40); at 0.008, 12, 118 and 70 (t'_{pk} 26, t_{B99} 50); with

ages $U\{4..80\}$, 126, 73 and 1 (t'_{pk} 29, t_{B99} 41); with $U\{4..52\}$, 186, 14 and 0. The column “disc.” of Table 19 shows that the criteria that discriminate against the null models (with O11 scored at six weeks) are untouched by the demographic variants (1.00 in every C1 arm with background mortality or spread ages; 0.96–0.99 in C1') and are lost only where O11 is lost (nest period, $s_{base} > 0$). Along the background-mortality curve the primary outcome is lost gradually: at $\mu_{bg} = 0.001$ per week, a tenth of the senescent hazard, O17p is still 0.83. In the lattice-size stage the failures at $L = 15$ (11 of 60) and $L = 20$ (6 of 60) are O5 marginal in 5 runs at each size, O11 marginal in 1 at each, and at $L = 15$ the anti-pattern A2 (recovery of natality) in 6 runs; at $L = 30$ one run fails O5 and at $L \geq 45$ none fails.

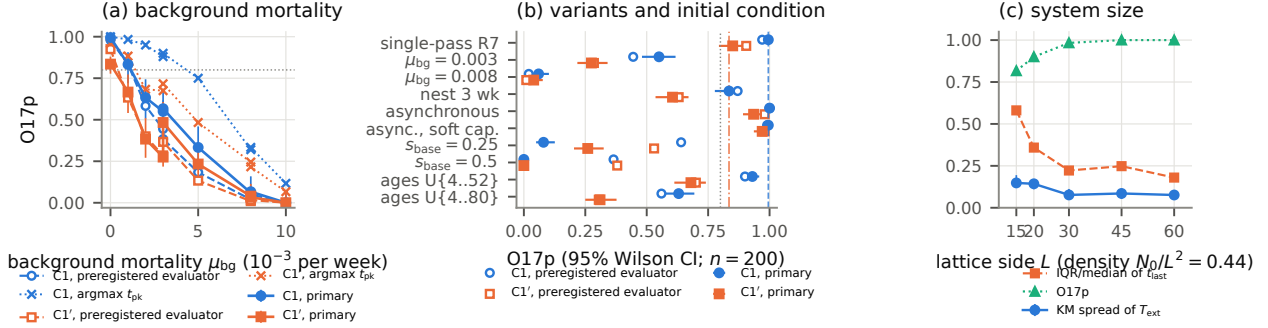


Figure 11: Robustness of the primary outcome ($n = 200$ per arm unless stated). (a) O17p against the background mortality μ_{bg} for C1 and C1' ($n = 60$ at the intermediate values), with the primary evaluator (filled), the preregistered evaluator (open) and with $\arg\max N$ as peak time (crosses). (b) Variants of C1 and C1': single-pass refuge admission, background mortality, nest period, asynchronous update with hard and soft capacity, pups born at $s_{base} = 0.25$ and 0.5 , and initial ages drawn from $U\{4..52\}$ and $U\{4..80\}$. Vertical lines: the reference ensembles and the 0.8 threshold. (c) Lattice side at fixed density ($n = 60$): KM spread of the extinction time, relative spread of the last birth and O17p.

C.8 Phase planes

All planes use 30 runs per point; the 40 points nearest to a Calhoun-like boundary (9 of P1, 12 of P1- a_w and 19 of P2) were re-run with 100 independent runs.

Socialisation: inheritance w and learning rate r (P1). With crowding damage ($\gamma = 0.1$, Fig. 12a), every one of the 56 points goes extinct ($P_{ext} = 1$), so the plane decides only *which* collapse occurs. O17p is 0.83–1.00 for $w \leq 0.3$ and $r \leq 0.10$. It falls in two directions, both through the primary O11: with strong inheritance ($w \geq 0.5$: 0.33 at $r = 0.02$, 0 at $w \geq 0.75$) the late cohorts inherit the competence of their mothers, and with fast learning ($r \geq 0.2$: 0.57 at $w = 0.2$, $r = 0.2$; 0 at $r = 0.3$ for $w \geq 0.2$) they acquire it from the few competent adults. These are point estimates with 30 runs per cell (Wilson half-width up to ± 0.17); the boundaries of the maps have no intervals of their own. Without crowding damage ($\gamma = 0$, panel b), colonies either explode or, in the subcritical corner, collapse endogenously after overshooting the capacity (Q7).

Plasticity window a_w and learning rate r (P1- a_w). For every finite window up to $a_w = 24$ weeks and every r from 0.02 to 0.3, extinction is certain (48 of 48 points at $P_{ext} = 1$). O17p degrades at the fastest learning rates: at $r = 0.2$ it is 0.10–0.13 for $a_w \leq 8$ and 0.40–0.60 for $a_w \geq 12$, and

at $r = 0.3$ it is at most 0.07; in every case O11 is the criterion that fails, more often the shorter the window. The loss of the persistent class with long windows and fast learning, which set the boundary with the preregistered evaluator (O13p 0.17 at $a_w = 24$, $r = 0.3$), is hidden behind O11 with the primary evaluator. Without a window ($a_w = \infty$, top row of Fig. 12c), recovery is again individual and reversible: P_{ext} falls from 1.00 at $r = 0.02$ to 0.47 at $r = 0.07$ and to 0 for $r \geq 0.2$, and the anti-pattern A2 (recovery of natality) rises from 0 to 1.

Attraction α and aversion κ (P2). O17p is at least 0.8 at 34 of the 48 points with $\alpha \geq 2$ and at 35 of the 56 points of the plane (Fig. 13a). It fails in three places:

- where the free space of the primary O4 is lost: at $\alpha \leq 2$ with $\kappa \geq 0.1$, at $\alpha = 3$ with $\kappa = 0.2$ and at $\kappa = 0.3$ for $\alpha \leq 5$, the empty fraction at the peak is below 0.10 (panel c);
- at $\kappa = 0.3$ with $\alpha \geq 5$, where in addition the crowded fraction falls to the 10% threshold of O13 (0.33–0.73);
- without aversion and with strong attraction ($\kappa = 0$, $\alpha \geq 5$; 0.67, 0.63 and 0.33), where O5 and O11 fail in a third of the runs; with 100 runs the point $(\alpha, \kappa) = (8, 0)$ gives 0.53 [0.43, 0.63].

The empty fraction at the peak increases with α and decreases with κ (Spearman +0.56 and -0.77 , exploratory, BH-significant; panel c). The peak density does the opposite. The coexistence band predicted for the model without refugees is not observed in C1 (Q5). Varying the aversion exponent at $\alpha = 4$ ($p = 2$ or 8) leaves O17p at or above 0.8 except at $\kappa = 0.3$ (0 and 0.30) and at $p = 2$, $\kappa = 0.2$ (0.47).

Refuges: fraction f_R , preference π and capacity c_R (P5). A persistent isolated class requires both a strong preference and a large enough total capacity: with $c_R = 2$, $\pi \geq 30$ and $f_{RCR} \geq 0.6$, or $\pi \geq 100$ and $f_{RCR} \geq 0.4$; with $c_R = 1$, $\pi \geq 100$ and $f_{RCR} \geq 0.4$, or $\pi = 1000$ and $f_{RCR} \geq 0.3$ (9 of 50 points with O13p ≥ 0.8 ; Fig. 13d,e). The bound $\Phi_{\text{BO}}^{\text{pers}} \leq \phi_R = f_{RCR}|G|/N_{\text{pk}}$ of Section 4.4 holds at all 50 points and is approached for $\pi \geq 100$ (panel f). C1 lies at $\phi_R = 0.21$, with $\Phi_{\text{BO}}^{\text{pers}} = 0.17$. Large capacity combined with the strongest preference degrades the senescent decline: O8 is 0.00–0.13 at $\pi = 1000$ with $f_R \geq 0.2$, $c_R = 2$, and O17p falls to 0.47 at $f_R = 0.4$ through O5.

Refined boundary points. Over the 40 refined points the 100-run estimate differs from the 30-run estimate by 0.07 on average (mean shift +0.03, maximum 0.30, correlation 0.93), the size expected from binomial sampling with 30 runs; 26 of the 40 30-run values lie within the 100-run Wilson interval. Five points crossed the 0.8 threshold, four of them upwards. The points were selected because their 30-run estimate lay near the boundary, so part of these shifts is regression to the mean. No conclusion of the planes changes.

C.9 Global sensitivity

Fig. 15 reports the elementary effects of the Morris screening, over 15 factors and ranges around C1, and the Sobol' indices of the 11 factors of the production design, each with its bootstrap interval and the empirical noise floor of Section 7. Table 21 and Fig. 14 give the estimators of the Calhoun-like region of the hypercubes.

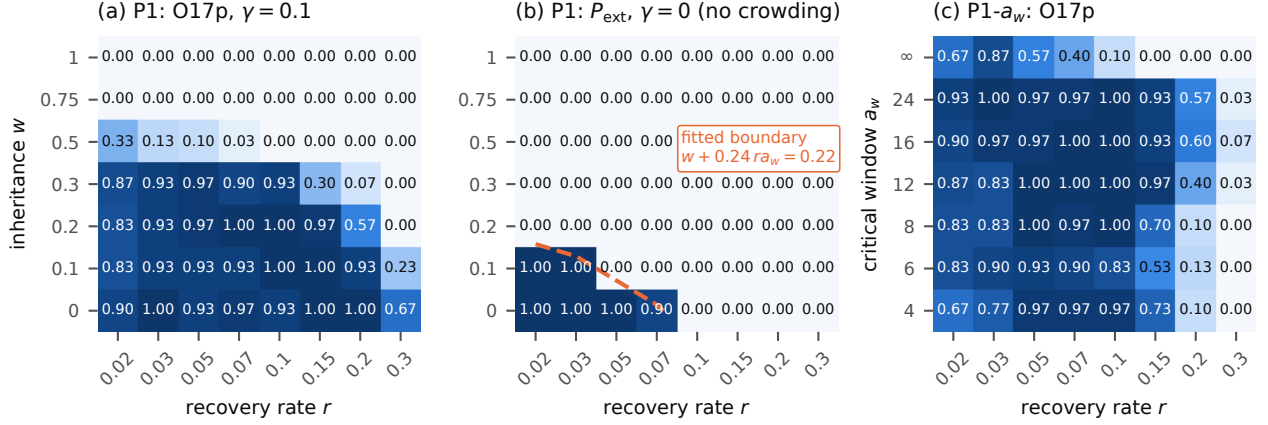


Figure 12: Socialisation planes around C1 (30 runs per cell, primary evaluator). (a) O17p in the plane of learning rate r and inheritance w , with crowding damage $\gamma = 0.1$; every cell goes extinct. (b) Extinction probability in the same plane without crowding damage ($\gamma = 0$); the other colonies explode. The dashed line is the fitted boundary $w + \hat{\varphi}ra_w = c$ of the generational map (Q7). (c) O17p in the plane of r and the critical window a_w . The top row ($a_w = \infty$) is C1 without R1.

Noise floor. The split-half contrast $e = (\bar{Y}_{\text{even}} - \bar{Y}_{\text{odd}})/2$ of the replicate means at each design point has expectation zero and the variance of the replicate noise, so the Morris and Sobol' estimators applied to it give the floor below which an index is indistinguishable from noise. A factor is distinguishable in the screening if its μ^* exceeds that of e by a factor of two, and relevant for a Sobol' index if $S_T - S_T^{\text{noise}} > 0.05$ and the lower confidence bound exceeds the floor; without the floor, synthetic data with three replicates and null factors give $S_T \approx 0.3$.

Screening. For O17p the largest elementary effects are those of the refuge capacity c_R ($\mu^* = 0.18$), the aversion κ (0.16), the community size m_c (0.15), the conception probability p_0 (0.09) and the attraction α (0.08); the preference π and the inheritance w follow (0.075). With the primary evaluator O17p is zero at most Morris points, so the elementary effects are small and their bootstrap intervals are as large as the effects. For every factor σ is about twice μ^* ; because each output is an average of 10 binary runs, σ includes replicate noise, and without a noise floor for σ this does not show that the factors act through interactions. The automatic selection averages the relative μ^* over seven metrics and retained m_c , κ , α , c_R , γ , w , η and r , together with the aversion exponent p and p_0 ; w , r and a_w were added by design, so the Sobol' design has 11 factors and holds four at their C1 values: f_R , π , a_{max} and δ . The indices below are conditional on these four.

Variance decomposition. The Saltelli design ($N = 256, 3328$ points, 11 factors) was run with 3 runs per point and then extended to 6 (19968 runs); the values below use the 6 runs. Doubling the runs halved the noise floors (median floor of S_T for O17p from 0.12 to 0.07) and changed the total indices of the continuous outputs by at most 0.01 (ρ_{pk} , f_0 , $\Phi_{\text{BO}}^{\text{pers}}$) and those of O13 and O13p by at most 0.05; for O17p the largest change is 0.10 (m_c from 0.39 to 0.31, γ from 0.26 to 0.16), and the number of factors above the floor rises from 7 to 10. With the primary evaluator O17p is zero at 78% of the points and at least 0.8 at 9% (mean 0.135): the Calhoun-like region is a minority of the box even with four parameters held at their C1 values. For O17p the total indices are c_R 0.49 ± 0.16 , w 0.43 ± 0.15 , κ 0.41 ± 0.14 , r 0.32 ± 0.12 and m_c 0.31 ± 0.12 , followed by η , γ , p , p_0 and α (0.12–0.17, against floors of 0.05–0.09); a_w (0.11) does not exceed its floor. The intervals of the five

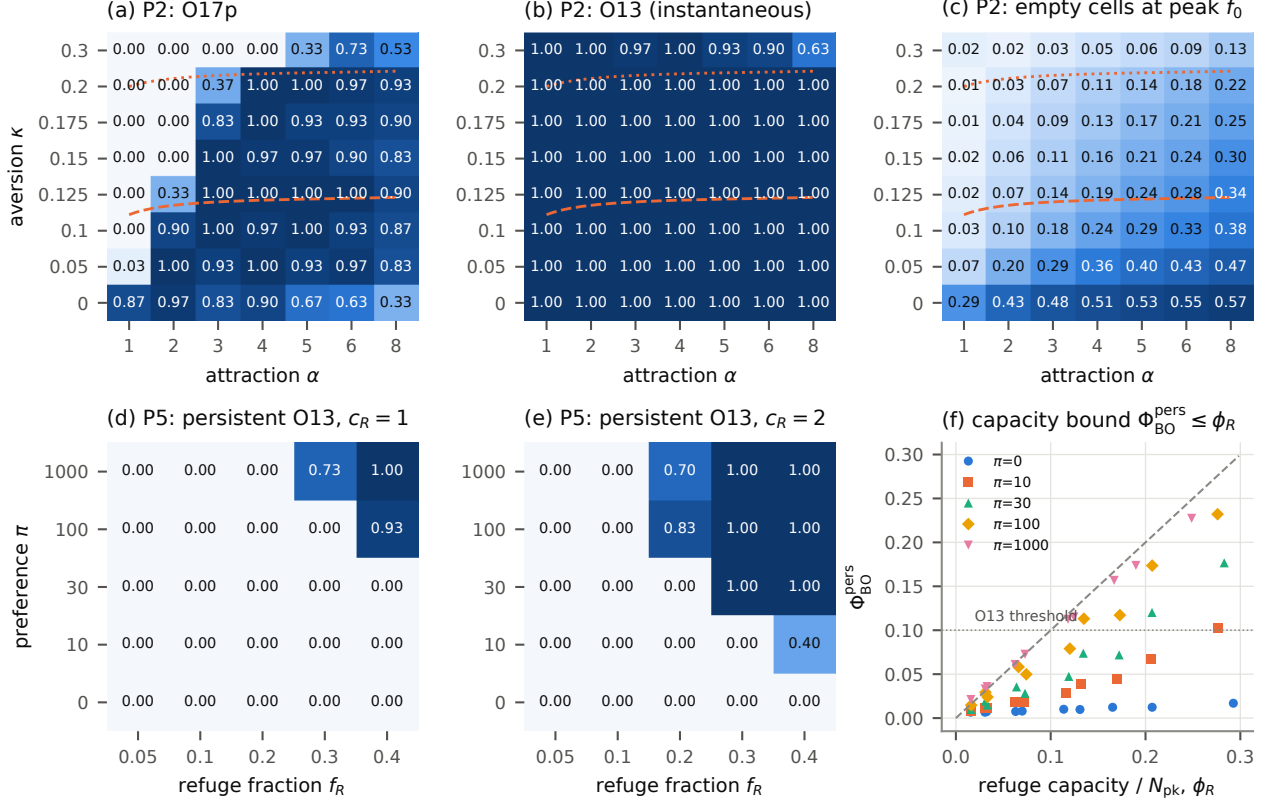


Figure 13: Spatial planes around C1 (30 runs per cell, primary evaluator). (a)–(c) Plane of attraction α and aversion κ (exponent $p = 4$): O17p, the instantaneous coexistence criterion O13, and the empty fraction at the peak. The dashed and dotted lines mark $n^* = m_c$ and $n^* = 4$, with $n^* = 1/\kappa - 1/\alpha$. (d, e) Persistent O13 in the refuge plane of fraction f_R and preference π , for capacities $c_R = 1$ and 2. (f) Persistent isolated fraction against the refuge capacity per individual at the peak, ϕ_R , for the 50 points of (d, e). The dashed line is the bound $\Phi_{BO}^{pers} = \phi_R$.

leading factors overlap, so they cannot be ordered. The first-order indices add up to 0.15 and the total indices to 2.8: O17p is almost entirely interactions, and 31% of its run-level variance is within points. In the Sobol' design of the preregistered plan, which excluded w , r and a_w , m_c , f_R and c_R led; with the 11 factors, the inheritance w and the aversion κ rank with the refuge capacity, in agreement with the hypercube regressions; with the free-space condition of O4, κ enters through O4 ($S_T = 0.55$), and w through O11 ($S_T = 0.75$, $S_1 = 0.49$). For the persistent class O13p the leading factors are c_R (0.62), m_c (0.48) and κ (0.30), and w is below the floor. Two of these are partly definitional: with $c_R = 3$ refuge residents exceed the isolation threshold of O13, so O13p averages 0.56 at $c_R = 2$ against 0.11 at $c_R = 1$ and 0 at $c_R = 3$; and m_c also defines the damage threshold, so the mean of O13 falls from 0.68 for $m_c \in (6, 9]$ to 0.39 for $m_c > 12$. The continuous outputs are almost additive and dominated by the spatial and learning parameters: for the peak density (sum of first-order indices 1.01) r 0.21, κ 0.19, η 0.15, γ 0.13, a_w 0.10 and α 0.10, every factor above a floor of 0.002; for the empty fraction (0.76) κ 0.44, m_c 0.28, α 0.16 and η 0.12; for Φ_{BO}^{pers} (0.89) c_R 0.64 and m_c 0.37, with no other factor above the floor. The indices of O7 and P_{ext} are degenerate (98% of the points share the same value; total indices above one with intervals wider than one) and are not reported: over the box, extinction is generic. For the median extinction time m_c (0.66), κ (0.32) and p_0 (0.30) exceed floors of about 0.1; with 3 runs per point only the first two did.

Latin hypercubes. A Firth logistic regression of O17p on the standardised factors shows lack of fit (Pearson dispersion 2–3), so standard errors are inflated by its square root. With this quasi-binomial correction and a false discovery rate of 10%, the factors significant in the first hypercube are κ (−1.4), f_R (+1.4), w (−1.4), r (−1.2), γ (+1.1), π (+1.0), $a_{\max}$ (+0.8), α (+0.7), m_c (−0.6) and a_w (+0.5); in the second, κ (−1.7), f_R (+1.1), m_c (−1.0), π (+0.9), p (+0.8), w (−0.7), $a_{\max}$ (+0.7), η (+0.6), p_0 (+0.5) and c_R (−0.4). The effects of κ , f_R , π , w , m_c and $a_{\max}$ replicate; those of r , γ , α and a_w appear only in the first hypercube and those of p , η , p_0 and c_R only in the second (the effect of c_R is non-monotone and a linear term does not describe it). With the preregistered evaluator the leading factors were m_c , f_R , π , w and κ ; the primary O4 and O5 add the aversion and the learning parameters.

Estimators of the Calhoun-like region. Table 21 gives, for the primary evaluator, the preregistered evaluator and the fixed class threshold alone, the naive fraction of points with $k/n \geq 0.8$, the mean of O17p over the box and the empirical-Bayes fraction, without and with the 40 extra runs at the 18 boundary points. The extra runs halve the number of uncertain points (posterior probability between 0.05 and 0.95) from 12 to 8 and leave the estimates within their intervals. Taking each option separately, the naive fraction is 2.0% with the primary evaluator, 3.8% with $\arg \max N$ as peak time, 5.0% with the end of the plateau, 4.2% with the preregistered O4, 2.5% with $m > m_c$ and 0% with O5 on the last sustained birth; the preregistered evaluator gives 7.2%.

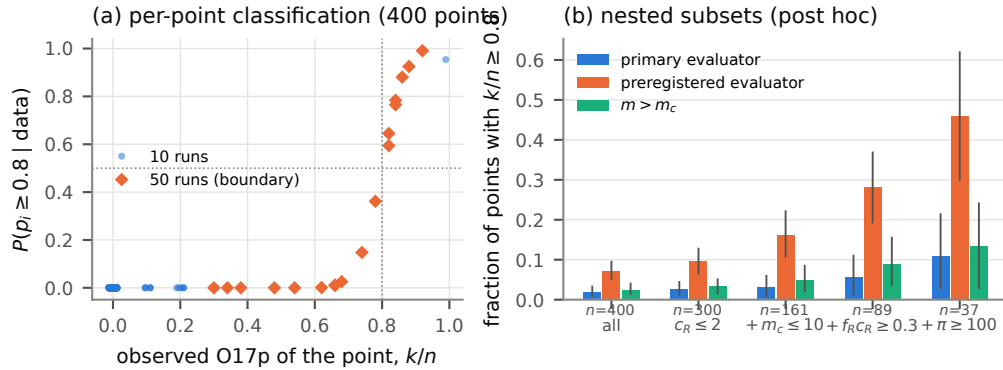


Figure 14: The Calhoun-like region of the hypercubes with per-point uncertainty (primary evaluator). (a) Posterior probability that a point has $p \geq 0.8$ against its observed fraction k/n ; the 18 boundary points with 50 runs are highlighted. (b) Fraction of points with $k/n \geq 0.8$ in nested subsets chosen after the runs ($c_R \leq 2$, then $m_c \leq 10$, $f_{RCR} \geq 0.3$ and $\pi \geq 100$), with bootstrap intervals, for the primary evaluator, the preregistered evaluator and the fixed class threshold alone.

C.10 Seeding density, lattice size and founders

Seeding density and lattice size (P4). The plane combines $L = 15, 20$ and 30 with $N_0 = 20$ – 800 animals of age 4 weeks placed at random, plus four founding pairs of age 7 weeks in one cell, with 30 runs per point and the other parameters of C1 (Fig. 16). The peak density is monotone in N_0/L^2 (Spearman 0.998) and rises from 0.07 to 0.94; points with the same density have nearly the same peak density whatever L , which supports, without testing, that the lattice size adds nothing at fixed density (the fixed-density stage of Section 8.5 tests it directly). O17p reaches 0.8 for $0.22 \leq N_0/L^2 \leq 0.89$. At low densities the persistent class and the failure of the late cohorts are lost and some colonies survive; at the highest density the peak approaches the capacity and O4

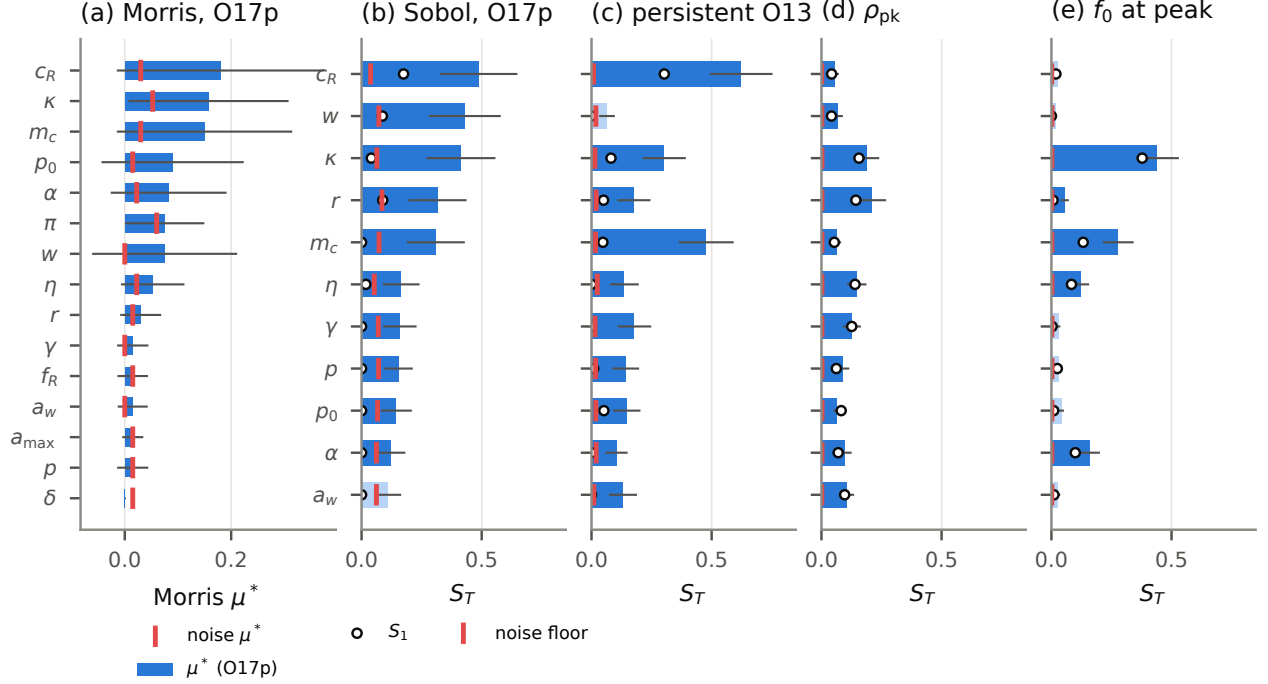


Figure 15: Global sensitivity around C1 (primary evaluator). (a) Morris μ^* of O17p with bootstrap intervals (15 factors, $r = 20$ trajectories, 10 runs per point with common random numbers). Red ticks: μ^* of the split-half noise contrast. (b)–(e) Sobol' total indices S_T (bars, bootstrap intervals), first-order indices S_1 (open circles) and noise floor (red ticks) for O17p, persistent O13, ρ_{pk} and f_0 . Saltelli design with $N = 256$ and 11 factors. Dark bars meet the relevance criterion $S_T - S_T^{noise} > 0.05$ and $S_T - CI > S_T^{noise}$.

fails. With four founding pairs the peak density is 0.07 ($L = 30$) to 0.17 ($L = 15$), and with mating in the Moore neighbourhood 0.18 at $L = 30$; no founder run is Calhoun-like.

Founder protocol (P6). Four pairs of 7-week-old animals start in one cell of the 30×30 lattice, with mating in the Moore neighbourhood, $T = 900$ and 30 runs per point. The plane varies the initial social state of the founders s_0 (0.2, 0.3, 0.5; a value below one represents adaptation to the new enclosure), the number of movement sub-steps per week k (1, 2) and the longevity a_{max} (120, 200; with $a_{max} = 200$ the fertile window is extended to 133 weeks). Table 22 lists the outcomes. The latency pattern O1 passes in up to 0.67 of the runs, but O1–O3 jointly with O17p pass in at most 0.43 [0.27, 0.61] ($s_0 = 0.2$, $k = 1$, $a_{max} = 200$). The best points need $a_{max} = 200$, outside the preset; with $a_{max} = 120$ no point exceeds 0.13. The peak density stays at 0.16–0.27.

C.11 Transplant experiments

Each of the 60 replicates per arm runs a mother colony and samples it at several times (Fig. 17, Tables 23 and 24): in phase B; at $1.75 t'_{pk}$ (phase D of the primary evaluator); at $1.75 t_{pk}$ with $t_{pk} = \arg \max N$ (the preregistered protocol); at week 95 in every arm, which removes the confound between rule and sampling time; and, not preregistered, at the first week in which four males and four females past the plasticity window have $0.2 \leq s \leq 0.5$ (the “band”). Four males and four females are moved to an empty lattice; success is reaching 80 individuals within 150 weeks. In the mixed

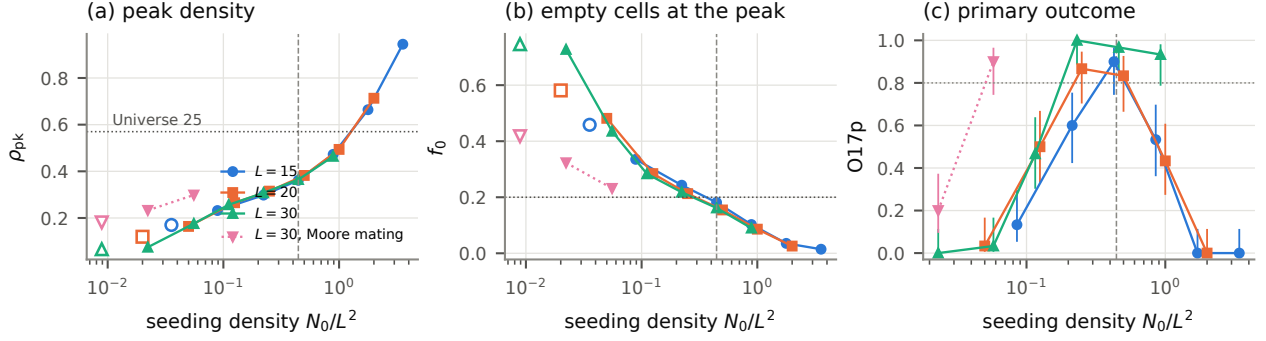


Figure 16: Seeding density and lattice size (P4; 30 runs per point). **(a)** Peak density, **(b)** empty fraction at the peak and **(c)** O17p (95% Wilson intervals) against the seeding density N_0/L^2 , for three lattice sizes with random seeding (filled markers) and four founding pairs in one cell (open markers); the triangles pointing down use mating in the Moore neighbourhood. The vertical dashed line is the default initial condition ($N_0 = 400$, $L = 30$); horizontal dotted lines mark the values of Universe 25 ($\rho_{pk} = 0.57$, about 20% of empty nest boxes) and the 0.8 threshold.

phases the animals of one sex are paired with healthy animals of the other. Arm X shares its mother colonies and its transplanted animals with C1 and places them in an enclosure without R1 and R2 with $r = 0.1$.

- *Healthy controls.* Animals taken in phase B found colonies in 77–98% of the trials in every arm.
- *Phase-D animals under R1.* Whenever R1 is present (C1, C1', C0, C2 and –R2), phase-D animals never found a colony at any of the three sampling times, nor with healthy partners (0/120 at week 95 in C1, C1', C0 and –R2; 3/120 in C2). Their social state is 0 (median), with all eight below 10^{-3} in 55–93% of the trials.
- *Without R1.* At week 60 the same test gives 0/60 in the arms without R1 too, because their colonies are still at the peak and the animals are young but fully deteriorated; at $1.75 t_{pk}$ (weeks 126–131) 10/60 recover alone in –R1 and 3/60 in –R1–R2, and with healthy partners 34/120 and 14/120 at week 95. In arm X the rates are 51/60 at week 60, 12/60 at week 97 and 5/60 at week 95: the recovery of the same animals is a matter of the fertile time left.
- *Intermediate animals.* Band animals (weeks 8–11, ages 12–15, $s = 0.33$ –0.40) found colonies in 5/60 (C1), 2/60 (C1'), 0/60 (C0), 7/60 (–R2) and 8/60 (C2) under R1, against 50/60 without R1, 46/60 without R1 and R2 and 57/60 in arm X. With healthy partners the arms with R1 reach 35–46/60 (band males with healthy females) and 7–32/60 (band females with healthy males); the asymmetry follows from the fecundity $s_f^2 s_m$. The equal-time comparisons between arms (Fisher, BH at 10%) separate the arms with R1 from those without it in the band phases and in the mixed phases at week 95, and nothing else.
- *Descendants of the band.* In the colonies founded by band animals, the adult descendants have a mean social state of 0.28–0.45 in C1, 0.26–0.42 in C1', 0.11–0.35 in C0 and 0.33–0.37 in arms X and –R1: in a small uncrowded colony the offspring learn the competence of their parents and stay there, with or without R1. The lineage criterion of Section 6 is therefore confirmed only in its first part, the founding bottleneck.

This reproduces Marsden’s observation, reported by Calhoun [5]: animals removed in mid-phase D failed to reproduce “even placing them with adequate sex partners” raised without crowding. The model reproduces it only with R1, and by construction. The evaluator rates O12 as PASS in every arm with R1 and in $-R1-R2$, and as MARGINAL in $-R1$ and arm X when sampled at $1.75 t_{pk}$.

Transplant grid. Over 12 points of the plane (α, κ) , with $\alpha \in \{2, 4, 6, 8\}$ and $\kappa \in \{0.1, 0.15, 0.2\}$ and 40 transplants per point and phase, no phase-D transplant founded a colony (0/480), nor did any mixed transplant (0/960). Phase-B transplants succeed in 0.63–1.00 of the trials, less often with stronger attraction (mean 0.97 at $\alpha = 4$ and 0.71 at $\alpha = 8$), and O12 passes at all 12 points.

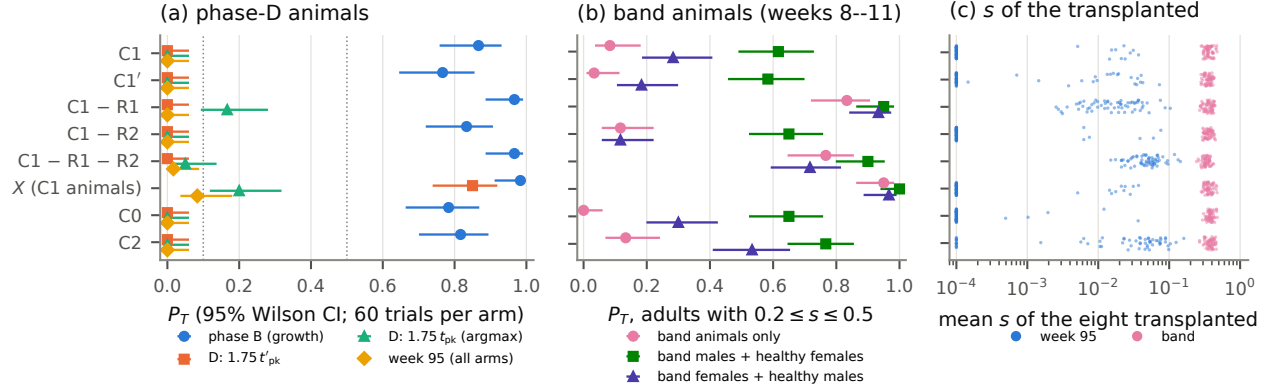


Figure 17: Transplant experiments (60 replicates per arm and phase). P_T is the probability that four males and four females moved to an empty lattice reach 80 individuals within 150 weeks. (a) Animals taken in phase B and in phase D at three sampling times: $1.75 t'_{pk}$ of each colony, $1.75 t_{pk}$ with $t_{pk} = \arg \max N$, and week 95 in every arm. (b) Adults past the plasticity window with $0.2 \leq s \leq 0.5$, alone and paired with healthy animals of the other sex. (c) Mean social state of the eight transplanted animals per trial, at week 95 and in the band. Arm X moves the animals of the C1 arm, the same individuals, into an enclosure without R1 and R2 with individual recovery at $r = 0.1$. Dotted lines in (a) mark the O12 thresholds (0.1 and 0.5).

C.12 C1 versus C2

Table 25 lists the confirmatory contrast; 13 of the 29 rows differ after Holm adjustment with the primary evaluator, 12 with the preregistered evaluator (O4 and O13p differ only with the primary O4 and the fixed class threshold). The 29 rows hold 27 distinct tests: O6, O7 and A2 (recovery of natality) are decided by the same four C2 colonies, which survive to T and recover natality, and in the production runs O17 and O17p differ (0.725 against 0.705). Nine rows are constant in both arms and enter the Holm sequence with $p = 1$, so the correction over 29 is conservative. Besides the losses in O8 and O9, natality collapse weakens in C2: O5 passes in 0.815 against 0.98, because its births are spread over a longer growth phase ($t'_{pk} = 62$ weeks, $t_{B99} = 86$). Four of its 200 colonies persist to T (A2 in 0.02). The decline is as long as in C1, $(T_{ext} - t_{pk})/t_{pk} \approx 1.9$ –2.1, so O8 fails because the mean age no longer rises at the required rate. The isolated class is slightly smaller in C2 ($\Phi_{BO}^{pers} = 0.13$ against 0.17), the peak higher (0.42 against 0.36), the empty fraction smaller (0.11 against 0.16) and extinction later (Kaplan–Meier median 218 against 169 weeks).

Fig. 18 shows the cohort structure behind O13b. In C2 the isolated class is born later than the crowded class in every run (median birth week 22 against 11), whereas in C1 the two classes are born

together (10 against 11 weeks; the isolated class is later in 4% of the runs): the isolated animals of **C1** are refugees of the early boom, of the same cohorts as the crowded ones. In both candidates births are front-loaded: half of the births of **C1** occur before $0.36 t'_{pk}$ and 93% before $0.8 t'_{pk}$, the start of the window used by the strict O13b criterion, which therefore cannot pass.

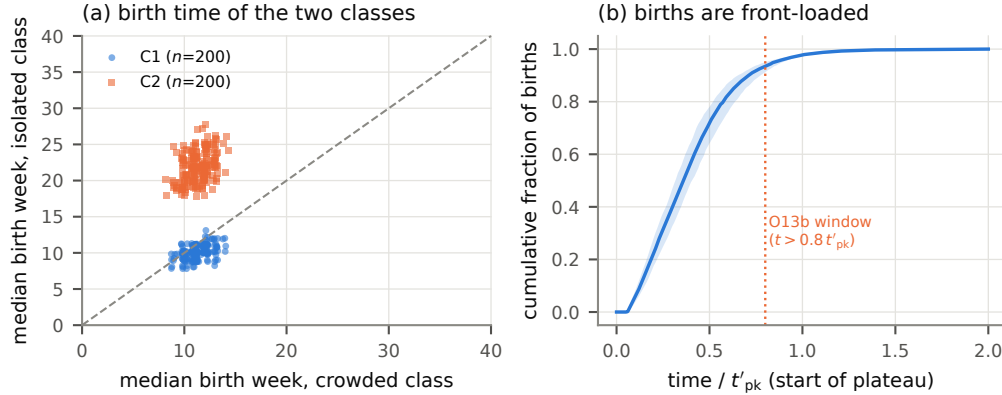


Figure 18: Cohort structure of the deteriorated classes. **(a)** Per-run median birth week of the persistent isolated class against that of the crowded class, at the time t^* used by O13b, for **C1** and **C2** ($n = 200$ each, jittered); the dashed line is the diagonal. **(b)** Cumulative fraction of births against time in units of t'_{pk} for the **C1** reference ensemble (median and 10–90% band over runs); the dotted line marks the start of the O13b window.

C.13 Ensembles

Table 26 lists every ensemble of the production program with its size, horizon and base seed, so that each number of the text can be traced to its source. Seeds are derived as `SeedSequence([base_seed, point, replicate])`; the base seeds of the production runs (25301–25336) differ from those of the runs of the preregistered design, so the two sets of runs are independent.

C.14 Deviations from the preregistration

D Calibration details

This appendix collects the development ensembles of Section 5. They used independent seeds and smaller samples than the production ensembles of Section 8, which supersede them wherever the two overlap.

D.1 Procedure

Transplants. The transplant patterns O12 and O12b are ensemble experiments.

- From each replicate colony, four reproductive males and four reproductive females are moved to an empty lattice: either from the middle of the growth window (control, phase B) or at $t = 1.75 t_{pk}$ (mid-phase D).
- In O12b, four phase-D animals of one sex are paired with four healthy animals ($s = 1$) of the other.
- A transplant “founds” a colony if it reaches 80 individuals within 150 weeks.

Ensembles. The calibration proceeded in stages:

1. screening of rule families: 23 points $\times$ 10 replicates, then 24 points (initial condition $\times \alpha \times \kappa$) $\times$ 10 replicates;
2. ablations of the selected rules: 10 points $\times$ 40 replicates;
3. one-at-a-time sensitivity around the selected point: 17 points $\times$ 30 replicates;
4. final ensembles: C0 with 120 replicates, 100 transplants and an $\alpha = 0$ control; C1 and C2 with 60 replicates and 60 transplants each;
5. exploratory tests of the founder protocol: 30–60 replicates per point.

Further details:

- replicates use independent seeds, `SeedSequence([base_seed, point, replicate]);`
- the horizon is $T = 600$ weeks, and every run of the final ensembles of C0 and C1 went extinct before week 220;
- a run of C0 costs about 1 s of CPU;
- the whole calibration used about 45 min on four cores.

D.2 Development ensembles of the candidates

Table 28 reports every pattern in the development ensembles of C0, C1 and C2. Two development estimates were optimistic: O9 in C1 (0.57 here, 0.45 in production) and O17 in C2 (0.95 here, 0.83 in production, where 7.5% of the colonies recover natality).

Magnitudes in the development ensemble of C0. The candidates also approach several magnitudes of Universe 25, none of which is a test (Section 5). Values are medians over the 120 runs of C0, with bootstrap intervals; the corresponding figure for Universe 25 is in parentheses.

- The peak lies at $N_{\text{pk}}/K_\gamma = 0.416$ [0.411, 0.420] of the damage-threshold capacity $K_\gamma = m_c|G|$, which is not a physical capacity (Universe 25: $2200/3840 = 0.57$, comparable only in sign).
- At the peak 18% of the cells are empty (20% of nest boxes).
- The population is still at $\nu_{8b} = 0.998$ of its peak $0.31 t_{\text{pk}}$ later (0.93). This is the plateau, a consequence of the absence of mortality before senescence.
- The decline lasts $(T_{\text{ext}} - t_{\text{pk}})/t_{\text{pk}} = 2.13$ [2.00, 2.18] (2.2 projected, ≥ 1.84 observed). It reaches at least the observed value in 78% [69, 84] of the runs. The agreement is not robust: the ratio depends on where t_{pk} falls on the plateau (Appendix D.9).

The adult social state falls below 0.5 at $0.28 t_{\text{pk}}$, long before the peak. In the $\alpha = 0$ control the clustering of births is weaker: the Gini coefficient of births over 4×4 blocks is 0.075, against 0.168 in C0 ($p = 2 \cdot 10^{-17}$; O14).

D.3 Ablations of C0

Table 29 removes the rules of C0 one at a time. Without refuges, each of the three rules is necessary.

- *R2*, switched off (full inheritance and individual recovery at $r = 0.01$). Without it, neither O11 nor O13 is ever reproduced. Viable pups come from the healthiest mothers and inherit their state, so the late cohorts are competent.
- *AV*. Without it the population collapses into a few huge pools that leave 76% of the lattice empty, which is the $[\neq \text{ORIG}]$ picture of “half the colony empty”, and the isolated class disappears.
- *R1*. Without it, adults recover when the population thins, and extinction (O7) and O13 drop to 0.75 and 0.50 at $\alpha = 4$. Removing R1 is tolerable only at $\alpha = 6$, with lower robustness.

The long-lived control (last row; no rules, $\alpha = 1.4$) fails O4: the peak reaches 0.83 of the capacity. They also fail O11 and O13.

Crowding is the trigger. Setting $\gamma = 0$ in C0 removes the collapse altogether: the population grows past $2K_\gamma$ and the mean social state stays near 0.42; among adults it stays near 0.6 (C1 with $\gamma = 0$), since the mean over all animals includes the pups, born with $s \leq 0.2$. The failure of socialisation under R2 therefore needs crowding damage to start. Calhoun’s emphasis on the social, rather than purely numerical, consequences of density is consistent with this division of labour [5].

The transplant control constrains the attraction. With $\alpha = 6$ the control transplants (healthy phase-B animals) fail to found a colony in 60% of the trials: strong aggregation of eight animals in one cell deteriorates them. This makes O12 not applicable, and is why $\alpha = 4$ was preferred to $\alpha = 6$.

D.4 Local sensitivity of C0

Table 30 varies each parameter of C0 in turn. O4, O5, O6, O7, O10 and O11 remain at ≥ 0.93 in every variation. The bottleneck is O13: the isolated fraction sits only about 0.02 above its threshold of 0.1. Two variations break it:

- faster learning ($r = 0.15$), which shortens the time the young spend deteriorated;
- a weaker selectivity of the aversion ($p = 2$), which also repels moderately healthy individuals.

Both homogenise the occupancy. C0 therefore lies inside the robust region but close to its border in r and κ . This is consistent with the narrow band of Fig. 1(c), where $n^* = 1/\kappa - 1/\alpha$ must stay between about 5.5 and 10.

With refuges the robust region is wider. In the production planes around C1 (Section 8.5):

- the primary outcome stays at ≥ 0.83 for $w \leq 0.3$ and $r \leq 0.10$, and falls through O11 at $w \geq 0.5$ and $r \geq 0.2$;
- it stays at ≥ 0.8 at 34 of the 48 points of the (α, κ) plane with $\alpha \geq 2$, and fails where the free space required by the primary O4 is lost.

This local robustness contrasts with the small Calhoun-like fraction of the global Latin hypercube, 2.0% of the points with the primary evaluator. That fraction is dominated by the refuge geometry and by thresholds that also enter the definitions of the classes, and it depends on the sampling box.

D.5 Deteriorated classes: persistence and cohort structure

C0 reproduces O13 only instantaneously. In C0 the criterion O13 is met: 12% of the adults are isolated ($m \leq 2$) and 14% crowded ($m > m_c$). But the two groups are snapshots of a single, homogeneously deteriorated population whose occupancy is overdispersed.

- Only 0.6% of the adults stay isolated for four consecutive weeks.
- The weekly autocorrelation of the occupancy experienced by an individual is 0.23.
- Isolated and crowded adults do not differ in age (46 versus 46 weeks), sex, social state or total affinity (132 versus 145).

Changing the affinity memory or formation rate, adding male territoriality, strengthening the aversion or restricting bond formation to competent pairs does not produce persistence (four-week isolated fraction ≤ 0.012 in every case). At a density of three to four animals per cell every individual touches neighbours almost every week, and no region is empty enough to stay in.

Refuges. Refuge cells of limited capacity, without a preference, remain essentially empty: they hold at most 5% of the population and leave the four-week isolated fraction unchanged. A deteriorated animal surrounded by acquaintances always prefers an occupied neighbouring cell.

With a preference, two conditions are needed:

- *The capacity must be hard.* With soft capacity, simultaneous moves produce stampedes into the same refuge: the mean crowding reaches 30–60 and the isolated class shrinks.
- *The refuges must hold enough animals.* The persistent isolated fraction is bounded by the refuge capacity relative to the adult population, $\phi_R = f_R c_R |G| / N_{\text{ad}}$. A class of 10% therefore needs $f_R c_R \gtrsim 0.3$.

With $f_R = 0.3$, $c_R = 2$ and $\pi = 100$ (C1), 15% of the adults form a persistent isolated class. 95% of them are in refuges and they remain isolated for 15–16 consecutive weeks on average. They receive no crowding damage and no social mortality, which matches the “physically healthy” beautiful ones of [5]. Persistent O13 holds in 60/60 runs, and O17 in 60/60 (Table 28).

Cohort structure. In C1 the persistent isolated animals are of the same age as the crowded ones (51 versus 51 weeks at the peak). They are refugees of the early boom, not a late, unsocialised cohort, and the pattern O13b fails in every run ($\Delta b = -0.02 t_{\text{pk}}$). Between t_{pk} and $t_{\text{pk}} + 20$, 17% of them are founders, 58–60% belong to the first generation and 22–27% to later ones. Most entered the refuges as pups: 62–67% first entered a refuge before the age of 12 weeks, and 15–20% before 3 weeks (8 exploratory runs, with the single-pass and with the exact hard capacity). Pups are independent agents from birth and, being born with $s \leq 0.2$, are strongly attracted to refuges; a nest period, in which pups stay in their mother’s cell, is a reported variant (Section 8.5).

Basing the retreat on the competence reached at the end of the critical window does not help. Under R2 almost every cohort after the first ends the window with $s < 0.1$, so there is no gradient between cohorts. Basing it on the competence of the mother at birth (C2) does create the gradient.

- The persistent isolated class is 11 weeks younger than the crowded class (51 versus 63 weeks).
- It has 28% fewer bonds (total affinity 97 versus 136) and a slightly higher social state.
- It belongs to the second half of all births in 58/60 runs.

- $\Delta b = +0.13 t_{\text{pk}}$, Mann–Whitney $p \approx 10^{-46}$.

This is the structure Calhoun describes, in which the beautiful ones came from “the last half of the population born” [5]. The price is a broader age distribution in phase D, which breaks the senescent rise of the mean age required by O8 (Table 28).

Why O13b cannot pass with its current threshold. O13b also requires that half of the isolated class be born after $0.8 t_{\text{pk}}$. With $N_0 = 400$ this is out of reach for any retreat mechanism, because of when births happen.

- The median birth time of the whole population is $0.18\text{--}0.24 t_{\text{pk}}$.
- Ninety percent of all births occur before $0.44 t_{\text{pk}}$.
- Only about 16 pups per run, 0.6% of all births, are born after $0.8 t_{\text{pk}}$.

In Universe 25, by contrast, growth continued almost to the peak, and about half of all births occurred after $\approx 0.8 t_{\text{pk}}$ (our estimate from the growth curve: N reached half its peak around day 435). In the model the “peak” is the top of a plateau: natality dies at $0.5\text{--}0.7 t_{\text{pk}}$, and the colony, made of young adults, hardly changes until senescence. We therefore also report the cohort criterion relative to the distribution of births (“last half” in Table 28). Under that reading C2 passes and C1 fails.

D.6 The founder protocol

Calhoun founded Universe 25 with four pairs of 48-day-old mice. We tested the corresponding protocol: four plus four individuals of age 7 weeks placed in the central cell. Table 31 summarises the results.

C0 with founders. The colony stays within about 10% of the lattice and deteriorates locally, giving a trivial peak of $\rho_{\text{pk}} \approx 0.02\text{--}0.03$. Under weaker aggregation ($\alpha = 1.4$, no R2) the colony grows as a slow front, at about 0.06 cells per week. It takes $t_{\text{pk}} \approx 170\text{--}215$ weeks to peak, against a local deterioration time of 20–30 weeks: a Damköhler number $\text{Da} \approx 10$.

More mobility. Several Moore steps per week (`move_substeps`) make things worse. Founders and their offspring disperse, males and females no longer meet in the same cell, and the colony fails by an encounter-limited Allee effect. With $N_0 = 400$, more mixing also homogenises the occupancy and destroys O13: O17 equals 0.93, 0.80, 0.30 and 0 for 1, 2, 4 and 8 sub-steps.

Mating in the Moore neighbourhood. With `mate_radius = 1`, two sub-steps and founders whose initial social state is reduced to 0.3 (a stand-in for the adaptation after transfer, phase A), O1–O3 become applicable and are frequently reproduced. However, the protocol is not yet robust:

- O17 is only 0.17 [0.09, 0.28];
- O5 and O13 fail in a third to a half of the runs;
- the peak comes too late relative to the life span ($t_{\text{pk}}/a_{\text{max}} \approx 1.4$, against 0.7 in Universe 25), so the senescent decline is too short and O8 fails.

The production plane of this protocol (initial social state $\times$ sub-steps $\times a_{\max}$, 30 runs per point, Appendix C.10) does not change the conclusion. Its best point ($s_0 = 0.2$, one sub-step, $a_{\max} = 200$) reproduces O1–O3 jointly with O17p in 0.43 [0.27, 0.61] of the runs, and with the preset longevity $a_{\max} = 120$ no point exceeds 0.13.

D.7 Generations and chronology

Table 32 follows every animal of C1 by generation: the founders are F0, and a pup belongs to the generation of its mother plus one. The realised fecundity is counted over the whole life of each female. The table uses the 200 runs of the reference ensemble: the same parameters and seeds were run again with a register of generations, and every run reproduces the trajectory of the ensemble (`scripts/analysis/generaciones_base.py`). Table 34 repeats the summary for the other members of the reference ensemble and for the variants of Section 8.5, re-running the first 20 replicates of each production arm with the same seeds (`scripts/analysis/gamma_v3.py`; every replicate reproduces the maximum population of its run). Table 33 gives the deterioration and generation times, and Table 35 compares the chronology with Universe 25.

A boom of one to two generations. The founders deteriorate by crowding during their own adult life: they reach the end of the window with $s = 0.90$, but have $s = 0.20$ at 26 weeks of age (week 22). They produce 69% of all births, about 6% of their potential fecundity. Their offspring are born at $s_{\text{nb}} = w s_f \simeq 0.19$, below the deterioration threshold, and gain only +0.04 by the end of the window, because their teachers are already deteriorated; without crowding damage the mean adult state stays near 0.6. The net reproductive number falls by a factor of about ten in a single step, from 3.8 to 0.40, and the third and later generations add 3.2% of the births. The generational gradient is monotone (s_{nb} 1, 0.19, 0.14, 0.10; R_{net} 3.8, 0.40, 0.11, 0.05), but it is almost complete between the founders and the first generation. The structure is the same in the other members of the reference ensemble and in every variant of Table 34: the mean generation of the births is 1.2–1.6, and the first generation with $R_{\text{net}} < 1$ is always F1. It is therefore not an artefact of the synchronous cohort. The long-lived control without new rules has a *deeper* boom (30 replicates of the null ensemble): 22% of the births belong to the third and later generations, the mean generation is 1.95 [1.90, 1.99] against 1.34, and R_{net} of F1 is 1.21 [1.17, 1.25] against 0.40 (medians with bootstrap intervals; every run of the control has $R_{\text{net}} > 1$ in F1, none of C1). The long-lived control also differs from C1 in α , so the clean contrast is C1 without its four rules (same $\alpha = 4$ and crowding damage): mean generation 1.59 and R_{net} of F1 0.66, against 1.34 and 0.40 (Table 33). The rules, R2 in particular ($-R2^*$: 1.62), shorten the generational depth, because pups no longer inherit the competence of the few healthy mothers.

Deterioration time and generation time. Table 33 measures how long deterioration takes. Three means of s are followed every week: over the adults (age $\geq a_{\min} = 6$, the S_{adult} of O10), over the animals past the critical window (age $\geq a_w = 12$) and over the founders. The adult mean is diluted: pups of the first generation are born with $s \leq 0.2$ and count as adults from the age of 6 weeks, while they are still inside the window. It falls below 0.5 in week 13, when the founders still have $\bar{s} = 0.66$ and are 39% of the adults. The animals past the window, which no longer learn and admit newborns only 12 weeks after their birth, and the founders, a closed cohort, cross 0.5 in week 16. We therefore define $t_{\text{det}} = t_{1/2} - t_{\text{on}}$ on the animals past the window: $t_{\text{det}} \approx 10$ weeks and $\Gamma = t_{\text{det}}/T_{\text{gen}} \approx 1.1$ in C1 (interquartile range over runs 1.0–1.29; founders 1.2), against 0.8 with the adult mean of the primary evaluator. Deterioration takes of the order of one generation time. The value depends on the definition of T_{gen} : with overlapping cohorts the interval between median

birth weeks is not a demographic generation time, and in the long-lived and rule-free null models it drops to 5 weeks (4 in some runs), below the age of first reproduction, which inflates their Γ . With the standard definition, the mean age of the mothers at birth ($T_{\text{gen}} = 14.3$ weeks in C1), $\Gamma \approx 0.7$ (interquartile range 0.63–0.74) in C1, 0.60–0.66 in C1', $-R2^*$ and the long-lived and rule-free null models, and 0.85 at $\gamma = 0.02$ (30 and 20 runs, from the same per-run data); Γ is of order one under either definition, and it does not order the depth of the boom under either.

Γ is not what keeps the boom shallow. Lowering the crowding damage from $\gamma = 0.1$ to 0.02 raises Γ from 1.2 to 1.6 (founders 1.2 to 1.8), but the mean generation of the births moves only from 1.35 to 1.42 and R_{net} of F1 from 0.39 to 0.52; the colonies still collapse and pass O17p in 17 of 20 runs. Pups born at $s_{\text{nb}} \approx 0.4$ ($s_0 = 0.25$) change the depth more at about the same Γ : 1.51 and $R_{\text{net}} = 0.62$ at $\gamma = 0.1$, 1.64 and 0.84 at $\gamma = 0.02$. Across the controls Γ does not order the depth either: $-R2^*$, with full inheritance, has $\Gamma = 1.2$ and a mean generation of 1.62; $\alpha = 0$ has 2.4 and 1.57; the long-lived control 1.6 and 1.95. The depth is set in the first step, F0→F1: pups are born below the deterioration threshold, conception ($\propto s_f s_m$) and pup survival ($\propto s_f$) fall with the parents' state, and, with R2, pups learn from neighbourhoods made mostly of other pups and of founders that are already deteriorating. Γ remains a heuristic for the comparison with Universe 25 (Section 4.4).

The phases are compressed. In terms of Calhoun's phases (Table 35), the model has a growth with competent adults (functional B) from week 3 to week 16: 13 weeks, about two doublings (from 400 to about 1900 animals) and two thirds of all births. In Universe 25 phase B lasted 30 weeks and about five doublings, and most births came later, in phase C. Measured with the adult mean of the primary evaluator, which pups dilute, functional B would end in week 13. Growth with deteriorated adults (C) lasts from week 16 to the start of the plateau at week 34; then the population neither grows nor declines for about 51 weeks, and the senescent decline (D) lasts from week 86 to week 169. The sequence A–B–C–D appears in the right order, but B and C are compressed, natality is front-loaded, and a plateau without births or deaths, absent from Universe 25, is inserted before D. O10b, which asks for a functional growth phase of half the time to the peak, passes in 2% of the runs with the adult mean of the primary evaluator (Section 8.1). Part of that failure is the same dilution that affected the preregistered O10. Measured on the animals past the window, the functional growth lasts $0.46 t'_{\text{pk}}$ (median), and O10b passes in 24% of the 200 runs and is MARGINAL in the rest (founders: 38%). It still falls short of Calhoun's phase B, which lasted $0.56 t_{\text{pk}}$ and 30 weeks. Measured on these animals O10 is unchanged: it passes in every run of C1 and, as with the adult mean, in three of the four null models ($-R2^*$, long-lived and no rules, 30 replicates each), so it remains generic. The same holds for the founders' mean.

Non-synchronous initial condition. In the production stage ($n = 200$, primary evaluator), initial ages drawn uniformly over 4–52 weeks give O17p = 0.93 [0.89, 0.96], and ages over 4–80 weeks, which include post-reproductive founders, give 0.63 [0.56, 0.69] (0.68 and 0.31 in C1'). Only O5 changes, and it becomes MARGINAL rather than failing: 14 MARGINAL and no FAIL with 4–52 weeks, 73 MARGINAL and one FAIL with 4–80 weeks. Its level clause is always met: $\nu_5 = N(t_{B99})/N_{\text{pk}}$ has a median of 0.997 and a minimum of 0.97 over the 200 runs of 4–80 weeks, so natality still ends at the population maximum. What fails is the timing clause. The older founders die earlier (plateau of 28 and 22 weeks), their deaths balance the last births, and the plateau starts earlier (t'_{pk} 30 and 29 weeks, against 34), while the last 1% of the births still comes at weeks 38–41. Hence $\Delta_5 = (t_{B99} - t'_{\text{pk}})/t'_{\text{pk}}$ rises from 0.17 to a median of 0.22 and 0.42, and lies between 0.52 and 0.97 in the MARGINAL runs (series of the production ensembles). The classes, O11, the low peak and the generational structure (Table 34) do not change: O11, O13 and O13p pass in every run of

both arms. The primary outcome therefore depends on the initial condition only through the timing of O5 relative to t'_{pk} , a generic criterion that carries no evidence for the rules (Section 8.5).

Pups born above the deterioration threshold. With $w = 0.2$ and $s_0 = 0$ every pup is born with $s \leq 0.2$, which makes part of O11 definitional. With $s_0 = 0.25$ or 0.5 pups are born at $s_{\text{nb}} \approx 0.4$ or 0.6 . In the production stage ($n = 200$) extinction, the low peak ($\rho_{\text{pk}} = 0.40$ and 0.44), the two classes and O5 still hold, but O11 passes in 8% and 0% of the runs, and so O17p is 0.08 and 0.00 (0.64 and 0.36 with the preregistered O11; Section 8.5). O11 reads the state at the age of maturity, 6 weeks, when these pups still have $s \approx 0.36$ – 0.47 (exploratory runs). At the end of the window crowding has brought them down: in 8 exploratory runs per value the last half of the births reaches $s(a_w) \approx 0.12$, and only 7% of its females breed. The failure of the crowded-born cohorts as O11 measures it is therefore largely definitional, tied to $s_0 = 0$; measured at the end of the window it survives, but it is then produced by crowding damage, as in the long-lived control, and not by birth below the threshold. The boom remains shallow: the mean generation of the births is 1.51 and 1.60, and R_{net} of F1 is 0.60 and 0.72 (20 replicates of each arm; Table 34).

D.8 Scales and the mapping to Universe 25

Table 36 lists the scales of C1 with their counterparts in Universe 25 and states which were calibrated. The mapping is the one of Section 4.2: a cell is a resting site, m_c the comfortable occupancy, and the refuges are the high, less preferred nest boxes.

D.9 The extinction time as an extreme-value statistic

Without mortality before senescence the colony dies when its last member dies, $T_{\text{ext}} = \max_i(b_i + \ell_i)$, with b_i the birth time and ℓ_i the lifetime of animal i . Table 37 decomposes it as $T_{\text{ext}} = t_{\text{last}} + (T_{\text{ext}} - t_{\text{last}})$, where t_{last} is the last surviving birth.

- The second term, the longest residual lifetime, is about 110 weeks in every model, with an interquartile range of 5–6 weeks: of the order of the inverse slope of the logistic hazard, $1/k \approx 7$ weeks.
- The spread of T_{ext} is therefore that of t_{last} (correlation 0.89–0.98). That spread is set by how abruptly natality stops, measured by the e-folding time of the tail of births, $\tau_B = (t_{B99} - t_{B90})/\ln 10$.
- The relative spread is small for two trivial reasons: the median contains a nearly deterministic lifetime of 110 weeks, and the numerator is set by τ_B . Without attraction ($\alpha = 0$) natality stops more abruptly ($\tau_B = 2.6$ weeks) and T_{ext} is *more* reproducible than in C1; without AV (C1') the tail of births is longer ($\tau_B = 8.7$) and so is the spread.

Dependence on the size of the system. Table 20 runs C1 at fixed density on lattices from $L = 15$ to $L = 60$ ($N_0 \propto L^2$, 60 runs each; production stage). The median of T_{ext} grows with the logarithm of the size, by about 6 weeks per unit of $\ln L^2$ (from 162 to 184 weeks), and so does t_{last} ; the residual lifetime $T_{\text{ext}} - t_{\text{last}}$ stays at 110–112 weeks. The relative spread is 0.15 for $L \leq 20$ and about 0.08 for $L \geq 30$; with 60 runs the intervals at $L \geq 30$ overlap, and the slow decrease expected for the maximum of many events is within them (exploratory runs give 0.05 at $L = 84$). The value 0.089 of the reference ensemble is therefore a value for $L = 30$. The peak density (0.35–0.37) and the persistent fraction (0.17–0.18) barely change; the empty fraction falls from 0.18 to 0.15, an effect

of the walls, and O17p rises from 0.82 at $L = 15$ to 1.00 at $L \geq 45$, so small lattices fail more often. In exploratory runs up to $L = 84$ (30 per size, single-pass refuge admission) no colony recovers (A2 cleared in all 150 runs, up to about 21 000 animals): we find no healthy pockets that could reseed a large colony.

E Supplementary discussion

Pups born above the deterioration threshold. With $w = 0.2$ and $s_{\text{base}} = 0$ every pup is born with $s \leq 0.2$, the threshold that the criteria use for “deteriorated”. This is why the preregistered forms of O10 and O11 pass in colonies that never deteriorate, and why the primary evaluator scores them on adults and on the late cohorts. With $s_{\text{base}} = 0.25$ or 0.5, O17p is 0.08 and 0.00 with the primary O11 ($n = 200$; 0.64 and 0.36 with the preregistered O11), entirely through O11, which the primary evaluator scores at sexual maturity (six weeks), when these pups are still above 0.2. At six weeks, then, the pass of O11 in C1 was partly definitional. Scored at the end of the plasticity window ($a_w = 12$ weeks) on the same runs, the late cohorts are below 0.2 whatever their birth state (O11 in 0.975 and 0.785, O17p in 0.965 and 0.76), so the definitional part disappears; but O11 then also passes in 0.81 of the long-lived runs and in 0.755 of the runs without rules. The late cohorts deteriorate by crowding damage during the window, with or without the rules, and their failure is not a contribution of the rules (Section 9).

Pups and refuges. Pups count towards local crowding and, born with a low social state, are strongly attracted to refuges: in weeks 8–16 between 77% and 89% of the refuge occupants are younger than 12 weeks, and 62–67% of the persistent isolated animals first entered a refuge before 12 weeks of age (Appendix D.5). With a nest period of three weeks, during which pups stay in their natal cell and neither move nor add to crowding, the boom becomes even shorter (83% of the births are to the founders), O17p is 0.835 [0.78, 0.88], borderline, and $\rho_{\text{pk}} = 0.31$. Because conception and birth coincide, conceptions without viable young (O15b) cannot be separated from infant mortality (O15).

The founder phase (O1–O3). With 400 initial animals, latency, early exponential growth and deceleration are not testable. From Calhoun’s four founding pairs a colony on the lattice spreads as a slow front, because of the slow mobility, and either stays confined (a trivial peak) or peaks far too late relative to the life span. An exploratory founder plane (stage p6, 12 points of 30 runs; Section D.6) uses Moore mating, a reduced initial social state as a stand-in for Calhoun’s “social turmoil” of phase A, and up to two movement sub-steps. It makes O1–O3 testable, and they pass at some points (O1 in up to 0.90, O2 in 0.83–1.00, O3 in up to 1.00), but no founder point is Calhoun-like: O17p is at most 0.37 [0.22, 0.54], and only with a life span of 200 weeks, outside every preset.

The persistent isolated class is not the beautiful ones (O13b). In C1 the persistent isolated animals carry no crowding damage, but they are refugees of the early boom: they do not differ in age or birth cohort from the crowded ones, about 60% are offspring of the founders, and most entered the refuges as pups. Calhoun’s beautiful ones were a late, unsocialised cohort, “the last half of the population born” [5]. We therefore call the class of C1 a “persistent isolated class” and reserve “beautiful ones” for the cohort-structured class of C2, which ties the retreat to maternal socialisation. Its isolated class is born later than the crowded class, and O13b reaches PASS or MARGINAL in every run. The preregistered window (born after $0.8t_{\text{pk}}$) still fails, because births

in the model are front-loaded (median birth time about $0.24t_{pk}$); read literally, Calhoun’s phrase refers to the second half of the births, and a post hoc reading relative to the median birth time passes in 58 of 60 development runs of **C2** and in 0 of 60 of **C1**. **C2** pays a demographic price: O8 fails in every run, O9 falls from 0.40 to 0.30, and the Calhoun-like fraction drops from 0.98 to 0.70. No candidate reproduces at the same time the cohort structure of the beautiful ones and the demographic patterns.

The coexistence band is refuted (Q5). We predicted that crowded and isolated classes coexist only when the preferred occupancy of a deteriorated animal, $n^* = 1/\kappa - 1/\alpha$, is close to m_c . This held in the calibration of **C0**, which has no refuges, and the prediction did not anticipate that refuges decouple the classes. In **C1** the instantaneous criterion is saturated over almost the whole (α, κ) plane, and the fitted dependence on $\log n^*$ is convex.

Fertility, infant mortality and sex bias (O15, O15b, O9). Conception and neonatal survival are both proportional to the mother’s social state, so they decline together. Total infant mortality (O15) is reproduced in 0.5% of the runs; conceptions without viable young (O15b) in 58%, whereas Calhoun reported about 320 days of them; and the female-biased remnant (O9) in 45%, because the model has no sex-specific mortality.

Comparison with Gwizdała and Duś. All comparisons with Universe 25 are qualitative and distributional, and some of Calhoun’s numbers were projected rather than observed, among them the extinction date. The agent-based model of [16] is quantitative: it uses a daily step and fitted life tables, calibrates the growth phase against the total and adult numbers on day 315, and produces the decline through deviation factors that switch on when the *total* population exceeds critical values (best fits around 300–400 animals), together with prescribed probabilities of litter abandonment and of becoming a “beautiful one”. Our model has no global threshold, treats deterioration and its transmission as state variables, validates against many patterns over ensembles with their discriminating power against null models, and backs its claims with preregistered tests, ablations and global sensitivity analysis. The irreversibility of the individual is an assumption of our model (R1), consistent with Marsden’s transplants, not a result.

Table 12: Production ensembles in full (same ensembles and precision as Table 5): essential patterns, transplants (founding transplants over trials, 60 replicates per arm; O12b pools both sexes; phase D at $1.75 t'_{\text{pk}}$ of each colony, and the third row at week 95 in every arm), anti-patterns, magnitudes, desirable patterns (fraction of passing runs), timings (medians over runs) and further magnitudes. “PASS or MARGINAL” for O13b counts the runs in which the evaluator finds the isolated class later-born at least marginally.

ID	Pattern	C1	C1'	C0	$\alpha=0$	C2	short-lived	long-lived
O4	peak below capacity, free space	1.00	1.00	1.00	0.00	0.90	0.87	0.08
O5	natality ends at high N	0.99	0.85	1.00	1.00	0.81	0.25	0.90
O6	no rebound	1.00	1.00	1.00	1.00	0.98	0.45	0.97
O7	extinction	1.00	0.99	1.00	1.00	0.98	0.47	0.98
O10	adult deterioration before peak	1.00	1.00	1.00	1.00	1.00	1.00	1.00
O11	late cohorts fail	1.00	0.97	1.00	1.00	0.99	0.00	0.00
O13	two deteriorated classes	1.00	1.00	0.92	1.00	1.00	0.00	0.00
O13p	persistent isolated class	1.00	1.00	0.00	0.00	0.94	0.00	0.00
O17	Calhoun-like run (O17)	0.99	0.83	0.92	0.00	0.72	0.00	0.00
O17p	primary outcome	0.99	0.83	0.00	0.00	0.70	0.00	0.00
	95% Wilson interval	0.97–1.00	0.78–0.88	0.00–0.02	0.00–0.02	0.64–0.76	0.00–0.04	0.00–0.04
O12	transplants found: phase B	52/60	46/60	47/60	–	49/60	–	–
	phase D ($1.75 t'_{\text{pk}}$)	0/60	0/60	0/60	–	0/60	–	–
	week 95, all arms	0/60	0/60	0/60	–	0/60	–	–
O12b†	D + healthy partners	0/120	0/120	0/120	–	2/120	–	–
A1–A4	anti-patterns triggered	none	none	none	none	A2: 0.02	A2: 0.55	A2: 0.03
	$\rho_{\text{pk}} = N_{\text{pk}}/K_{\gamma}$	0.36	0.33	0.42	0.71	0.42	0.77	0.85
	empty cells at peak, f_0	0.16	0.52	0.19	0.03	0.11	0.30	0.59
	persistent isolated, $\Phi_{\text{BO}}^{\text{pers}}$	0.17	0.16	0.01	0.09	0.13	0.00	0.00
	median T_{ext} (weeks, KM)	169	182	166	155	218	–	186
	spread of T_{ext} (KM IQR/median)	0.09	0.12	0.08	0.05	0.14	–	0.16
O8	senescent decline	0.62	0.21	0.82	0.97	0.00	0.09	0.37
O8b	slow initial decline	1.00	1.00	1.00	1.00	1.00	1.00	1.00
O9	old, female-biased remnant	0.45	0.29	0.44	0.49	0.30	0.21	0.37
O10b	functional growth phase	0.02	0.01	0.18	1.00	0.00	1.00	0.19
O11b	few late-born females conceive	1.00	1.00	1.00	1.00	1.00	1.00	0.97
O13b	isolated class later-born	0.00	0.00	0.00	0.00	0.00	0.00	0.00
O14	aggregation with free space	1.00	1.00	1.00	0.00	1.00	1.00	1.00
O15	infant mortality $\rightarrow 1$	0.01	0.00	0.00	0.01	0.01	0.00	0.01
O15b	conceptions without viable young	0.57	0.70	0.46	0.28	0.45	0.15	0.09
O13b	PASS or MARGINAL	0.01	0.04	0.04	0.00	1.00	0.16	0.63
	t'_{pk} (weeks, median)	34	35	33	31	62	31	45
	$t_{\text{pk}} = \arg \max N$ (median)	55	61	53	44	80	35	66
	plateau length (median)	51	50	54	59	28	9	45
	t_{B99} (median)	40	44	38	33	86	980	55
	95% CI of the KM median	167–171	180–185	165–167	154–156	217–222	–	181–190
	$P_{\text{ext}}(T)$	1.00	0.99	1.00	1.00	0.98	0.47	0.98
	CV of T_{ext} (extinct runs)	0.07	0.16	0.07	0.06	0.12	0.45	0.20
	$(T_{\text{ext}} - t_{\text{pk}})/t_{\text{pk}}$ (extinct)	2.13	2.15	2.16	2.60	1.88	2.78	2.02

Table 13: The minimal candidate $C1'$ ($= C1$ without AV). Top: reference ensemble ($n = 200$, $T = 600$); p is the two-sided Fisher test against $C1$. Middle: the AV arm of the preregistered factorial ($n = 120$); p is the one-sided test of Q2 before Holm adjustment. Bottom: single ablations of $C1'$ ($n = 40$); p is the one-sided Fisher test against the $C1$ –AV arm of the factorial. O17p with 95% Wilson interval; other columns are fractions of passing runs, and f_0 is the mean fraction of empty cells at the peak.

Arm	n	T	O17p	p	O5	O11	O13p	O8	f_0
C1	200	600	0.99 [0.97, 1.00]		0.99	1.00	1.00	0.62	0.16
$C1' = C1 - AV$	200	600	0.83 [0.78, 0.88]	1.0×10^{-9}	0.85	0.97	1.00	0.21	0.52
C1 (factorial)	120	600	0.99 [0.95, 1.00]		0.99	1.00	1.00	0.72	0.16
$C1 - AV$ (factorial)	120	600	0.79 [0.71, 0.85]	1.1×10^{-7}	0.82	0.97	1.00	0.16	0.52
$C1' - R1$	40	600	0.00 [0.00, 0.09]	$< 10^{-16}$	0.03	0.03	0.10	0.00	0.50
$C1' - R2$	40	600	0.00 [0.00, 0.09]	$< 10^{-16}$	0.90	0.00	0.97	0.53	0.51
$C1' - R7$	40	600	0.00 [0.00, 0.09]	$< 10^{-16}$	0.95	0.82	0.00	0.60	0.76

Table 14: The confirmatory family with the primary evaluator (left) and with the preregistered evaluator (right), on the same runs. Holm correction over seven tests within each column set; the p of Q6 is quasi-binomial. The preregistered evaluator takes the transplant of Q1 at $1.75 t_{pk}$ with $t_{pk} = \arg \max N$, as preregistered.

	primary evaluator				preregistered evaluator			
	Estimate	p_{Holm}	Rej.	Met	Estimate	p_{Holm}	Rej.	Met
Q1	0/60 vs. 51/60 ($P_T = 0.85$ [0.74, 0.92]); discordant pairs 51:0	1.3×10^{-15}	yes	yes	0/60 vs. 12/60 ($P_T = 0.20$ [0.12, 0.32]); discordant pairs 12:0	7.3×10^{-4}	yes	no
Q2	0.99 vs. $-R1$ 0.21, $-R2$ 0.00, $-AV$ 0.79, $-R7$ 0.00	2.2×10^{-7}	yes	no	0.97 vs. $-R1$ 0.23, $-R2$ 0.17, $-AV$ 0.90, $-R7$ 0.00	0.067	no	no
Q3	200/200 vs. 47/100; CV 0.066 [0.059, 0.071]	$< 10^{-16}$	yes	yes	200/200 vs. 47/100; CV 0.066 [0.059, 0.071]	$< 10^{-16}$	yes	yes
Q4	slope 0.996 ± 0.014 ; $\bar{r} = -2.0 \pm 0.3$ wk	$< 10^{-16}$	yes	yes	slope 0.996 ± 0.014 ; $\bar{r} = -2.0 \pm 0.3$ wk	$< 10^{-16}$	yes	yes
Q5	$b_2 = +185.3 \pm 64.3$ (convex); no interior maximum	1.00	no	no	$b_2 = +69.5 \pm 14.2$ (convex); no interior maximum	1.00	no	no
Q6	$\hat{\theta} = -0.15$ [-0.20, -0.05] (quasi-binomial, $\phi = 3.1$)	$< 10^{-16}$	yes	no	$\hat{\theta} = 0.30$ [0.15, 0.50] (quasi-binomial, $\phi = 1.8$)	4.0×10^{-5}	yes	yes
Q7	120/120 vs. 0/30 (30/30 explode)	$< 10^{-16}$	yes	yes	120/120 vs. 0/30 (30/30 explode)	$< 10^{-16}$	yes	yes

Table 15: Transition of P_{ext} at $\gamma = 0$ in $\Lambda = r a_w$, by inheritance weight w and window a_w (20 runs per point). Columns: the last grid value of Λ with $P_{\text{ext}} \geq 1/2$ and the first with $P_{\text{ext}} < 1/2$, each with its count of extinct runs, which bracket $\Lambda_{1/2}$; its log-linear interpolation; and the exponent $\theta_{\text{loc}} = 1 - \log[\Lambda_{1/2}(a')/\Lambda_{1/2}(a_w)]/\log(a'/a_w)$ between this window and the next ($a' = 12$ for $a_w = 6$, $a' = 24$ for $a_w = 12$): the range allowed by the two brackets, then the value from the interpolated $\Lambda_{1/2}$. The preregistered logistic fit is separated (51 of 54 points at $P_{\text{ext}} \in \{0, 1\}$) and is not reported as inference (text).

w	a_w	last Λ , $P_{\text{ext}} \geq \frac{1}{2}$	first Λ , $P_{\text{ext}} < \frac{1}{2}$	$\Lambda_{1/2}$	θ_{loc} to next window
0	6	0.6 (20/20)	0.84 (0/20)	0.71	(0.00, 1.00); 0.51
0	12	0.84 (19/20)	1.2 (0/20)	0.99	(0.49, 1.51); 0.99
0	24	0.84 (20/20)	1.2 (0/20)	1.00	–
0.05	6	0.48 (20/20)	0.6 (0/20)	0.54	(0.19, 1.00); 0.60
0.05	12	0.6 (20/20)	0.84 (0/20)	0.71	(0.51, 1.49); 1.00
0.05	24	0.6 (20/20)	0.84 (0/20)	0.71	–
0.1	6	0.36 (20/20)	0.48 (0/20)	0.42	(0.26, 1.00); 0.70
0.1	12	0.48 (14/20)	0.6 (0/20)	0.51	(0.68, 1.32); 0.88
0.1	24	0.48 (20/20)	0.6 (5/20)	0.56	–

Table 16: O17p by arm and evaluator (fraction of runs). Primary: t'_{pk} , t_{B99} , O4 with free space, $m > 8$, O10 on adults, O11 on the second half of the births. Prereg.: the full preregistered evaluator. The other columns change one option of the primary evaluator: peak time at $\arg \max N$ or at the end of the plateau; crowded class at $m > m_c$; O5 on the last sustained birth; the preregistered O4.

Arm	primary	prereg.	arg max	plateau end	$m > m_c$	O5 on t_{last}	O4 prereg.
C1	0.99	0.98	1.00	1.00	0.99	0.48	0.99
C1'	0.83	0.93	0.96	0.96	0.83	0.36	0.83
C2	0.70	0.88	0.73	0.81	0.70	0.23	0.78
C1 – AV	0.79	0.90	0.96	0.97	0.79	0.44	0.79
C1 – R1	0.21	0.23	0.22	0.22	0.21	0.12	0.21
C1 – R2	0.00	0.17	0.00	0.00	0.00	0.00	0.00
C1 – R7	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C1, $\alpha = 0$	0.00	0.01	0.00	0.00	0.00	0.00	0.00
$\gamma = 0$	0.00	0.00	0.00	0.00	0.00	0.00	0.00
–R2* ($w = 1$)	0.00	0.15	0.00	0.00	0.00	0.00	0.00
long-lived	0.00	0.00	0.00	0.00	0.00	0.00	0.00
no rules	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C1, single-pass R7	0.99	0.97	1.00	1.00	0.99	0.51	0.99
C1, $\mu_{\text{bg}} = 0.003$	0.55	0.45	0.88	0.99	0.55	0.04	0.55
C1, $\mu_{\text{bg}} = 0.008$	0.06	0.02	0.32	0.72	0.06	0.00	0.06
C1, nest 3 wk	0.83	0.87	0.90	0.90	0.83	0.47	0.83
C1, asynchronous	1.00	1.00	1.00	1.00	1.00	0.71	1.00
C1, async., soft cap.	0.99	0.99	0.99	0.99	0.99	0.72	0.99
C1, $s_{\text{base}} = 0.25$	0.08	0.64	0.07	0.08	0.08	0.04	0.08
C1, $s_{\text{base}} = 0.5$	0.00	0.36	0.00	0.00	0.00	0.00	0.00
C1, ages U{4..52}	0.93	0.90	1.00	1.00	0.93	0.36	0.93
C1, ages U{4..80}	0.63	0.56	0.97	1.00	0.63	0.10	0.63
C1', $\mu_{\text{bg}} = 0.003$	0.28	0.28	0.68	0.89	0.28	0.02	0.28
C1', nest 3 wk	0.60	0.63	0.69	0.69	0.60	0.34	0.60
C1', asynchronous	0.94	0.98	0.99	0.99	0.94	0.39	0.94
C1', $s_{\text{base}} = 0.25$	0.26	0.53	0.30	0.30	0.26	0.17	0.26
C1', ages U{4..52}	0.68	0.70	0.97	0.99	0.68	0.23	0.68
C1', ages U{4..80}	0.31	0.30	0.77	0.94	0.31	0.03	0.31

Table 17: Discriminating power of all criteria, essential (top, as in Table 6 of the main text) and desirable (bottom); primary evaluator, fraction of all runs that pass, NA counting as failure. Columns: the reference ensemble (C1, C1', $\alpha = 0$, C0), the four null models with the initial condition of C1 ($\gamma = 0$; -R2*: $w = 1$ without social learning; long-lived control; no rules: C1 without its four rules), $n = 200$ each, and the short-lived control ($n = 100$). “Generic”: passes in at least two of the four null models. “Discriminates”: upper Wilson bound below 0.8 and Newcombe interval of the difference with C1 excluding zero (long: long-lived control; none: no rules; short: short-lived control).

	C1	C1'	$\gamma=0$	$\alpha=0$	C0	-R2*	long	none	short	generic	discriminates against
O4	1.00	1.00	0.00	0.00	1.00	0.99	0.17	1.00	0.87	yes	$\gamma=0$, $\alpha=0$, long
O5	0.99	0.85	1.00	1.00	1.00	0.97	0.89	0.91	0.25	yes	short
O6	1.00	1.00	0.00	1.00	1.00	1.00	0.99	1.00	0.45	yes	$\gamma=0$, short
O7	1.00	0.99	0.00	1.00	1.00	1.00	0.99	1.00	0.47	yes	$\gamma=0$, short
O10	1.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00	yes	$\gamma=0$
O11	1.00	0.97	0.00	1.00	1.00	0.00	0.00	0.00	0.00	no	$\gamma=0$, -R2, long, none, short
O13	1.00	1.00	0.00	1.00	0.92	1.00	0.00	0.00	0.00	no	$\gamma=0$, long, none, short
O13p	1.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	no	$\gamma=0$, $\alpha=0$, C0, long, none, short
O17p	0.99	0.83	0.00	0.00	0.00	0.00	0.00	0.00	0.00	no	$\gamma=0$, $\alpha=0$, C0, -R2, long, none, short
O10b	0.02	0.01	1.00	1.00	0.18	0.05	0.20	0.12	1.00	no	none
O8	0.62	0.21	0.00	0.97	0.82	0.69	0.39	0.64	0.04	no	C1', $\gamma=0$, long, short
O8b	1.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00	yes	$\gamma=0$
O9	0.45	0.29	0.00	0.49	0.44	0.51	0.39	0.45	0.18	no	C1', $\gamma=0$, short
O11b	1.00	1.00	1.00	1.00	1.00	0.99	0.95	0.93	1.00	yes	none
O14	1.00	1.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	yes	$\alpha=0$

Table 18: Discriminating power of the redefined criteria with the preregistered evaluator (same ensembles and columns as Table 6). The preregistered O10 passes in every model; the preregistered O11 passes vacuously without crowding damage.

	C1	C1'	$\gamma=0$	$\alpha=0$	C0	-R2*	long	none	short	generic	discriminates against
O4	1.00	1.00	0.00	1.00	1.00	1.00	0.17	1.00	0.87	yes	$\gamma=0$, long
O5	1.00	0.99	0.00	1.00	1.00	1.00	0.96	0.99	0.16	yes	$\gamma=0$, short
O10	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	yes	none
O11	0.98	0.94	1.00	1.00	0.98	0.15	0.09	0.21	0.00	no	-R2, long, none, short
O13p	1.00	1.00	0.00	0.01	0.00	1.00	0.00	0.00	0.00	no	$\gamma=0$, $\alpha=0$, C0, long, none, short
O17p	0.98	0.93	0.00	0.01	0.00	0.15	0.00	0.00	0.00	no	$\gamma=0$, $\alpha=0$, C0, -R2, long, none, short

Table 19: Robustness variants ($n = 200$ per arm). O17p with 95% Wilson interval and Newcombe interval of the difference with the reference ensemble; the same arm with the preregistered evaluator and with $\arg \max N$ as peak time; “disc.”: O17p restricted to the criteria that discriminate against the null models (O11 scored at six weeks, O13 and O13p pass and no anti-pattern is triggered), which leaves out the generic O4–O7 and O10; the criterion with the lowest pass rate when $\text{O17p} < 0.95$ (NA counted as failure); median peak density, plateau length (weeks within 2% of the maximum) and Kaplan–Meier median of T_{ext} .

Variant	O17p	Δ vs. reference	prereg.	$\arg \max$	disc.	fails	ρ_{pk}	plateau	T_{ext}
CI , reference	0.99 [0.97, 1.00]	–	0.98	1.00	1.00	–	0.36	51	169
single-pass R7 admission	0.99 [0.97, 1.00]	+0.00 [-0.02, +0.02]	0.97	1.00	1.00	–	0.37	52	168
$\mu_{\text{bg}} = 0.003$	0.55 [0.48, 0.62]	-0.44 [-0.51, -0.37]	0.45	0.88	1.00	O5 0.55	0.33	17	174
$\mu_{\text{bg}} = 0.008$	0.06 [0.03, 0.10]	-0.94 [-0.96, -0.89]	0.02	0.32	1.00	O5 0.06	0.30	10	182
nest period 3 wk	0.83 [0.78, 0.88]	-0.16 [-0.22, -0.11]	0.87	0.90	0.90	O11 0.90	0.31	53	171
asynchronous, hard	1.00 [0.98, 1.00]	+0.01 [-0.01, +0.03]	1.00	1.00	1.00	–	0.43	57	159
asynchronous, soft	0.99 [0.97, 1.00]	+0.00 [-0.02, +0.02]	0.99	0.99	0.99	–	0.42	57	158
$s_{\text{base}} = 0.25$	0.08 [0.05, 0.13]	-0.92 [-0.95, -0.86]	0.64	0.07	0.08	O11 0.08	0.40	53	164
$s_{\text{base}} = 0.5$	0.00 [0.00, 0.02]	-0.99 [-1.00, -0.97]	0.36	0.00	0.00	O11 0.00	0.44	53	165
ages U{4..52}	0.93 [0.89, 0.96]	-0.06 [-0.11, -0.03]	0.90	1.00	1.00	O5 0.93	0.36	28	167
ages U{4..80}	0.63 [0.56, 0.69]	-0.36 [-0.43, -0.30]	0.56	0.97	1.00	O5 0.63	0.34	22	173
CI' , reference	0.83 [0.78, 0.88]	–	0.93	0.96	0.97	–	0.32	50	182
single-pass R7 admission	0.85 [0.79, 0.89]	+0.02 [-0.06, +0.09]	0.91	0.95	0.96	O5 0.89	0.33	49	180
$\mu_{\text{bg}} = 0.003$	0.28 [0.22, 0.34]	-0.56 [-0.63, -0.47]	0.28	0.68	0.96	O5 0.28	0.29	19	183
$\mu_{\text{bg}} = 0.008$	0.04 [0.02, 0.08]	-0.79 [-0.84, -0.73]	0.01	0.24	0.96	O5 0.04	0.26	9	182
nest period 3 wk	0.60 [0.54, 0.67]	-0.23 [-0.31, -0.14]	0.63	0.69	0.69	O11 0.69	0.30	52	181
asynchronous, hard	0.94 [0.89, 0.96]	+0.10 [+0.04, +0.16]	0.98	0.99	0.99	O5 0.94	0.38	56	173
asynchronous, soft	0.97 [0.94, 0.99]	+0.14 [+0.08, +0.19]	0.98	1.00	1.00	–	0.38	56	173
$s_{\text{base}} = 0.25$	0.26 [0.20, 0.32]	-0.57 [-0.65, -0.49]	0.53	0.30	0.31	O11 0.32	0.38	50	175
$s_{\text{base}} = 0.5$	0.00 [0.00, 0.02]	-0.83 [-0.88, -0.77]	0.38	0.01	0.00	O11 0.01	0.43	47	174
ages U{4..52}	0.68 [0.61, 0.74]	-0.15 [-0.24, -0.07]	0.70	0.97	0.99	O5 0.69	0.33	26	182
ages U{4..80}	0.31 [0.25, 0.38]	-0.52 [-0.60, -0.44]	0.30	0.77	0.99	O5 0.31	0.30	21	187

Table 20: Lattice side L at fixed seeding density $N_0/L^2 = 0.44$ ($n = 60$ per size, $T = 600$). Medians over runs; t_{last} is the last surviving birth, $D_{\text{last}} = T_{\text{ext}} - t_{\text{last}}$; the spread is the interquartile range over the median of the Kaplan–Meier estimate of T_{ext} , with bootstrap interval.

L	N_0	n	O17p	ρ_{pk}	f_0	$\Phi_{\text{BO}}^{\text{pers}}$	t_{last}	IQR	D_{last}	T_{ext}	spread
15	100	60	0.82 [0.70, 0.89]	0.35	0.18	0.18	50	29	110	162	0.148 [0.121, 0.194]
20	178	60	0.90 [0.80, 0.95]	0.36	0.17	0.17	62	22	112	175	0.143 [0.110, 0.169]
30	400	60	0.98 [0.91, 1.00]	0.37	0.16	0.17	58	13	110	170	0.076 [0.053, 0.112]
45	900	60	1.00 [0.94, 1.00]	0.37	0.15	0.17	66	16	111	176	0.085 [0.052, 0.113]
60	1600	60	1.00 [0.94, 1.00]	0.37	0.15	0.17	75	14	110	184	0.076 [0.050, 0.103]

Table 21: Estimators of the Calhoun-like fraction of the two 15-factor hypercubes (400 points, 10 runs each), relative to the box and to the sampling scales. Naive: fraction of points with $k/n \geq 0.8$; volume: mean of O17p over the box; EB: mean posterior probability of $p_i \geq 0.8$ under a beta prior centred on a logistic emulator, with dispersion ϕ fitted by beta-binomial maximum likelihood. Bootstrap intervals over points with re-fit; “uncertain”: points with posterior probability between 0.05 and 0.95. The extra runs are 40 additional runs at the 18 points with $0.4 \leq k/n \leq 0.95$ under the primary evaluator.

Evaluator	Runs	naive	volume	EB	ϕ	uncertain
primary	10 per point	0.022 [0.010, 0.040]	0.041 [0.027, 0.060]	0.015 [0.006, 0.029]	1.22	12
primary	+40 at 18 points	0.020 [0.007, 0.036]	0.040 [0.026, 0.059]	0.018 [0.007, 0.032]	0.82	8
preregistered	10 per point	0.077 [0.052, 0.105]	0.110 [0.082, 0.137]	0.066 [0.045, 0.090]	0.63	30
preregistered	+40 at 18 points	0.072 [0.050, 0.098]	0.110 [0.083, 0.137]	0.069 [0.048, 0.092]	0.60	26
$m > m_c$	10 per point	0.028 [0.013, 0.045]	0.054 [0.039, 0.075]	0.020 [0.009, 0.034]	1.00	17
$m > m_c$	+40 at 18 points	0.025 [0.013, 0.043]	0.054 [0.038, 0.073]	0.021 [0.010, 0.035]	0.79	12

Table 22: Founder protocol (P6): four pairs in one cell with mating in the Moore neighbourhood, 30 runs per point. s_0 is the initial social state of the founders and k the number of movement sub-steps per week. O1–O3 is the fraction of runs that pass the latency, growth and deceleration patterns jointly; O17p with 95% Wilson interval.

s_0	k	$a_{\max}$	O1	O1–O3	O17p	O13p	P_{ext}	ρ_{pk}	f_0
0.2	1	120	0.53	0.47	0.13 [0.05, 0.30]	0.64	0.63	0.18	0.43
0.2	1	200	0.67	0.63	0.43 [0.27, 0.61]	0.80	0.93	0.23	0.33
0.2	2	120	0.57	0.23	0.13 [0.05, 0.30]	0.54	1.00	0.16	0.51
0.2	2	200	0.63	0.43	0.23 [0.12, 0.41]	0.74	1.00	0.18	0.46
0.3	1	120	0.47	0.47	0.10 [0.03, 0.26]	0.72	0.67	0.20	0.38
0.3	1	200	0.43	0.43	0.33 [0.19, 0.51]	0.83	0.97	0.24	0.30
0.3	2	120	0.57	0.40	0.17 [0.07, 0.34]	0.54	1.00	0.21	0.37
0.3	2	200	0.37	0.30	0.23 [0.12, 0.41]	0.80	1.00	0.27	0.24
0.5	1	120	0.00	0.00	0.00 [0.00, 0.11]	0.60	0.53	0.20	0.38
0.5	1	200	0.03	0.03	0.03 [0.01, 0.17]	0.90	0.87	0.24	0.29
0.5	2	120	0.10	0.10	0.03 [0.01, 0.17]	0.80	1.00	0.26	0.24
0.5	2	200	0.23	0.23	0.10 [0.03, 0.26]	0.93	1.00	0.27	0.23

Table 23: Transplants that found a colony, by arm and phase (successes/trials; 60 replicates per arm). Phase D is sampled at $1.75 t'_{\text{pk}}$ (median sampling week in parentheses), then at $1.75 t_{\text{pk}}$ with $t_{\text{pk}} = \arg \max N$, and at week 95 in every arm; “+healthy (95)” pools the two mixed phases at week 95 (h.: healthy partners of the other sex). The band samples eight adults past the plasticity window with $0.2 \leq s \leq 0.5$ at the first week in which they exist (median week in parentheses), alone or with healthy partners of the other sex.

Arm	B	D	$1.75 t'_{\text{pk}}$	week 95	+healthy (95)	band	band M + h. F	band F + h. M
C1	52/60	0/60 (60)	0/60	0/60	0/120	5/60 (9)	37/60	17/60
C1'	46/60	0/60 (63)	0/60	0/60	0/120	2/60 (8)	35/60	11/60
C1 – R1	58/60	0/60 (86)	10/60	0/60	34/120	50/60 (9)	57/60	56/60
C1 – R2	50/60	0/60 (65)	0/60	0/60	0/120	7/60 (10)	39/60	7/60
C1 – R1 – R2	58/60	0/60 (82)	3/60	1/60	14/120	46/60 (9)	54/60	43/60
X (C1 animals)	59/60	51/60 (60)	12/60	5/60	41/120	57/60 (9)	60/60	58/60
C0	47/60	0/60 (58)	0/60	0/60	0/120	0/60 (11)	39/60	18/60
C2	49/60	0/60 (110)	0/60	0/60	3/120	8/60 (10)	46/60	32/60

Table 24: Social state and age of the transplanted animals, by arm and phase: median over trials of the per-trial median s of the eight animals / maximum s over all trials; median over trials of the youngest and oldest animal (weeks). Last column: mean social state of the adult descendants (age $\geq a_w$) in the colonies founded by band animals, pooled over the three band phases and weighted by their number (in parentheses).

Arm	D ($1.75 t'_{pk}$)		week 95		band		descendants
	s med./max	ages	s med./max	ages	s med./max	ages	$\bar{s}$ (n)
C1	0.000 / 0.36	37–64	0.000 / 0.30	65–80	0.358 / 0.50	13–13	0.31 (10261)
C1'	0.000 / 0.60	40–66	0.000 / 0.49	60–80	0.354 / 0.50	12–12	0.27 (8156)
C1 – R1	0.003 / 0.37	46–78	0.004 / 0.46	56–80	0.361 / 0.50	13–13	0.36 (39369)
C1 – R2	0.000 / 1.00	40–67	0.000 / 0.61	59–80	0.362 / 0.50	14–14	0.19 (10131)
C1 – R1 – R2	0.025 / 0.80	47–78	0.028 / 0.81	52–80	0.330 / 0.46	13–13	0.23 (31128)
X (C1 animals)	0.000 / 0.36	37–64	0.000 / 0.30	65–80	0.358 / 0.50	13–13	0.32 (38964)
C0	0.000 / 0.60	36–58	0.000 / 0.29	66–80	0.396 / 0.50	15–15	0.11 (7448)
C2	0.000 / 0.63	30–79	0.000 / 0.71	28–79	0.398 / 0.50	14–14	0.35 (17350)

Table 25: Confirmatory contrast between C1 and C2 ($n = 200$ each, $T = 700$; primary evaluator). Binary patterns are compared with two-sided Fisher tests, continuous magnitudes with Mann–Whitney tests, and T_{ext} with a log-rank test with censoring. Holm correction is applied over the 29 comparisons. Entries are fractions of passing runs, or medians, with the number of applicable runs in parentheses. The last column gives the two fractions and the verdict with the preregistered evaluator. Omitted rows show no difference: O10, O13 and O14 pass in every run of both arms; O15, A1, A3 and A4 are identical in both arms; O8b, O11b and strict O13b are constant.

Pattern / magnitude	C1	C2	p	p_{Holm}	Differs	prereg. evaluator
O4	1.000 (200)	0.900 (200)	1.2×10^{-6}	2.2×10^{-5}	yes	1.00 / 1.00 no
O5	0.980 (200)	0.815 (200)	2.3×10^{-8}	4.7×10^{-7}	yes	1.00 / 0.90 yes
O6	1.000 (200)	0.980 (200)	0.12	1.00	no	1.00 / 0.98 no
O7	1.000 (200)	0.980 (200)	0.12	1.00	no	1.00 / 0.98 no
O8	0.645 (200)	0.000 (196)	$< 10^{-16}$	$< 10^{-16}$	yes	0.65 / 0.00 yes
O9	0.400 (200)	0.300 (200)	0.0462	0.74	no	0.40 / 0.30 no
O11	1.000 (200)	0.995 (197)	0.50	1.00	no	0.99 / 0.99 no
O13p	1.000 (200)	0.945 (200)	8.5×10^{-4}	0.014	yes	1.00 / 1.00 no
O15b	0.650 (200)	0.452 (197)	8.2×10^{-5}	0.0015	yes	0.65 / 0.45 yes
O17	0.980 (200)	0.725 (200)	4.0×10^{-14}	8.7×10^{-13}	yes	0.99 / 0.88 yes
O17p	0.980 (200)	0.705 (200)	1.6×10^{-15}	3.8×10^{-14}	yes	0.99 / 0.88 yes
O13b (PASS or MARG.)	0.000 (200)	1.000 (200)	$< 10^{-16}$	$< 10^{-16}$	yes	0.00 / 1.00 yes
A2	0.000 (200)	0.020 (200)	0.12	1.00	no	0.00 / 0.02 no
ρ_{pk} (median)	0.361 (200)	0.42 (200)	$< 10^{-16}$	$< 10^{-16}$	yes	yes
f_0 (median)	0.163 (200)	0.113 (200)	$< 10^{-16}$	$< 10^{-16}$	yes	yes
Φ_{BO}^{pers} (median)	0.174 (200)	0.133 (200)	$< 10^{-16}$	$< 10^{-16}$	yes	yes
$(T_{\text{ext}} - t_{pk})/t_{pk}$ (extinct)	2.06 (200)	1.8 (196)	3.9×10^{-14}	8.7×10^{-13}	yes	yes
$T_{\text{ext}} - t_{\text{last}}$ (extinct)	110 (200)	110 (196)	0.37	1.00	no	no
T_{ext} (KM median; log-rank)	169 (200)	218 (200)	$< 10^{-16}$	$< 10^{-16}$	yes	yes

Table 26: Ensembles of the production program (`results_v2/f2b_<stage>`). Points: design points or arms; per point: runs or transplant replicates; T : horizon in weeks; seed: base seed of the stage.

Stage	Content	points	per point	runs	T	seed	used for
base	reference ensemble: C1, C1', C0, $\alpha = 0$	4	200	800	600	25301	production
c1c2	C1 and C2	2	200	400	700	25318	QC
ref	short-lived and long-lived controls	2	100	200	1000	25302	Q3, controls
nulos	four null models	4	200	800	600	25330	discrimination
abl1	single ablations and variants	9	120	1080	600	25303	Q2
abl2	double to quadruple ablations, $\gamma = 0$	12	40	480	600	25304	ablations
trans	8 transplant arms, 11 phases	8	60	480	300	25306	Q1, O12
amax	$a_{\max} \in \{90, \dots, 150\}$	5	60	300	600	25305	Q4
rob	demographic and update variants	16	200	3200	700	25331	robustness
ci	non-synchronous initial condition	4	200	800	700	25333	robustness
size	$L \in \{15, \dots, 60\}$ at fixed density	5	60	300	600	25334	size
q7aw	a_w varied at $\gamma = 0$	54	20	1080	600	25335	Q7
p1	plane $(r, w) \times \gamma$	112	30	3360	600	25309	Q7, P1
p1aw	plane (r, a_w)	56	30	1680	600	25310	Q6
p2	plane $(\alpha, \kappa), p$	72	30	2160	600	25308	Q5, P2
p5	refuge plane	50	30	1500	600	25311	P5
lhs	Latin hypercube 1	200	10	2000	400	25312	global
lhs2	Latin hypercube 2	200	10	2000	400	25313	global
lhsx	boundary points of the hypercubes	18	40	720	400	25336	global
morris	Morris screening	320	10	3200	400	25307	global
sobol	Saltelli design, 11 factors	3328	6	19968	400	25314	global
robmu	μ_{bg} curve	12	60	720	700	25332	robustness
p4	seeding density and lattice	24	30	720	600	25315	P4
trgrid	transplants on (α, κ)	12	40	480	300	25316	O12
refine	40 boundary points of the planes	40	100	4000	600	25317	planes
p6	founder protocol	12	30	360	900	25319	O1–O3

Table 27: Deviations from the preregistered plan (`docs/preregistered_plan.md`, dated 1 October 2026, 22:02 UTC, before the production runs) and post hoc analyses. “Before production” means after the calibration but before the production runs; “post hoc” means after the runs had been seen. The last block lists the changes that are not preregistered but were fixed, with their decision rules, before the production runs (`docs/production_plan.md`, 2 October 2026, 05:34 UTC).

Change	When	Affects	Treatment
O13p (persistent isolated class) defined and made part of the primary outcome, together with R7	before production, after seeing that C0 has no persistent class	O17p, Q2, necessity of R7	declared as circular for R7 (Section 9)
O6: rebound count also detects slow regrowth still rising at censoring	post hoc (2 Oct., 02:44)	one C1 run of 200 under the preregistered design; none in the production runs	primary evaluator
O13b: reading of the window against the median birth week (Fig. 18)	post hoc	the interpretation of O13b in C1 and C2	labelled post hoc
Robust spread of T_{ext} (Kaplan–Meier interquartile range over median)	post hoc (the plan had $\text{CV} \leq 0.10$ for Q3)	the claims about the spread of the extinction time	labelled post hoc; the equivalence of C1' with the long-lived control found under the preregistered design does not hold in the production runs
Ensembles of C1' and the C1–C1' contrast	post hoc	minimality, the role of AV	labelled post hoc
Subsets of the hypercubes (nested conditions on c_R , m_c , $f_R c_R$, π)	post hoc	interpretation of the Calhoun-like fraction	descriptive only
Re-scoring with other class thresholds θ and ensemble thresholds τ (Appendix C.6)	post hoc	conditionality of minimality	descriptive only
Cure model for the extinction time	post hoc	survival summaries	no test reported
Holm p -value of Q6 computed from the binomial likelihood	as planned	Q6	the quasi-binomial value enters the Holm sequence; no decision changes
<i>Not preregistered, fixed before the production runs</i>			
Hard refuge capacity enforced strictly (pups and rejected movers counted); the single-pass admission kept as a variant	before the production runs	every stage with refuges; all stages run with seeds different from those of the preregistered design	both admission rules reported side by side (no change in O17p)
Primary evaluator redefined: peak time at the start of the plateau (t'_{pk}); end of natality in O5 at 99% of the births; O4 also requires free space at the peak ($f_0 \geq 0.10$, an informal target of the calibration); crowded class $m > 8$, independent of m_c ; O10 on the adult social state; O11 on the second half of the births, not applicable when the late cohorts are empty; NA counts as failure in O17p	fixed after exploratory runs with the primary evaluator and before the production runs	every pattern normalised by t_{pk} , O4, O5, O10, O11, O13, O13p, O17p in every stage; the verdicts of Q1 (prediction), Q2 (null) and Q6 (prediction)	the full preregistered evaluator, and each option changed separately, reported next to the primary one (Appendix C.6)
O5 birth reference t_{B99} instead of the last sustained birth t_{last}	after exploratory runs had shown that t_{last} with t'_{pk} fails O5 in most C1 runs (0.48 in the production runs, Table 16); a choice informed by the result	Q2: it is the only option on which the rejection of its null hypothesis depends (with the preregistered evaluator 5 of 7 null hypotheses are rejected and Q2 is not; with the primary one, 6 of 7)	both birth references reported for every arm; the count with the preregistered evaluator given next to the primary one (Section 8.4)
Confirmatory test of the equivalence of the exponents of r and a_w at $\gamma = 0$ (TOST, margin $[2/3, 3/2]$, stage a_w varied)	before the production runs	Q7 with a_w varied	the design cannot carry it out as inference (51 of 54 points at $P_{\text{ext}} \in \{0, 1\}$, separated logistic); replaced by a non-parametric summary per window (Table 15); the planned estimates are kept, labelled, in <code>results_numbers.json</code>
O8 (senescent decline) of C1: 0.71 under the preregistered design, 0.62 in the production runs (0.64 with the single-pass admission and the seeds of the production runs)	observed after the production runs	a desirable pattern; no decision	reported; not a conclusion
Transplant of Q1 taken at $1.75 t'_{\text{pk}}$ instead of $1.75 t_{\text{pk}}$, sampling at t_{pk}	before the production runs	Q1, O12	both times reported; comparisons between arms only

Table 28: Development ensembles of C0, C1 and C2: pattern pass fractions (fraction of PASS runs, 95% Wilson interval; superseded by Table 5). $N_0 = 400$, $T = 600$ weeks. Transplants: $P_T(\cdot)$ is the fraction of transplants that found a colony. “Persistent O13” requires the isolated class to stay isolated for ≥ 4 consecutive weeks ($\Phi_{\text{BO}}^{\text{pers}} \geq 0.1$ and $\Phi_{\text{CR}} \geq 0.1$). “Last half” is the fraction of runs in which the median birth time of the persistent isolated class is later than the median birth time of all births. A dash means not measured. The O13b entries for C1 come from an independent 60-run ensemble.

Pattern		C0 ($n = 120$)	C1 ($n = 60$)	C2 ($n = 60$)
O4 peak below capacity	E	1.00 [0.97, 1]	1.00 [0.94, 1]	1.00 [0.94, 1]
N_{pk}/K ; empty cells at peak		0.42; 0.18	0.37; 0.16	0.40; 0.13
O5 natality collapses at high N	E	1.00 [0.97, 1]	1.00 [0.94, 1]	0.98 [0.91, 1]
O6 no rebound	E	1.00 [0.97, 1]	1.00 [0.94, 1]	0.97 [0.89, 0.99]
O7 extinction	E	1.00 [0.97, 1]	1.00 [0.94, 1]	0.97 [0.89, 0.99]
O10 deterioration before peak	E	1.00 [0.97, 1]	1.00 [0.94, 1]	1.00 [0.94, 1]
O11 crowded-born cohorts fail	E	1.00 [0.97, 1]	1.00 [0.94, 1]	1.00 [0.94, 1]
O12 transplant: $P_T(\text{B})$	E	0.82 [0.73, 0.88]	0.83 [0.72, 0.91]	0.87 [0.76, 0.93]
$P_T(\text{D})$		0/100 [0, 0.04]	0/60 [0, 0.06]	0/60 [0, 0.06]
O13 two deteriorated classes	E	0.98 [0.93, 0.99]	1.00 [0.94, 1]	1.00 [0.94, 1]
persistent O13		0 ($\Phi_{\text{BO}}^{\text{pers}} = 0.006$)	1.00 [0.94, 1]	1.00 [0.94, 1]
O17 Calhoun-like runs	E	0.98 [0.93, 0.99]	1.00 [0.94, 1]	0.95 [0.86, 0.98]
O8 senescent decline	D	0.75 [0.67, 0.82]	0.72 [0.59, 0.82]	0.00 [0, 0.06]
O8b slow initial decline	D	1.00 [0.97, 1]	1.00 [0.94, 1]	0.92 [0.82, 0.96]
O9 old, female-biased remnant	D	0.35 [0.27, 0.44]	0.57 [0.44, 0.68]	0.17 [0.09, 0.28]
O11b few late-born females conceive	D	0.99 [0.95, 1]	0.96 [0.88, 0.99]	1.00 [0.94, 1]
O12b mixed transplant	D	0/200 [0, 0.02]	0/120 [0, 0.03]	0.07 [0.03, 0.13]
O13b isolated class is later-born	D	–	0 [0, 0.06]	0 [0, 0.06] (MARG 60/60)
$\Delta b/t_{\text{pk}}$; last half		–	–0.02; 0/60	+0.13; 58/60 [0.89, 0.99]
O14 aggregation with free space	D	0.95 [0.90, 0.98]	1.00 [0.94, 1]	1.00 [0.94, 1]
O15 infant mortality $\rightarrow 100\%$	D	0.02 [0.00, 0.06]	0 [0, 0.06]	0.03 [0.01, 0.11]
O15b conceptions without viable young	D	0.33 [0.25, 0.41]	0.58 [0.46, 0.70]	0.63 [0.50, 0.74]
A1–A4 anti-patterns triggered		0/120 [0, 0.03]	0/60 [0, 0.06]	A2: 2/60; others 0

Table 29: Ablations of C0 (40 replicates per row; $N_0 = 400$, longevity preset). Fractions of PASS runs; O17 with its 95% Wilson interval. $\rho_{\text{pk}} = N_{\text{pk}}/K$; f_0 = fraction of empty cells at the peak; Φ_{BO} , Φ_{CR} = isolated and crowded deteriorated adults (maximum over t_{pk} , $t_{\text{pk}} + 10$, $t_{\text{pk}} + 20$).

Variant	O4	O7	O11	O13	O17	ρ_{pk}	f_0	Φ_{BO} / Φ_{CR}
C0 ($\alpha = 4$, $\kappa = 0.15$, R1, R2)	1.00	1.00	0.98	0.98	0.95 [0.83, 0.99]	0.42	0.19	0.12 / 0.15
C0, $\alpha = 6$	1.00	1.00	0.95	1.00	0.95 [0.83, 0.99]	0.35	0.32	0.13 / 0.17
C0, $\kappa = 0.10$	1.00	1.00	0.95	0.98	0.93 [0.80, 0.97]	0.37	0.33	0.11 / 0.31
C0 without R2	1.00	0.95	0.00	0.05	0.00 [0, 0.09]	0.51	0.12	0.08 / 0.18
C0 without AV	1.00	1.00	0.68	0.00	0.00 [0, 0.09]	0.32	0.76	0.05 / 0.87
C0 without R1	1.00	0.75	1.00	0.50	0.40 [0.26, 0.55]	0.46	0.16	0.10 / 0.18
C0, $\alpha = 6$, without R1	1.00	0.98	1.00	0.85	0.82 [0.68, 0.91]	0.38	0.30	0.11 / 0.21
C0, no inheritance ($w = 0$, $r = 0.15$)	1.00	1.00	1.00	0.68	0.68 [0.52, 0.80]	0.46	0.15	0.11 / 0.16
Long-lived control ($\alpha = 1.4$)	0.25	1.00	0.10	0.00	0.00 [0, 0.09]	0.83	0.58	0.04 / 0.91

Table 30: One-at-a-time sensitivity of C0 (30 replicates per row). O17 = fraction of Calhoun-like runs with 95% Wilson interval. In every row the other essential patterns stay at ≥ 0.93 .

Variation	O13	O17	Variation	O13	O17
C0	0.97	0.97 [0.83, 0.99]	$a_{\max} = 90$	1.00	0.93 [0.79, 0.98]
$\alpha = 3$	0.70	0.70 [0.52, 0.83]	$a_{\max} = 150$	0.97	0.93 [0.79, 0.98]
$\alpha = 5$	1.00	1.00 [0.89, 1]	$\gamma = 0.07$	0.80	0.80 [0.63, 0.90]
$\kappa = 0.125$	1.00	0.97 [0.83, 0.99]	$\gamma = 0.15$	0.97	0.90 [0.74, 0.97]
$\kappa = 0.175$	0.70	0.67 [0.49, 0.81]	$w = 0.1$	0.97	0.97 [0.83, 0.99]
$p = 2$	0.10	0.10 [0.03, 0.26]	$w = 0.3$	0.90	0.87 [0.70, 0.95]
$p = 8$	0.93	0.93 [0.79, 0.98]	$a_w = 8$	0.97	0.93 [0.79, 0.98]
$r = 0.07$	1.00	1.00 [0.89, 1]	$a_w = 16$	0.87	0.87 [0.70, 0.95]
$r = 0.15$	0.17	0.17 [0.07, 0.34]			

Table 31: Founder protocol (four pairs of 7-week-old animals in one cell). Fractions of PASS runs, with 95% Wilson intervals where $n \geq 30$; counts for the exploratory row with $n = 8$. “No latency” founders start with $s = 1$; “adaptation” founders start with $s = 0.3$. Same-cell rows use $s = 1$.

Variant	n	ρ_{pk}	O1	O2	O3	O5	O13	O17
C0, same-cell mating, 1 sub-step	30	0.02	0.43	0.33	1.00	0.83	0.55	0.13 [0.05, 0.30]
C0, same-cell mating, 4 sub-steps	30	0.01	0.10	1.00	1.00	0.33	0.00	0 [0, 0.11]
C1, Moore mating, 2 sub-steps, no latency	8	0.27	0/8	8/8	8/8	5/8	5/8	0/8
C1, Moore mating, 2 sub-steps, adaptation	60	0.24	0.58 [0.46, 0.70]	0.97 [0.88, 0.99]	0.71 [0.59, 0.81]	0.65 [0.52, 0.76]	0.54 [0.42, 0.66]	0.17 [0.09, 0.28]

Table 32: Generations in C1 (reference ensemble, 200 runs). Births: mean over runs and share of all births. s_{nb} : social state at birth; $s(a_w)$: at the end of the critical window, after which s can only decrease; R_{net} : daughters per female of that generation; “bred”: fraction of females with at least one surviving pup. Founders are 400 animals of age 4 weeks with $s = 1$. Later generations add 0.1% of the births.

Generation	Births	Share	Median birth week	s_{nb}	$s(a_w)$	$s(a_w) \geq 0.5$	Pups per female	R_{net} (bred)
F0	400	–	–	1	0.90	91%	7.6	3.82 (73%)
F1	1529	69.2%	10	0.19	0.23	20%	0.80	0.40 (16%)
F2	611	27.6%	19	0.14	0.13	6%	0.22	0.11 (6%)
F3	67	3.0%	28	0.10	0.11	2%	0.11	0.05 (4%)
F4	4	0.2%	38	0.07	0.08	0	0.06	0.03 (2%)

Table 33: Deterioration time and generation time (medians over runs; n runs). t_{on} : first week with 5% of the individuals in cells with $m > m_c$. $t_{1/2}$: first week with mean $s < 0.5$ over the adults (age $\geq a_{\text{min}} = 6$, as in O10), over the animals past the critical window (age $\geq a_w = 12$, the definition used in the text) and over the founders (F0). T_{gen} : interval between the median birth weeks of F1 and F2. $\Gamma = (t_{1/2} - t_{\text{on}}) / T_{\text{gen}}$, with medians of numerator and denominator; with the mean age of the mothers at birth as T_{gen} (14.3 weeks in C1) $\Gamma \approx 0.7$, 0.60–0.66 in C1', –R2*, long-lived and no rules, and 0.85 at $\gamma = 0.02$, $s_0 = 0$, so Γ is of order one under either definition. $\langle g \rangle_B$: mean generation of the births. Upper block: replicates of the reference ensemble and of the null ensemble (same seeds; –R2*: $w = 1$ without social learning; none: C1 without its four rules). Lower block: C1 with crowding damage γ and newborn base s_0 varied (other seeds). Script: `scripts/analysis/gamma_v3.py`.

Variant	n	t_{on}	$t_{1/2}$: $a \geq a_{\text{min}}$ / $a \geq a_w$ / F0	T_{gen}	Γ : $a \geq a_{\text{min}}$ / $a \geq a_w$ / F0	$\langle g \rangle_B$; R_{net} F1	O17p
C1	200	6	13 / 16 / 16	9	0.8 / 1.1 / 1.2	1.34; 0.40	0.995
C1'	30	5.5	12 / 15 / 15	9	0.8 / 1.0 / 1.1	1.34; 0.36	0.87
C0	30	8	15 / 17 / 18	7	0.9 / 1.3 / 1.4	1.36; 0.45	0
C1, $\alpha = 0$	30	13	22 / 24 / 26	5	1.8 / 2.4 / 2.6	1.57; 0.89	0
–R2* (null)	30	7	14 / 15 / 15	6.5	1.1 / 1.2 / 1.2	1.62; 0.69	0
Long-lived (null)	30	11	19 / 19 / 20	5	1.6 / 1.6 / 1.8	1.95; 1.21	0
No rules (null)	30	8	15 / 15 / 15	5	1.4 / 1.6 / 1.6	1.59; 0.66	0
$\gamma = 0.1$, $s_0 = 0$	20	6	13 / 16 / 16	8.5	0.9 / 1.2 / 1.2	1.35; 0.39	0.95
$\gamma = 0.05$, $s_0 = 0$	20	5	14 / 17 / 17.5	8	1.1 / 1.4 / 1.5	1.37; 0.43	1.00
$\gamma = 0.02$, $s_0 = 0$	20	6	15 / 18 / 20	8	1.3 / 1.6 / 1.8	1.42; 0.52	0.85
$\gamma = 0.1$, $s_0 = 0.25$	20	6	14 / 15 / 15	7	1.1 / 1.3 / 1.3	1.51; 0.62	0.05
$\gamma = 0.05$, $s_0 = 0.25$	20	6	14 / 15.5 / 16	7	1.1 / 1.4 / 1.4	1.54; 0.66	0
$\gamma = 0.02$, $s_0 = 0.25$	20	6	16 / 17 / 17.5	7	1.3 / 1.6 / 1.6	1.64; 0.84	0

Table 34: Generational structure (medians over runs; n runs). Share of births of F1, F2 and later generations; mean generation of the births $\langle g \rangle_B$; net reproductive number of the founders and of F1; $t_{1/2}$: first week with mean $s < 0.5$ over the adults ($a \geq a_{\text{min}}$) and over the animals past the window ($a \geq a_w$; Table 33); plateau: weeks with $N \geq 0.98N_{\text{pk}}$. Upper block: the reference ensemble ($t_{1/2}$ for $a \geq a_w$ from 30 replicates for C1', C0 and $\alpha = 0$). Lower block: the first 20 replicates of each production arm of Section 8.5 (30 for the long-lived null model), re-run with the same seeds and a register of generations; their O17p over $n = 200$ is in that section.

Variant	n	F1 / F2 / F3+ (%)	$\langle g \rangle_B$	R_{net} F0 / F1	$t_{1/2}$: $a \geq a_{\text{min}}$ / $a \geq a_w$	Plateau
C1	200	69 / 28 / 3.2	1.34	3.8 / 0.40	13 / 16	51
C1'	200	70 / 26 / 4.0	1.34	3.4 / 0.37	12 / 15	50
C0	200	68 / 29 / 3.1	1.35	4.4 / 0.44	15 / 17	54
C1, $\alpha = 0$	200	50 / 44 / 6.2	1.56	5.9 / 0.87	22 / 24	59
C1, initial ages 4–52 weeks	20	68 / 29 / 3.5	1.36	3.8 / 0.42	12 / 14.5	28.5
C1, initial ages 4–80 weeks	20	62 / 33 / 4.9	1.43	3.4 / 0.52	12.5 / 15	22
C1, nest period of 3 weeks	20	79 / 19 / 1.1	1.22	3.7 / 0.25	12 / 15	51.5
C1, $\mu_{\text{bg}} = 0.003$ per week	20	68 / 28 / 3.3	1.36	3.8 / 0.41	13 / 16	19
C1, $\mu_{\text{bg}} = 0.008$ per week	20	65 / 30 / 5.0	1.40	3.7 / 0.48	14 / 17	11
C1, asynchronous update	20	67 / 29 / 3.0	1.36	4.5 / 0.45	14 / 16	57
C1, $s_0 = 0.25$	20	58 / 34 / 7.6	1.51	3.7 / 0.60	14 / 15	52
C1, $s_0 = 0.5$	20	52 / 38 / 11	1.60	3.6 / 0.72	14 / 15	54
Long-lived, no new rules (null)	30	35 / 42 / 22	1.95	4.9 / 1.21	19 / 19	45.5

Table 35: Chronology of C1 (reference ensemble, 200 runs; medians, interquartile range in brackets) and of Universe 25 (day/7; [5]). t'_{pk} : first week with $N \geq 0.98N_{\text{pk}}$. “Functional B”: growth while the animals past the critical window ($a \geq a_w$), which newborns do not dilute, keep a mean $s \geq 0.5$ (Table 33); the founders cross 0.5 in the same week. The adult mean ($a \geq a_{\text{min}}$, as in O10) crosses 0.5 three weeks earlier because pups count as adults while still inside the window.

Event	C1 (week)	$/t_{\text{pk}}$	Universe 25 (week)	$/t_{\text{pk}}$
First litters (end of A)	3 [0]	0.06	15	0.19
Mean $s < 0.5$, adults $a \geq a_{\text{min}}$	13 [1]	0.24	—	—
Mean $s < 0.5$, $a \geq a_w$ (end of functional B)	16 [1]	0.28	45 (end of B)	0.56
Half of all births	13 [1]	0.23	≈ 62 (our estimate)	≈ 0.8
90% / 99% of all births	25 [2] / 40 [5]	0.45 / 0.75	—	—
Start of the plateau, t'_{pk}	34 [4]	0.63	no plateau	—
$t_{\text{pk}} = \arg \max N$	55 [10]	1	80	1
Last surviving birth	59 [16]	1.00	86	1.07
Last conception	61 [16]	1.02	131	1.64
End of the plateau (first deaths)	86 [1]	1.54	—	—
Extinction	169 [15]	3.07	254 (projected)	3.18

Table 36: Scales of C1 and of Universe 25 [5]. Last column: calibrated in this work (C), taken from Calhoun (U), a modelling choice of the core model (M), or emergent (E). D_{eff} in cells² per week. [†]Our estimate. [‡] f_R and π are calibrated; $c_R = 2$ is the isolation threshold of O13, not the capacity of a nest box. [§]Used informally in the calibration and part of the primary O4 ($f_0 \geq 0.10$; Section 5). The model’s t_{det} is measured on the animals past the critical window (Table 33); for Universe 25 it is the duration of phase C, a different operational definition, so the two values of Γ compare only in order of magnitude.

Quantity	Model (C1)	Universe 25	
Time step	1 week	—	M
Resting sites	900 cells	256 nest boxes	M
Comfortable occupancy	$m_c = 8$	15 per box	M
K_γ (sites $\times$ occupancy)	7200 (5580 with refuges)	3840	—
Isolation sites	270 refuges, 540 places	high boxes, 25–50% [†]	C [‡]
Initial population	400, age 4 weeks	8, age 7 weeks	M
Peak; N_{pk}/K_γ	2600; 0.36	2200; 0.57	E
Occupancy per site at the peak	2.9	8.6	E
Empty sites at the peak	16%	20%	E [§]
Movement	≤ 1 cell/week; $D_{\text{eff}} \approx 0.13$	enclosure crossed daily	C (α)
Contact neighbourhood	3×3 cells	cell (0.36 m ²), whole floor	M
Maturity / menopause / life	6 / 80 / 120 weeks	6–7 / 80 / 114 weeks	U
Litter; interval between litters	6.55; ≥ 3.3 weeks	6–8; 3–4 weeks [†]	M
Gestation; weaning	none; none	3 weeks; 3 weeks	—
Critical window	12 weeks	youth	C
Doubling time during growth	6–6.5 weeks	4.7–7.9 (B), 21 (C) weeks	E
Generation time T_{gen}	≈ 9 weeks	≈ 10 weeks [†]	E
Deterioration time t_{det} ($a \geq a_w$)	10 weeks	≈ 35 weeks (phase C) [†]	E
$\text{Da} = t_{\text{mix}}/t_{\text{det}}$	$\approx 1.7 \times 10^2$	$\lesssim 4 \times 10^{-3}$	E
$\Gamma = t_{\text{det}}/T_{\text{gen}}$	1.1	$\approx 3.5^{\dagger}$	E

Table 37: Decomposition of the extinction time (reference ensemble, 200 runs per model; medians, interquartile range in brackets; one run of **C1'** is censored).

Model	T_{ext}	IQR/median	t_{last}	$T_{\text{ext}} - t_{\text{last}}$	$\text{corr}(T_{\text{ext}}, t_{\text{last}})$	τ_B
C1	169 [15]	0.089	59 [16]	111 [5]	0.91	6.9
C1'	182 [22]	0.121	71 [22]	110 [6]	0.98	8.7
C0	166 [14]	0.084	56 [13]	111 [5]	0.92	6.1
C1, $\alpha = 0$	155 [7]	0.045	45 [10]	111 [6]	0.89	2.6